



Five-Year Review Report

Pursuant to CERCLA

Fourth Five-Year Review Report

Northernair Plating Superfund Site
Cadillac, Wexford County, Michigan



Prepared by:

U.S. Environmental Protection Agency
Region 5
Chicago, Illinois

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In conjunction with:

Michigan Department of Natural Resources and Environment
Lansing, Michigan

Approved By:

for Richard C. Karl, Director
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9/17/10
Date

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List of Acronyms

ARAR	Applicable or Relevant and Appropriate Requirement
CERCLA	Comprehensive Environmental Response, Compensation, and Liability Act
CFR	Code of Federal Regulations
Cr(VI)	Hexavalent Chromium
ESD	Explanation of Significant Difference
GAC	Granular Activated Carbon
IC	Institutional Control
LDFA	Local Development Finance Authority
MCL	Maximum Contaminant Level
MDEQ	Michigan Department of Environmental Quality
MDNR	Michigan Department of Natural Resources
MRL	Method Reporting Level
NCP	National Contingency Plan
NPL	National Priorities List
O&M	Operation and Maintenance
OU	Operable Unit
PRP	Potentially Responsible Party
PTAS	Packed Tower Air Stripping
RA	Remedial Action
RCRA	Resource Conservation and Recovery Act
RD	Remedial Design
RI/FS	Remedial Investigation/Feasibility Study
ROD	Record of Decision
RPM	Remedial Project Manager
SVE	Soil Vapor Extraction

SWRAU	Site-wide Ready for Anticipated Use
TCE	Trichloroethylene
TCL	Target Cleanup Level
UAO	Unilateral Administrative Order
UCL	Upper Confidence Level
USEPA	United States Environmental Protection Agency
VOC	Volatile Organic Compound

EXECUTIVE SUMMARY

The Northernnaire Plating Superfund Site is located in Cadillac, Michigan, in an industrial park approximately one square mile in area. Over 40 manufacturing plants, including the Northernnaire Plating Superfund Site ("Northernnaire" or the "Site") as well as another Superfund site, the Kysor Industrial Corp. Site, are located in the industrial park. The primary contamination of concern at the Northernnaire Site was chromium contamination in groundwater. Groundwater samples from the city of Cadillac municipal well field, located 1,300 feet northeast of the Northernnaire Site, have not shown hexavalent chromium or total chromium above detection limits since 2001. The historical chromium groundwater plume from the Northernnaire Site is not considered a probable threat to the municipal well field.

The Northernnaire Site was listed on the National Priorities List (NPL) on September 8, 1983. The cleanup of the Site included a removal action in 1983, a source control remedy (Operable Unit 1), implemented from 1988 to 1989, and an ongoing groundwater remedial action (Operable Unit 2) that began in 1996.

The removal action in 1983 addressed the disposal of containers, drums, and process wastes inside a building on the Site, decontamination of the building interior, and excavation and removal of a contaminated sewer line. Operable Unit 1 (OU1) focused on removing contaminated soil found on the Site. The main component of Operable Unit 2 (OU2) is the extraction, treatment and monitoring of groundwater.

The remedy for the Northernnaire Site is protective of human health and the environment. No current exposure pathways exist and the remedy is functioning as planned. The soil removal in OU1 served to eliminate the risks associated with direct contact with, and ingestion of, contaminated soil. It also eliminated areas of significant soil contamination that would otherwise act as a continuing source of groundwater contamination and would work counter to the groundwater extraction and treatment system.

The OU2 groundwater treatment system is achieving discharge limits and performance standards for hexavalent and total chromium. The groundwater extraction and treatment system began operating in 1996. For almost nine years, since May 2001, the target cleanup level (TCL) for hexavalent chromium (50 ug/l), as established in the OU2 Record of Decision (ROD), has been achieved in all ten compliance monitoring wells and in the chromium groundwater extraction well. The additional extraction and monitoring well, tested since 2006 also show no chromium contamination above the TCLs.

The remedy for the Northernnaire Plating Site required in the OU2 ROD continues to be protective of human health and the environment in both the short and long term. Cleanup goals have been achieved throughout the plume of chromium-contaminated groundwater.

This is the fourth Five-Year Review report for the Northernnaire Site. The report covers the Northernnaire Site only. A separate Five-Year Review report was prepared for the Kysor Industrial Corp., Site.

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Five-Year Review Summary Form

SITE IDENTIFICATION		
Site name (from WasteLAN): Northernaire Plating		
EPA ID (from WasteLAN): MID0020883609		
Region: 5	State: MI	City/County: City of Cadillac, Wexford County
SITE STATUS		
NPL status: ☒ Final ☐ Deleted ☐ Other (specify)		
Remediation status (choose all that apply): ☐ Under Construction ☒ Operating ☐ Complete		
Multiple OUs?* ☒ YES ☐ NO	Construction completion date: 09/23/1996	
Has site been put into reuse? ☒ YES ☐ NO		
REVIEW STATUS		
Lead agency: ☒ EPA ☐ State ☐ Tribe ☐ Other Federal Agency		
Author name: Sheila A. Sullivan		
Author title: RPM	Author affiliation: U.S. EPA, Region 5	
Review period:** 09/30/2005 to 07/15/2010		
Date(s) of site inspection: May 21, 2010		
Type of review: ☒ Post-SARA ☐ Pre-SARA ☐ NPL-Removal only ☐ Non-NPL Remedial Action Site ☐ NPL State/Tribe-lead ☐ Regional Discretion		
Review number: ☐ 1 (first) ☐ 2 (second) ☐ 3 (third) ☒ (fourth)		
Triggering action: ☐ Actual RA On-site Construction at OU # _____ ☐ Actual RA Start at OU# _____ ☐ Construction Completion ☒ Previous Five-Year Review Report ☐ Other (specify)		
Triggering action date (from WasteLAN): 09/30/2005		
Due date (five years after triggering action date): 09/30/2010		

* ["OU" refers to operable unit.]

** [Review period should correspond to the actual start and end dates of the five-year review in WasteLAN.]

Five-Year Review Summary Form, cont'd.

Issues:

There are no issues associated with the Northernair remedial action. Long-term remedial action goals for the Site have been met. Hazardous substances above health-based limits are no longer present on the Site above levels that allow for unlimited use and unrestricted exposure. Future Five-Year Reviews will no longer be necessary.

Recommendations and Follow-up Actions:

None

Protectiveness Statement(s):

OU I – Soil

The OUI remedy for soil removal at the Northernair Site is protective of human health and the environment in both the short and long term. The removal eliminated the risks associated with direct contact with, and ingestion of, contaminated soil. It also eliminated areas of significant soil contamination that would otherwise act as a continuing source of groundwater contamination and would work counter to the groundwater extraction and treatment system.

OU II

The remedy for the Northernair Plating Site required in the OU2 ROD for groundwater remediation is protective of human health and the environment in the short and long term. Cleanup goals have been achieved throughout the plume of chromium-contaminated groundwater.

Site-wide

The entire Northernair Plating Site is protective of human health and the environment in both the short term and long term.

Other Comments:

None

Environmental Indicator Data:

Date of last Regional Review of Human Environmental Exposure Indicator (from CERCLIS):
5/19/2010

Human Exposure Survey Status (from CERCLIS): **Long-term Human Health Protection Achieved**

Date of Last Regional Review of Groundwater Migration Indicator (from CERCLIS): **5/19/2010**

Groundwater Migration Survey Status (from CERCLIS): **Contaminated Groundwater Migration Under Control**

Ready for Reuse Determination Status (from CERCLIS): **Site-wide Ready for Reuse**

FIVE-YEAR REVIEW REPORT

NORTHERNAIRE PLATING SUPERFUND SITE WEXFORD COUNTY, MICHIGAN

I. INTRODUCTION

Authority and Purpose

The purpose of a five-year review is to determine whether the remedy at a site is protective of human health and the environment. The methods, findings, and conclusions of reviews are documented in five-year review reports. In addition, five-year review reports identify issues found during the review, if any, and identify recommendations to address them.

The United States Environmental Protection Agency (EPA) is preparing this Five-Year Review report pursuant to CERCLA §121 and the National Contingency Plan (NCP). CERCLA §121 states:

[i]f the President selects a remedial action that results in any hazardous substances, pollutants, or contaminants remaining at the site, the President shall review such remedial action no less often than each five years after the initiation of such remedial action to assure that human health and the environment are being protected by the remedial action being implemented. In addition, if upon such review it is the judgment of the President that action is appropriate at such site in accordance with section [104] or [106], the President shall take or require such action. The President shall report to the Congress a list of facilities for which such review is required, the results of all such reviews, and any actions taken as a result of such reviews.

EPA interpreted this requirement further in the NCP. 40 CFR §300.430(f)(4)(ii) states:

[i]f a remedial action is selected that results in hazardous substances, pollutants, or contaminants remaining at the site above levels that allow for unlimited use and unrestricted exposure, the lead agency shall review such action no less often than every five years after the initiation of the selected remedial action.

EPA Region 5 conducted the Five-Year Review of the remedy being implemented at the Northernnaire Plating Superfund Site in Wexford County, Michigan. This review was conducted by the EPA Remedial Project Manager, Sheila Sullivan, and Jeffrey Dietlin, city of Cadillac utilities, from September 2005 through June 2010. Assistance was provided by Nick Swiger and Steve Cunningham of the MDNRE Cadillac district office, and William Bolio of the MDNRE Lansing office. This report documents the results of the review. The final review report will be placed in the EPA site files and at the local repositories for the Northernnaire Site at the Cadillac-Wexford County Public Library, 411 South Lake Street, Cadillac, Michigan, and the Cadillac City-Municipal Complex, 200 North Lake Street, Cadillac, Michigan. This is the fourth Five-Year Review for the Northernnaire Superfund Site.

The triggering action for this statutory review is the last Five-Year Review completed on September 30, 2005. This Five-Year Review is required due to the fact that hazardous substances, pollutants, or contaminants remained at the Site above levels that allowed for unlimited use and unrestricted exposure (UU/UE). Currently no contamination remains at the Site above levels that allow UU/UE; therefore this will serve as the last Northernaire Five-Year Review.

II. CHRONOLOGY

Table 1 - Chronology of Site Events

EVENT	DATE
Northernaire Plating, Inc., begins operation	1971
Hexavalent chromium detected in two private drinking wells approximately two blocks downgradient of Northernaire Site	1978
MDNR orders Northernaire to conduct an RI/FS but Northernaire states no funds are available	1978
Northernaire ceases operation	May 1981
Children receive chemical burns from hazardous waste stored outside facility	1981
MDNR inspects Site	July 19, 1982
USEPA and MDNR conduct emergency removal	July 1983 to August 1983
Final NPL listing	September 8, 1983
U.S. EPA refers case to USDOJ for cost recovery	March 13, 1984
State-led, federally-funded RI/FS	1984 to 1985
State completes focused feasibility study	July 22, 1985
ROD (OU1) for source control signed	September 11, 1985
OU1 remediation	October/November 1988; March 1991
Court judgment granting all removal response costs	May 6, 1988
FS for Cadillac area groundwater completed	August 1988
ROD (OU2) for groundwater cleanup signed	September 29, 1989
Administrative Order (AO) for remedial design of OU2 signed	May 16, 1990
Explanation of Significant Difference (ESD) signed	March 3, 1994

EVENT	DATE
Two separate Unilateral Administrative Orders (UAOs) for remedial action (RA) issued (Kysor Industrial Corp. and Four Winns/A.H. Joynt)	January 30, 1995
Third UAO for RA issued (Nothernaire Plating)	April 11, 1995
On-site construction for OU2 begins	June 29, 1995
First Five-Year Review completed	September 28, 1995
OU2 Remedial Action begins	1996
Second Five-Year Review completed	July 26, 2000
Third Five-Year Review completed	September 30, 2005
Site-wide Ready for Anticipated Use (SWRAU) memo signed	January 31, 2008
Fourth Five-Year Review Site inspection conducted	May 21, 2010

III. BACKGROUND

Physical Characteristics

The Northernnaire Plating Site ("Northernnaire Site" or the "Site") is located on the northwestern corner of Sixth Street and Eighth Avenue, at 1002 Sixth Street, in Cadillac, Michigan (see Figure 1). The Site is about thirteen acres in area. The center of the city of Cadillac, which has a population of about 10,000, is about one-half mile to the southeast. The city of Cadillac is the county seat for Wexford County.

The plating facility that formerly operated on the Northernnaire Site consisted of a 100-foot by 50-foot prefabricated metal building that occupied 12.75 acres of land. The building that housed the plating operations still stands, however, the interior is empty. The facility is currently owned and operated by a trucking company.

The Northernnaire Site is located in one of the city of Cadillac's industrial parks. In EPA documents, the area has historically been referred to as the "Cadillac Industrial Park" or the "Cadillac Industrial Park Area." The Industrial Park occupies a one-square-mile area and is generally bounded by Thirteenth Street to the north, Mitchell Drive (Route 131) to the east, Wright Street to the south, and Leeson Avenue to the west (see Figure 2). Northernnaire is located in the southwest quadrant of the Industrial Park.

In addition to more than forty manufacturing facilities, the Industrial Park also includes garages and storage areas for the city of Cadillac, the city of Cadillac municipal well field, and a number of private residences. The city of Cadillac municipal well field is 1,300 feet northeast of the Northernnaire Site (see Figure 3). Facilities in the Industrial Park that were issued one of the EPA Unilateral Administrative Orders (UAOs) for remedial action (RA) for the Northernnaire/Kysor

Sites were: Northern Plating, Four Winns' Cruiser Division, A.H. Joynt, and Kysor Industrial.¹ Two other related sites identified as having trichloroethylene (TCE) and/or other chlorinated solvents in either soil or sumps at their plants were the Four Winns' facility located on Frisbie Street and the Leo Ingraham property. Addressing the contamination at these two facilities was included in the work intended by the UAOs.

Of these facilities, Northern Plating was identified as the only source of the hexavalent chromium contamination in groundwater. The primary contaminants at the remaining facilities are volatile organic compounds (VOCs). The design and construction of the groundwater extraction and treatment ("pump-and-treat") remedy for both the Northern Plating and the Kysor Sites was conducted jointly. The operation of the remedy is also handled jointly. There were a number of reasons this approach was taken, including the proximity of all the facilities and the commingling of the chromium and VOC plumes.

A number of other facilities in the Industrial Park are known or possible sources of groundwater contamination.

- The Paulstra/CRC plant and Mitchell Corporation are sources of groundwater contamination. Paulstra has completed the remediation of its TCE releases from its facility.
- Mitchell Corporation Company declared bankruptcy in 2006 and the MDNRE² has been operating the extraction system to treat the tetrachloroethylene (PCE) plume emanating from their property.
- The Rexair facility east of the Kysor Site is responsible for a TCE plume which appears to extend into the subdivision to the north of the Cadillac Industrial Park. Data have indicated that the Rexair extraction well system is not adequately capturing its plume, leading to commingling with the Kysor Site plume and resulting in the Kysor remediation system capturing and treating part of the contamination arising from Rexair since 1997. A consent judgment requires Rexair to pump 8-9 purge wells to treat its on-site contamination.

Features near the Site and nearby facilities are shown in Figure 4. Also, a diagram prepared by a contractor for MDEQ that depicts some of the plumes in the area, is in Figure 5.

The closest surface water bodies to the Site are the Clam River and Lake Cadillac. Both are approximately three-quarter-miles to the southeast. The effluent from the groundwater treatment facility discharges via an outfall to the Clam River. Some of the surface water drainage from the Industrial Park flows to a water retention pond located near Leeson Avenue at the western boundary of the park; however, the majority of it flows south and east towards the Clam River.

¹ Throughout this report, references to the "Kysor Industrial Corp., Site," the "Kysor Industrial Site," the "Kysor Site" all refer to the Superfund site, which includes a number of different facilities.

² The Michigan Department of Natural Resources (MDNR) was created in 1968 as the successor of the Michigan Department of Conservation. In 1995, the environmental regulatory programs of the MDNR were transferred to the Michigan Department of Environmental Quality (MDEQ). In 2010, the MDEQ was merged back with the MDNR and became the Michigan Department of Natural Resources and Environment (MDNRE).

Studies of the area have shown three distinct aquifers at or near the Northernnaire Site. The three aquifers are termed the “shallow,” “intermediate,” and “well field” or “deep” aquifers. The shallow and intermediate aquifers consist of sand with some silty clay and gravel. Clay layers separate the three aquifers from one another; however, it appears there may be some hydraulic connection between the shallow and intermediate aquifer north of the Site.

Prior to groundwater remediation, groundwater flow in the shallow aquifer was to the north/north-northeast, while flow in the intermediate aquifer was to the northwest/north-northwest. Regional groundwater flow in the deep city of Cadillac well field aquifer was to the north/northwest, however pumping of the city wells has altered the deep groundwater flow to the south/southeast. VOC contamination found in the deep, municipal well field aquifer indicates that it is also hydraulically connected to the intermediate and shallow aquifers. The clay layers between the aquifers are absent near the Cadillac municipal well field. The city wells are screened at 230 to 330 feet and draw from a deep, predominantly sand aquifer. Flow in the shallow aquifer has also been impacted by operation of the remedial action and is currently generally to the north. Flow in the intermediate aquifer has likewise been impacted by the remedial action and is more to the northwest.

Land and Resource Use

The vicinity of the Northernnaire Site is predominantly industrial, but there are several residences, a trailer park, a baseball diamond and municipal garages also in the area. The land and resource use of primary concern near the Site is groundwater that is used as a potable water supply. The city of Cadillac municipal well field is about 1,300 feet to the northeast. The well field includes seven municipal wells and is the sole community drinking water source for the city’s 10,000 residents (see Figure 2). In addition, the residential area to the north of the Industrial Park, the North Park and Pine View subdivisions, are part of the charter township of Haring.³ A number of the residences in the North Park area obtain their drinking water from private wells in the intermediate aquifer.

The Clam River currently receives the effluent from the groundwater treatment facility. According to the application submitted to Michigan Department of Natural Resources and Environment (MDNRE) for National Pollutant Discharge and Elimination System (NPDES) permitted discharge limitations, the river is protected for the following uses: agricultural use, navigation, industrial water supply, cold-water fish, partial body contact recreation, and total body contact recreation. Currently, the main uses are industrial water supply and sport fishing.

History of Contamination

The Northernnaire plant was a former plating facility that operated from 1971 to 1981. Northernnaire Plating provided custom chrome and nickel plating services to automobile manufacturers and other industries. Types of electroplating at the plant included flash chrome, hard chrome, bright nickel, bright chrome, sulfonate nickel, black oxide, and rack and barrel

³ Throughout this document, references to the “North Park” subdivision also include the Pine View subdivision, which is adjacent to North Park, but on a separate plat of land. Both the North Park and Pine View subdivisions are in Haring Township and subject to the same contamination issues and private well use concerns.

zinc. The wastes at Northernair included metal complexing agents, acids and heavy metals. No historical records or reports indicated use of significant amounts of solvents at the Northernair facility.

The waste handling practices at the facility that led to the release of hazardous waste to surrounding soil and to area groundwater included discharging process wastes to a poorly-sealed private sewer line and to drywells. Specifically, from 1971 through 1978, when MDNR revoked Northernair's wastewater discharge permit, Northernair discharged waste waters containing cadmium and chromium to a private line that led to the municipal sewer system. The poorly-constructed and inadequately-sealed underground pipes leading from the facility to the municipal line allowed waste to be discharged directly to the ground. A drywell connected to the facility sewer piping was also a source of releases. A floor drain inside the building released wastes to a second dry well not connected to the sewer lines.

Although MDNR revoked the discharge permit and plugged the discharge pipe leading to the municipal sewer in 1978, wastes continued to seep into the ground via the plant's drywells until operations ceased in 1981.

Initial Response

In 1978, MDNR received complaints from home owners near the Northernair Site about their drinking water. Samples collected by the district health department showed hexavalent chromium present in the tap water of two homes. Levels detected were 2,800 micrograms-per-liter (ug/l) or parts-per-billion (ppb), and 3,800 ug/l, which were significantly above the Maximum Contaminant level (MCL)⁴ of 50 ug/l. The city of Cadillac extended city water service to these homes.

It should be noted that chromium can exist in various forms, the most common of which are trivalent or Cr(III) and hexavalent or Cr(VI).⁵ Analyses for total chromium can indicate the likely presence of chromium in other forms besides Cr(VI). Of these two common forms, hexavalent chromium is the most toxic to human health and was the focus of the RA in the EPA Record of Decision (ROD). Therefore, in regard to the Northernair Plating facility, Cr(VI) is the focus of this report.

During the course of subsequent investigations, hexavalent chromium was detected in groundwater at a concentration of 930 ug/l at a depth of 150 feet below ground level (bgl). The study done in 1985 indicated that the groundwater had been contaminated as far as 1,400 feet north of the Northernair Site.

In 1981, children playing around drums outside the Northernair Site suffered chemical burns. When MDNR inspected the Site in May 1981, they transferred all the barrels into the

⁴ Maximum Contaminant Levels (MCLs) are promulgated under the federal Safe Drinking Water Act, 42 U.S.C. Sect. 300ff et seq. MCLs specify safe levels for drinking water contaminants measured at the tap from public water supplies.

⁵ Throughout this report, a distinction is made between total chromium and hexavalent chromium. Hexavalent chromium is also expressed as its chemical symbol Cr(VI). These two representations for hexavalent chromium are used interchangeably throughout this report.

Northernair building. Based on reports of possible gas vapors in the building, EPA and MDNR inspected the Site in March 1983. No vapor readings were detected, and no drums or tanks inside the facility appeared to be leaking.

During the Site inspection, discolored soil was seen in the catch basin twenty feet north of the building. Two pipes leading into the catch basin were not well-sealed and the bottom of the catch basin was open to the ground. The function of the second pipe appeared to be to redirect waste water to municipal sewer lines once the catch basin was almost full. This second pipe drained northeast where it was connected to the sanitary sewer line.

EPA conducted a removal action on July 5, 1983. EPA containerized and removed waste from the Site which included: 3,450 gallons of acids; 5,402 gallons of cyanide wastes; 160 drums of cyanide wastes; and 5,000 gallons of waste hypochlorite solution. After decontaminating the inside of the building, liquid and sludge wastes were removed from process tanks and shipped to RCRA disposal facilities. Tanks were also decontaminated. The contaminated sewer line to the north of the building was then excavated and removed.

Because Northernair Plating was not able to fund an investigation, MDNR conducted a state-led, federally-funded remedial investigation/feasibility study (RI/FS) in 1984 and 1985. The RI did not detect any cyanide contamination; however, hexavalent chromium was detected in two groundwater wells near the facility. The two wells were both screened in the intermediate aquifer. VOCs were detected at similar concentrations in wells both upgradient and downgradient of the Northernair Site.

The second part of the investigation involved a broader look at the groundwater contamination in the Cadillac Industrial Park Area. The State-led, federally-funded study, "Cadillac Area Groundwater Contamination Remedial Investigation," was completed in August 1988.

Basis for Taking Action

Hazardous substances that were released into groundwater and soil at the Northernair Site included hexavalent chromium, cadmium, and cyanide. Hexavalent chromium was identified in the soil on the Kysor property at 28 micrograms-per-kilogram (ug/kg) or parts-per-billion (ppb), and total chromium was found on the Ingraham property soil at 360 milligrams-per-kilogram (mg/kg) or parts-per-million (ppm). The Ingraham property is located approximately 1,000 feet north of the Northernair Site and was used by Kysor Industrial as a disposal area for drums and solvents (see Figure 2). In addition, the Northernair Site may have contributed TCE and other VOCs to the Industrial Park groundwater contamination; however, the amount of VOC contamination from the Northernair Plating facility was never determined. Hexavalent chromium was measured in the groundwater at 770 ug/L, well above its MCL.

The primary exposure pathway of concern at the Site is via ingestion of groundwater. Evaluation of the potential exposure and risk showed that a resident drinking groundwater from either the shallow or the intermediate aquifer would be exposed to excess lifetime cancer risks in the range

of 1×10^{-5} to 1×10^{-3} .⁶ The high end of this risk range (1×10^{-3}) exceeds what EPA considers to be an acceptable excess lifetime cancer risk.

Additional concerns included the fact that: 1) the city of Cadillac municipal well field is situated 1,300 feet north of the Northernnaire Site; 2) TCE, though not a primary contaminant of concern at the Northernnaire Site, had been detected in Cadillac city well #7 (CW-7) on numerous occasions above its MCL of 5 ug/l; and 3) a number of residents in the North Park subdivision, to the north of the Industrial Park in which the Site is located, still use their private drinking wells as either drinking water or for other domestic uses.

IV. REMEDIAL ACTIONS

Remedy Selection

The primary goals of the remedial actions at the Northernnaire Site were to:

- Eliminate any human exposure to residual hazardous waste disposed of or contaminated materials at the Site, and;
- Address all potential risks to human health and/or impacts to the environment.

A ROD for OU1 was signed by EPA in September 1985. The ROD called for the following actions:

- Excavate soils and remaining sediments surrounding the sewer line to meet soil cleanup requirements of 50 mg/kg for chromium and 10 mg/kg for cadmium and dispose to an off-site facility;
- Decontaminate the floor inside the Northernnaire building; and
- Remove the portion of the concrete floor inside the building where the drywell is located and, based on soil sample results from the drywell, excavate and dispose of all soils that do not meet the soil cleanup criteria.

The OU2 ROD for the Northernnaire Site was signed on September 29, 1989. It was intentionally written to address the groundwater contamination at two Superfund sites -- the Northernnaire Plating Site and the Kysor Industrial Site, which is another Superfund site located in the Cadillac Industrial Park Area. As described above, the predominant contaminant of concern at the Northernnaire Site was hexavalent chromium, while the contaminants of concern at the Kysor Site included a number of VOCs in the groundwater beneath the Cadillac Industrial Park that were migrating to the north into Haring Township. The OU2 ROD has two separate cover sheets,

⁶ Risks are probabilities that are expressed in scientific notation (e.g., 1×10^{-6}). An excess lifetime cancer risk of 1×10^{-6} indicates that an individual experiencing the reasonable maximum exposure estimate has a one-in-one-million chance of developing cancer solely as a result of site-related exposure. This is referred to as an "excess lifetime cancer risk" because it would be in addition to the cancer risk individuals face from other activities. The chance of an individual developing cancer from all other causes has been estimated to be as high as one in three. EPA's generally acceptable risk range for site-related exposures is one-in-ten-thousand (1×10^{-4}) to 1×10^{-6} .

referred to as the “Declaration for the Record of Decision,” but the remainder of the document serves as a single ROD for both the Northernnaire Plating and the Kysor Industrial Sites. After the 1989 ROD was signed, the Sites were more often referred to as the “Northernnaire/Kysor” Sites rather than by their individual names. The OU2 ROD served as the second and final action for the Northernnaire Site and as the first and final action for the Kysor Industrial Site.

This Five-Year Review focuses primarily on the status of the cleanup of hexavalent chromium in groundwater in the Cadillac Industrial Park. A separate Five-Year Review of August 2010 for the Kysor Industrial Site provides more information about the cleanup of VOCs in groundwater.

The OU2 ROD called for the following RA at the Northernnaire Site:

- Install a groundwater extraction and treatment system to remove groundwater contamination from the area surrounding the Site;
- Conduct groundwater monitoring to assess the quality of area groundwater; and
- Impose access and use restrictions.
- Construct a fence
- Construct a force main to discharge treated groundwater to the Clam River
- Install vapor extraction system for VOCs caused by the Kysor Industrial Site

While the ROD for the Kysor and Northernnaire Sites was essentially the same, the Kysor Site required the additional requirement to address soil contamination, via removal or otherwise, at the Ingraham property, where Kysor had disposed of industrial waste from its Wright Street plant.

The most critical cleanup level in the Northernnaire Site OU2 ROD was for hexavalent chromium. The OU2 ROD set the groundwater target cleanup level (TCL) at 50 ug/l. Groundwater cleanup levels in the ROD for the VOC contamination in groundwater are shown below.

Compound	Target Cleanup Level (ug/l)
1,1,1-trichloroethane	200
trans-1,2-dichloroethene	70
1,1-dichloroethylene	5
1,2-dichloroethane	5
methylene chloride	5
tetrachloroethylene	1
trichloroethylene	5
xylenes	440
toluene	40

Remedy Implementation

The OU1 RA took place in October and November 1988. Due to complications arising from disposal of concrete and debris inside the plant building, the final portion of the remedy was completed in early 1991.

A Unilateral Administrative Order (UAO) for the remedial design (RD) of OU2 was issued in May 1990. Nine parties, including Northernair Plating Company, were named in the Order. The other parties were Top Locker Enterprises, Inc., R.W. Meyer, Inc./Meyer Construction, Co., Willard S. Garyood, Kysor of Cadillac, Four Winns Company, Four Star Corp., Jomar Company, and Leo Ingraham, Sr. The RD addressed the work required in the OU2 ROD for both the Northernair and the Kysor Industrial Sites, and was finalized in March 1995.

In 1994, EPA signed an Explanation of Significant Difference (ESD) to document its decision to expand the remedy to include the contaminated groundwater from the Northernair/Kysor Site that had migrated into the North Park subdivision area. The cleanup established in the OU2 ROD, and to which the ESD applies, is the remediation of the releases from the facilities to which EPA issued a UAO. Through a funding arrangement put in place by the city of Cadillac, other facilities in the Cadillac Industrial Park are also contributing to the costs of the cleanup.

Based on the findings of the RI activities, the areas of contamination to be addressed by this RA were:

- A VOC plume in the shallow aquifer containing up to 115 mg/l of VOCs;
- A less concentrated VOC plume (up to 12 mg/l) in the intermediate aquifer;
- A hexavalent chromium plume at the base of the intermediate aquifer and in the shallow aquifer; and
- An area of soil contaminated with VOCs.

The systems and system components designed to remediate these areas were:

- A groundwater extraction system consisting of 17 wells (originally 18 wells; one well (S-7) is no longer used due to low yield), one of which delivers water to a granular activated carbon (GAC) unit for chromium cleanup and then to the packed tower air stripping (PTAS) system for VOC removal, while the other 16 wells deliver extracted groundwater directly to the PTAS system for VOC removal (see Attachment 2, Table 4).
- Associated pipelines to convey VOC and VOC/chromium-contaminated groundwater to the treatment system
- A discharge pipe to convey treated groundwater to the Clam River
- A PTAS system to remove VOCs from groundwater
- A GAC system to remove hexavalent chromium from groundwater
- A treatment building to house the treatment system components
- A soil vapor extraction (SVE) and treatment system to remediate soils contaminated with VOCs at the Kysor property

At the Site, there is a dedicated chromium extraction well (I-5) which pumps chromium-contaminated groundwater via a separate piping system, which does not include VOCs, to the treatment plant. The water entering the treatment plant from this well goes through GAC treatment for chromium removal and is then routed to the air stripping system. Water entering the plant from the other 16 VOC extraction wells is sent directly to the air stripping towers. The 17 extraction wells and the groundwater piping layout and outfall location into the Clam River are shown in Figure 6.

The treatment system for chromium includes two GAC contactors, a pH control system, and associated valves and piping. The piping system is configured so that either a single contactor may be used or the two may be used in series. After concentrated sulfuric acid is added using an in-line mixer, the flow is sent to the GAC contactors for chromium treatment. Prior to discharge, the pH of the carbon-treated effluent is adjusted by adding a sodium hydroxide solution. The effluent from the carbon treatment is then conveyed directly to the influent line for the air stripping system.

The effluent from the PTAS system discharges to the Clam River. The treatment facility design includes vertical turbine pumps, also referred to as cooling water make-up pumps, which can pump up to 800 gallons-per-minute (GPM) to the Co-generation Power Plant located 4,000 feet west of the treatment facility. Revenue from the sales of cooling water to the Co-generation Plant is used to help fund the costs of operating the treatment plant.

An initial list of discharge limitations provided by MDNR was included in the OU2 ROD. During RD, however, the Respondents went through the formal application process with the MDNR Surface Water Quality Division to obtain final discharge limits. A Substantive Requirements Document (SRD) prepared by MDNR in 1994 and the new discharge permit issued in 1996 provided the discharge monitoring requirements (DMRs) for the treatment plant. The purpose of the SRD and the discharge permits was to establish requirements and limits for discharging treated groundwater via an outfall to the Clam River. For Superfund sites, the SRD can substitute for actual permits, such as an air emissions permit or an NPDES permit. The 1994 SDR and 1996 permit provide the following limits:

Compound	Daily Maximum (ug/l)
1,1,1-trichloroethane	5
1,2-dichloroethane	5
tetrachloroethene	5
trichloroethene	5

Compound	Monthly Average (ug/l)
hexavalent chromium	8.3
total chromium	59

Sampling was generally required on a weekly basis. In 2002, a reissued permit increased the DMR monthly averages for hexavalent chromium from 8.3 ug/l to 12 ug/l, and for total chromium from 59 ug/l to 92 ug/l. DMRs are requirements that are expected to be met by the start of treatment and throughout the cleanup. Compliance with DMRs is based on samples collected at the Clam River outfall pipe, after the treated groundwater travels through approximately 4,000 feet of underground piping. However, the goal of the RA is to achieve the groundwater TCLs in the ROD. When the TCLs are met, the cleanup is considered to be complete. The TCLs are compared to results of groundwater samples as the water exits the treatment systems. The groundwater TCL (50 ug/l) for hexavalent chromium is less stringent than its DMR of 8.3 ug/l.

Construction Implementation and Operation and Maintenance

On-site construction began in June 1995. The groundwater pump-and-treat system began operating in September 1996, and the first round of the quarterly monitoring program was completed in November 1996.

Typically, Respondents to a UAO incur the full costs and responsibility for constructing the remedy and conducting O&M. For the Kysor Industrial and Northernnaire Plating Sites, a public-private sector partnership was formed to address the cleanup. Although the city of Cadillac was not identified as a potentially responsible party (PRP) and is not a Respondent to the UAO, it has taken the lead in constructing and operating the remedy. Using State legislation, the city formed a Local Development Financing Authority (LDFA) in the early 1990s to facilitate construction of the remedy and assist with the capital expenditures of building the treatment system. The development project that was the genesis of the LDFA was the construction of the Co-generation Power Plant. Revenue generated through tax increment funding (TIF) authorities, and funds from \$7.4 million in bonds issued by the city, helped to finance the remedial construction.

The city established a Special Assessment District to finance the operation and maintenance of the treatment plant. Annual costs include running the plant, conducting monitoring, and any unexpected costs. The area covered by the Special Assessment District coincides to an area of the Cadillac Industrial Park. All non-exempt property owners, including the city of Cadillac, within the Special Assessment District pay a yearly special assessment in addition to their property taxes.

Institutional Controls and Access Restrictions

Institutional controls (ICs) are required to ensure the protectiveness of the remedy. ICs are non-engineered instruments, such as administrative and/or legal controls, that help minimize the potential for exposure to contamination and protect the integrity of the remedy. Compliance with ICs is required to assure long-term protectiveness for any areas that do not allow for unlimited use or unrestricted exposure (UU/UE).

Decision Document:

For the Northernnaire Plating and Kysor Industrial Sites, the September 1989 ROD for OU2 required that access and use restrictions be imposed. Specifically, restrictions on groundwater use for drinking water purposes in the shallow and intermediate aquifers were to be placed on the Kysor Industrial and Northernnaire Plating Sites where groundwater contamination is located.

The Northernnaire Site RA has addressed all Site contamination from Northernnaire. No part of the Northernnaire Site has contamination that does not allow unlimited use or unrestricted exposure; therefore, no ICs are needed for the Northernnaire Site. Because the Kysor Site plume is not yet completely remediated, ICs are still needed to prevent exposures to contaminated groundwater associated with the Kysor Site. These ICs have been partially implemented but need to be strengthened as discussed in the August 2010 Five-Year Review for the Kysor Site.

Land and Resource Use:

The majority of facilities in the Cadillac Industrial Park, except for the Four Star plant which is inactive and the Ingraham property lot, which has since been cleaned up and will be used as a storage facility and parking lot, are still operating and future use of these properties is predicted to be industrial. Certain areas of the Cadillac Industrial Park, such as along Leeson Avenue, also contain commercial establishments.

The land and resource use of primary concern near the Northernnaire Site is groundwater that is used as a drinking water source. The city of Cadillac well field is the sole community drinking water source for the city's 10,000 residents. The residential area to the north of the Industrial Park, commonly referred to as the North Park subdivision, contains a number of residences that still use private wells for drinking water (see Figure 7).

The effluent from the groundwater treatment facility discharges via an outfall to the Clam River. According to the application submitted to MDNR for NPDES permitted discharge limitations, the river is protected for agricultural use, navigation, industrial water supply, cold-water fish, partial body contact recreation, and total body contact recreation. The main uses are recreation and sport fishing.

Operation and Maintenance (O&M)

Routine maintenance of the wells, extraction wells and the groundwater treatment plant is performed by the city of Cadillac as part of the LDFA system. This system continues to operate due to the remaining groundwater contamination associated with the Kysor Site. Some of the routine maintenance includes checking items and replacing oil, belts, heat tape, worn valves and packing; cleaning the roof-top heating unit; pulling and cleaning extraction wells if needed, and checking wells for freezing, damage, secured locks, and extraction well failure. Daily bench logs are kept current. Logs and data are electronically stored. One particular monitoring well, F-10S, has not been sampled lately due to the field crew not being able to locate it. It is suspected that the well may have been paved over by a property owner. One of the shallow VOC extraction wells, S-7, is not operational. These are not issues of concern for the Northernnaire Site.

During the initial two years of operation, sampling frequency was quarterly. Annual monitoring began in October 1998, although several wells are still monitored on a quarterly basis. The current sampling program to monitor the progress in cleaning up the hexavalent chromium in the shallow and intermediate aquifers includes:

- Quarterly sampling of 10 monitoring wells and three VOC fringe monitoring wells
- Annual sampling of 10 monitoring wells and shallow and intermediate purge wells
- Annual sampling of the I-5 extraction well, i.e., the well specifically designed to extract the chromium-contaminated groundwater
- Monthly sampling at the treatment plant of the influent, midpoint and effluent flows
- Weekly sampling of effluent for discharge monitoring requirements (DMRs)
- Annual static water level measurements at about 100 wells

Extraction well I-5 is set in the intermediate aquifer and has a total depth of 182 feet bgl. The well screen in I-5 is 25 feet in length with the interval being 157 to 182 feet bgl. One of the 10 monitoring wells for chromium (MW-5) is in the shallow aquifer, and the remaining nine, along with I-5, are in the intermediate aquifer. MW-5 has since been dropped from the monitoring program; however, the LDFA has been monitoring the shallow and intermediate extraction wells on a quarterly basis since March 2006. All of the wells above are sampled not only for dissolved total chromium and hexavalent chromium, but also for the nine VOCs identified as contaminants of concern in soil and groundwater for the Kysor Site (see Attachment 2, Table 4).

Costs and Operation

Approximate annual costs of O&M for both the Northern Plating and Kysor Industrial Sites are shown in the table below. The average amount spent per year is approximately \$222,000. The estimate for annual O&M expenditures in the 1989 ROD was \$125,000. Capital and construction costs of the remedy were less than expected (about \$1.3 million) and close to the amount projected in the OU2 ROD. Approximately 60 percent of the 2010 and proposed 2011 budget respectively, is allotted to construction and utilities costs.

Table 2 - City of Cadillac Annual Operations/O&M Costs

Dates		Total Cost
From	To	
7/2000	6/2001	\$160,000
7/2001	6/2002	\$210,000
7/2002	6/2003	\$240,000
7/2003	6/2004	\$240,000
7/2004	6/2005	\$226,266
7/2005	6/2006	\$227,556
7/2006	6/2007	\$289,124
7/2007	6/2008	\$244,717
7/2008	6/2009	\$242,682
7/2009	6/2010	\$380,800
7/2010	6/2011	\$380,400 (<i>Proposed Budget</i>)

Note: The city of Cadillac fiscal year runs from July 1st through June 30th

The increase in total annual costs between the 2008-2009 cycle and the last two cycles of 2010-2011 (\$380,400) is due to the upgrading of the LDFA telemetry system in 2010. The project will be funded through TIF funds spread out over two years and will cost about 100K.

V. PROGRESS SINCE THE LAST FIVE-YEAR REVIEW

The protectiveness statement in the previous Five-Year Review for the Northernnaire Site indicated that the remedy was protective of human health and the environment in the short term as no exposures were occurring. Long-term protectiveness would not be achieved until: 1) ICs were implemented to restrict the installation and use of private wells in the subdivision north of the Site, and the adequacy of these existing restrictions, was confirmed; 2) confirmation was achieved that the extraction system was capturing all contaminated groundwater from the Industrial Park Site; and 3) cleanup goals in the monitoring network that were representative of the entire groundwater plume was achieved.

Table 3 - Progress Since the Last Five-Year Review

Recommendations from Previous Review	Party Responsible	Action Taken
Add chromium to the list of analytes for monitoring and extraction wells where the plume may have migrated beyond the capture zone. This may include wells beyond the current definition of the plume to the north, east and west and in the deep aquifer.	Respondent	The LDFA monitors for chromium in wells located north of 13 th Street and beyond the chromium plume. This testing began in 2006 and, in addition to the monitoring wells, includes the shallow and intermediate purge wells, as well as the VOC fringe wells.
Develop an Institutional Controls Study Plan to evaluate the city ordinance currently in place and to devise and implement a plan to prevent exposure of residents in the North Park subdivision to contaminated groundwater from the Northernnaire/Kysor Sites.	Respondents	An IC Study Plan was not completed; however residential areas were investigated to determine what ICs were appropriate due to contamination from the Kysor Site. The private wells in the North Park and Pine View subdivision were inventoried in July 2006. In early 2008, EPA proposed an ordinance to Haring Township to restrict the use of private wells in North Park and Pine View subdivisions and to require residents to abandon private wells in the affected area and connect to the township water system. The proposed ordinance was rejected. EPA tested 16 wells of concern identified from the survey on May 2008. No well contamination above health-based limits was identified in the residential wells. The Wexford County DPW is currently proposing to Haring Township a new ordinance. This proposal is not related to the Northernnaire Site, but addresses the area VOC contamination problems.
Re-evaluate the LDFA plume capture zone, and review design and current operation of the extraction well system to determine if the system is effective and/or can be optimized.	USEPA/ Respondents	EPA evaluated the capture zone and issued a report in April 2007 summarizing data collected through then. Preliminary analysis showed that the plume capture system is effective.
Test the influent to the air stripping towers for chromium.	Respondent	This testing has been performed by the LDFA since March 6, 2006.

VI. FIVE-YEAR REVIEW PROCESS

Administrative Components

MDNRE and the city of Cadillac LDFA were notified of the initiation of the Five-Year Review in February 9, 2010. The preparation of the Northernnaire Plating Site Five-Year Review was led by Sheila Sullivan of EPA, with assistance and review provided by William Bolio, Nick Swiger and Steve Cunningham of the MDNRE. EPA was the lead agency for the review. Both the Northernnaire and Kysor Sites were reviewed concurrently.

The components of the Five-Year Review schedule include:

- Community Notification
- Document Review
- Data Review
- Site Inspections
- Report Development and Review

Community Involvement

A Public Notice announcing that a Five-Year Review of the Northernnaire/Kysor Sites was to be conducted and was published on March 13, 2010 in the Cadillac News (see Attachment 3). The city has been very satisfied with the progress of the cleanup and has requested that MDNRE and EPA evaluate its proposal to shut down the chromium treatment system.

Document Review

This Five-Year Review consisted of a review of relevant documents including RODs, the ESD, investigatory reports and studies, correspondence, O&M records, hydrogeological studies, city of Cadillac summary description of LDFA and annual budgets, expert opinion reports, annual evaluation reports, and monitoring data (see Attachment 4).

Data Review

Descriptions of data related to the Northernnaire Plating Site are provided below.

Types of Monitoring and Compliance Status

Four types of groundwater samples for hexavalent chromium are collected as part of the Northernnaire Site monitoring program:

- Monitoring and extraction well samples
- Treatment plant samples–GAC chromium treatment
- Treatment plant samples–PTAS system
- Clam River discharge samples

In summary, applicable performance standards have been achieved in all four types of monitoring since May 2001 or earlier. The compliance status for each of the four sampling components is listed below.

- Monitoring Well Samples: The system has been in compliance since May 2001. As of the sampling round in March 2005, 10 of the 10 monitoring wells in the sampling program had achieved the cleanup level for hexavalent chromium. Since the last Five-Year Review in 2005, the monitoring wells remain in compliance. In addition, all nine VOC fringe wells (F-6, F-7, FD-2, FD-3, CMS-5-W, SLW-7, SLW-34, WMW-6D, and WMW-6I) do not show detectable levels of hexavalent or total chromium. The shallow and intermediate extraction wells showed some chromium detections that were well below the TCLs.

Of the shallow extraction wells ("S" wells), the highest total chromium level (32 ug/l) was found in well S-3 in June 2006. Well S-3 is located about 1,000 feet northwest of the Northern Plating facility and 400 feet west of the plume edge. The highest hexavalent chromium level (9.4 ug/l) was found in well S-4 in March 2007. Well S-4 is located about 500 feet directly north of the facility. All detected concentrations of chromium (total and hexavalent) in shallow extraction wells over the past five years have been below the cleanup goal of 50 ug/l for hexavalent chromium.

Of the intermediate wells, the highest total chromium level (43 ug/l) was found in MW-13 in June 2001; the highest hexavalent chromium level (40 ug/l) was also found in MW-13 in June 2001. MW-13 is located about 1,700 feet north and downgradient of the Northern Plating facility and 350 feet south of Thirteenth Street. All detected concentrations of chromium (total and hexavalent) in intermediate wells over the past five years have been below the cleanup goal of 50 ug/l for hexavalent chromium.

- Treatment Plant Samples–GAC Chromium Treatment: The system has been in compliance since 1996 and continues as such. Concentration of hexavalent chromium has been below the 50 ug/l TCL at the point the groundwater from the extraction well (I-5) enters the treatment plant – that is, before it flows through the carbon treatment contactor – since the end of 1996.
- Treatment Plant Samples–PTAS System: The system has been in compliance since 1996. The concentration of hexavalent chromium exiting the air stripping system has been 10 ug/l or below for the entire O&M period (since September 1996). The concentration of total chromium has been below 10 ug/l since 2000.
- Clam River Discharge Samples: The system has been in compliance since 2001. The limit for the concentration of hexavalent chromium in the effluent is a monthly average of 8.3 ug/l. Samples are collected weekly. This limit has not been exceeded on a per sample basis since August 2001. Since March 2003, effluent results have been nondetect for hexavalent chromium. Since 2005, the average total chromium effluent concentration has been 1.64 ug/l and the average hexavalent chromium has been nondetect (see Attachment 1, Graphs 1-2).

Sampling Data and Capture of Chromium-Contaminated Groundwater

Attachment 1, Graphs 3-33 provides plots of all of the wells monitored for chromium. The trend is presented for these wells over the past 10 years of sampling. In some cases, the plots begin in 2006 when sampling began at the additional wells, i.e., the VOC extraction and VOC fringe wells.

During the period from 1996 to 2002, it was estimated that over 260 lbs of Cr(VI) were removed from the aquifer by the combined Kysor/Northernnaire groundwater extraction wells. It is very unlikely that the decline in Cr(VI) concentrations – and in particular, mass recovery – is due to Cr(VI) migration beyond the extent of hydraulic capture since the extent of capture encompasses the probable extent of the Cr(VI) contamination (see Attachment 1, Graphs 34-35).

According to the data, the majority of Cr(VI) recovered at the Site was from the VOC extraction wells, namely I-2, I-3 and possibly I-11, as opposed to the chromium extraction well I-5. This is because the extraction system for the VOC plume capture is so much more extensive than that for the chromium plume. As a consequence, any chromium in the groundwater recovered at these VOC wells has bypassed the GAC pre-treatment to remove Cr(VI), going directly to the air strippers. Nonetheless, the concentration of Cr(VI) in water discharged to the Clam River did not exceed the NPDES discharge limits. A definite conclusion cannot be reached with regard to the migration of Cr(VI) because recent sampling has indicated concentrations of Cr(VI) at or below the method reporting level (MRL) in most wells. It is not even currently possible to define a capture zone for Cr(VI) due to the absence of samples exceeding the cleanup level. This suggests successful remediation of groundwater contaminated with Cr(VI).

In 2005/2006, the LDFA began an expanded sampling program to support the conclusion that cleanup of Cr(VI) has been achieved, and to close out the Northernnaire Site RA in accordance with EPA guidance. During 2006, sampling and analysis for Cr(VI) was performed at several wells to confirm that cleanup of Cr(VI) has been successfully achieved. These wells lie within or are proximal to, and slightly downgradient of the schematic extent illustrated in Figure 8. The expanded sampling program also included some lower intermediate aquifer wells based on the detection at well I-3. All wells were sampled at least four times in order to confidently establish that, if concentrations are greater than the MRL, the upper confidence limit is still below the TCL and the concentration trend is not upward. Under the expanded monitoring program implemented in 2006, the additional sampled wells do not show exceedances of the Cr(VI) TCL, or the Cr(VI) MRL of 50 ug/l.

The 2005 Five-Year Review indicated that on some occasions the concentrations of chromium exiting the plant have been higher than what is detected in groundwater entering the plant from the chromium extraction well. However, this is no longer the case. The GAC influent and plant discharge are both nondetect for hexavalent chromium. More recently, since February 2010, data was obtained for the tower influent and tower midpoint--the sampling port between the two air strippers. These results show no difference throughout the air stripping treatment stages, and hexavalent chromium is nondetect. Concentrations exiting the plant have at all times been well below the Cr(VI) groundwater TCL (see Attachment 1, Graphs 1-2). Data show that I-5 carbon influent, tower influent, tower midpoint and plant discharge values for chromium are nondetect.

The ROD estimated that Cr(VI) contamination would meet the TCL within four years after system startup. Given that the LDFA groundwater remedy began operating in 1996, the estimated completion time of the remediation would be 2000. This projection is fairly consistent with groundwater sampling which indicates that since May 2001 all 10 Cr(VI) monitoring wells have shown no detections, and that sampling since 2006 at a larger number of monitoring and extraction wells indicated no concentrations approaching the TCL. It is likely that the majority of Cr(VI) has been effectively removed via groundwater extraction by I-5 and the combined VOC wells. Data to date has indicated that any Cr(VI) remaining in the groundwater is present at concentrations well below the TCL.

The data collected to date shows that the LDFA groundwater remedy created a sufficient capture zone to contain and recover groundwater contaminated above TCLs for Cr(VI). The results obtained indicate no concentration above the TCL. This suggests that cleanup has been achieved.

Site Inspection

The Five-Year Review site inspection of the Northernair Plating Site was conducted on May 21, 2010 by the EPA Remedial Project Manager and city of Cadillac officials. The purpose of the inspection was to assess the progress of remedy implementation, ensure records and Site documents were available and current, inspect treatment units to verify they were operational and have no significant problems, and view general Site conditions and areas of other groundwater releases and plumes in the Industrial Park. The intent was to collect information to assess the protectiveness of the remedy and try to foresee any future remedy implementation problems and needs.

Most issues related to the remedy for the Northernair Site were identified at the time of the Site inspection or prior to it. The treatment plant appeared to be very well-maintained. A new operating control system had been installed to manage the PTAS system. No significant problems were noted (see Attachment 5).

The city of Cadillac has been studying several potential locations for a new city well field. This is needed due to the age of the current system and the need to expand. The city originally had seven wells (CW-1 through CW-7). In the 1980s, the casing ruptured in CW-5 and the well had to be abandoned. CW-7, located about 350 feet north/northwest of the city well field, is no longer used due to TCE contamination. The city currently operates CW-1, CW-2, CW-3, CW-5 and CW-6. Due to the necessary isolation distance, the city cannot add new wells to the existing system. The selected location for the new well field is two miles southeast of the existing well field and southwest of the 55/East Division and 131 interchange. This location is downgradient of the Kysor/Northernair Sites and the Cadillac Industrial Park, as well as the groundwater contamination plumes associated with these areas (see Figure 9). The city projects that the well field construction will be half-completed by 2012, and that the current well field will be abandoned by 2014.

VII. TECHNICAL ASSESSMENT

Question A: *Is the remedy functioning as intended by the decision documents?*

Yes. Based on a review of relevant documents, data from monitoring wells and the treatment plant, applicable or relevant and appropriate requirements (ARARs), risk assumptions, and the results of the Site inspection, EPA believes that the remedy is functioning as intended in the OU2 ROD, as modified by the ESD. The groundwater monitoring programs are sufficient.

The statistical analyses show that none of the wells with data for Cr(VI) exceed the cleanup standard of 50 ug/l. In addition, trend analysis showed no wells with a statistically significant upward trend in Cr(VI) concentration. VOC wells that are most likely to have recovered or are currently recovering Cr(VI) are S-4, S-6, I-2, I-3, I-5, I-6, and I-11. These wells lie along, or close to, the potential paths of migration that are illustrated in Figure 8. Significant extraction of Cr(VI) at shallow well S-4 may have occurred at an earlier time. Given the locations and rates of groundwater extraction within the intermediate aquifer downgradient of Cr(VI) extraction well I-5, it appears unlikely that Cr(VI) could migrate beyond the VOC extraction wells. Therefore, the implemented Northernaire RA has achieved the remedial action objective for Cr(VI) groundwater contamination. The remedy is protective in the short and long term.

Question B: *Are the exposure assumptions, toxicity data, cleanup levels, and remedial action objectives (RAOs) used at the time of the remedy selection still valid?*

Yes. The assumptions and information on which the OU2 ROD was based are still valid. There have been no changes in the physical conditions at the Site that would affect the protectiveness of the remedy.

Changes in Standards and To-Be-Considered Requirements

There have been no other changes in the ARARs and no new standards or to be considered requirements affecting the protectiveness of the remedy.

Changes in Exposure Pathways, Toxicity, and Other Contaminant Characteristics

The exposure assumptions used to develop the Human Health Risk Assessment included exposure to chromium-contaminated groundwater via ingestion. There has been no change in this exposure pathway. There have also been no changes in the toxicity factors for the contaminants of concern that were used in the baseline risk assessment. No changes to these assumptions appear to be needed. Furthermore, there has been no change to the standardized risk assessment methodology that could affect the protectiveness of the remedy.

Question C: *Has any other information come to light that could call into question the protectiveness of the remedy?*

No. Although the data suggest that a portion of the chromium plume is being extracted by one or more of the VOC extraction wells and is by-passing the chromium treatment system, this has not

affected the progress of the cleanup and since data also indicate that the chromium cleanup goals have been met and discharge concentrations are nondetect.

Technical Assessment Summary

Based on a review of relevant documents, data, ARARs, risk assumptions, and the results of the Site inspection, the remedy for the Northernnaire Site is functioning as intended by the decision documents and is protective in the short term and long term. No exposure pathways are complete, and none of the samples collected from the monitoring wells, extraction wells and VOC fringe wells show detectable chromium. In addition, none of the private wells in the North Park subdivision have had detectable chromium levels.

VIII. ISSUES

There are no issues associated with the Northernnaire RA. Long-term remedial action goals for the Site have been met. Hazardous substances above health-based limits are no longer present on the Site above levels that allow for UU/UE.

IX. RECOMMENDATIONS AND FOLLOW-UP ACTIONS

None

X. PROTECTIVENESS STATEMENT

OU I – Soil

The OUI remedy for soil removal at the Northernnaire Site is protective of human health and the environment in both the short and long term. The removal eliminated the risks associated with direct contact with, and ingestion of, contaminated soil. It also eliminated areas of significant soil contamination that would otherwise act as a continuing source of groundwater contamination and would work counter to the groundwater extraction and treatment system.

OU II – Groundwater

The remedy for the Northernnaire Plating Site required in the OU2 ROD for groundwater remediation is protective of human health and the environment in the short and long term. Cleanup goals have been achieved throughout the plume of chromium-contaminated groundwater.

Site-wide

The entire Northernnaire Plating Site is protective of human health and the environment in both the short term and long term.

XI. NEXT REVIEW

This is the last Five Year Review for the Northernnaire Plating Site.

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- Figure 6 – Diagram of Extraction Well Discharge and Piping
- Figure 7 – Residential Well Users
- Figure 8 – Schematic of Hexavalent Chromium Location
- Figure 9 – City of Cadillac Water System Improvements

Attachments

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- Attachment 2 – Additional Tables 4 and 5
- Attachment 3 – Advertisement of Five-Year Review Start
- Attachment 4 – List of Documents Reviewed
- Attachment 5 – Site Inspection Checklist
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FIGURES

Figure 1 – Site Location Map

Figure 2 – Cadillac Industrial Park Area

Figure 3 – Location of Cadillac City Well Field

Figure 4 – Industrial and Commercial Facilities in Area

Figure 5 – Plume Map

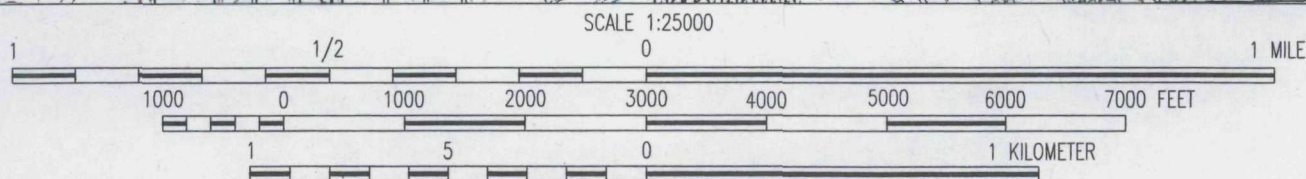
Figure 6 – Diagram of Extraction Well Discharge and Piping

Figure 7 – Residential Well Users

Figure 8 – Schematic of Hexavalent Chromium Location

Figure 9 – City of Cadillac Water System Improvements

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CONTOUR INTERVAL 3 METERS / SUPPLEMENTAL CONTOUR INTERVAL 1.5 METERS
NATIONAL GEODETIC VERTICAL DATUM OF 1929



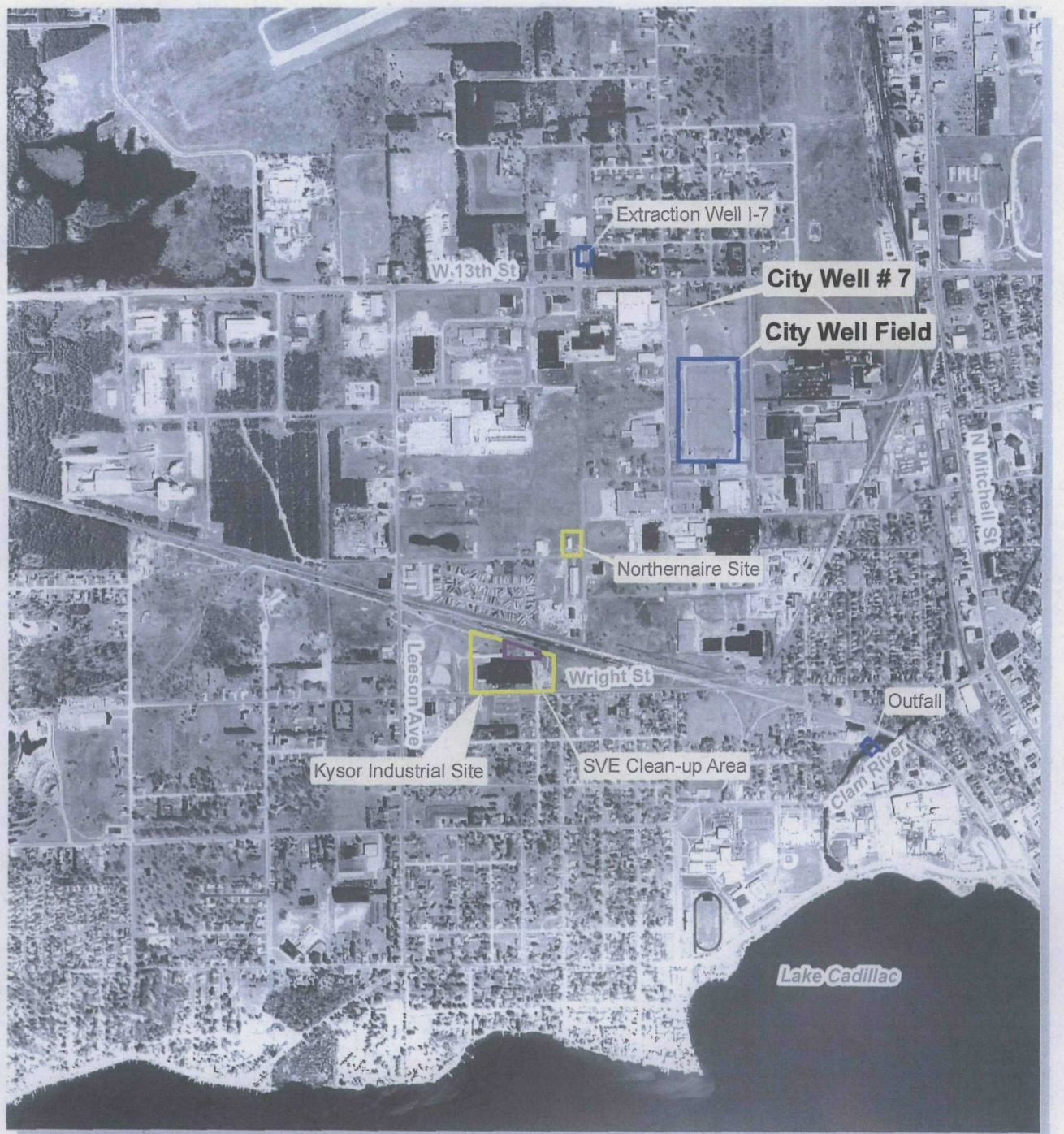
NOTE: MAP ADAPTED FROM USGS IN COOPERATION WITH THE STATE OF MICHIGAN.
CADILLAC NORTH, PROVISIONAL EDITION (1983), DATUM IS MEAN SEA LEVEL, PHOTO
COMPILED (1981), FIELD CHECKED (1981), EDITED (1983) QUADRANGLE 7.5 MINUTE SERIES.

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The Group, Inc.
2365 Haggerty Road South Canton, Michigan 48188
Telephone: (734) 397-3100

FIGURE 1

DATE	DRAWN BY	DESIGNED BY	PROJECT NO.
3/08	CJB	CJB	DEQA3A

Location of City of Cadillac Well Field



0.25 0.125 0 0.25 Miles

Plot created by Andrea Porter U.S. EPA Region 5 on 9/27/2005

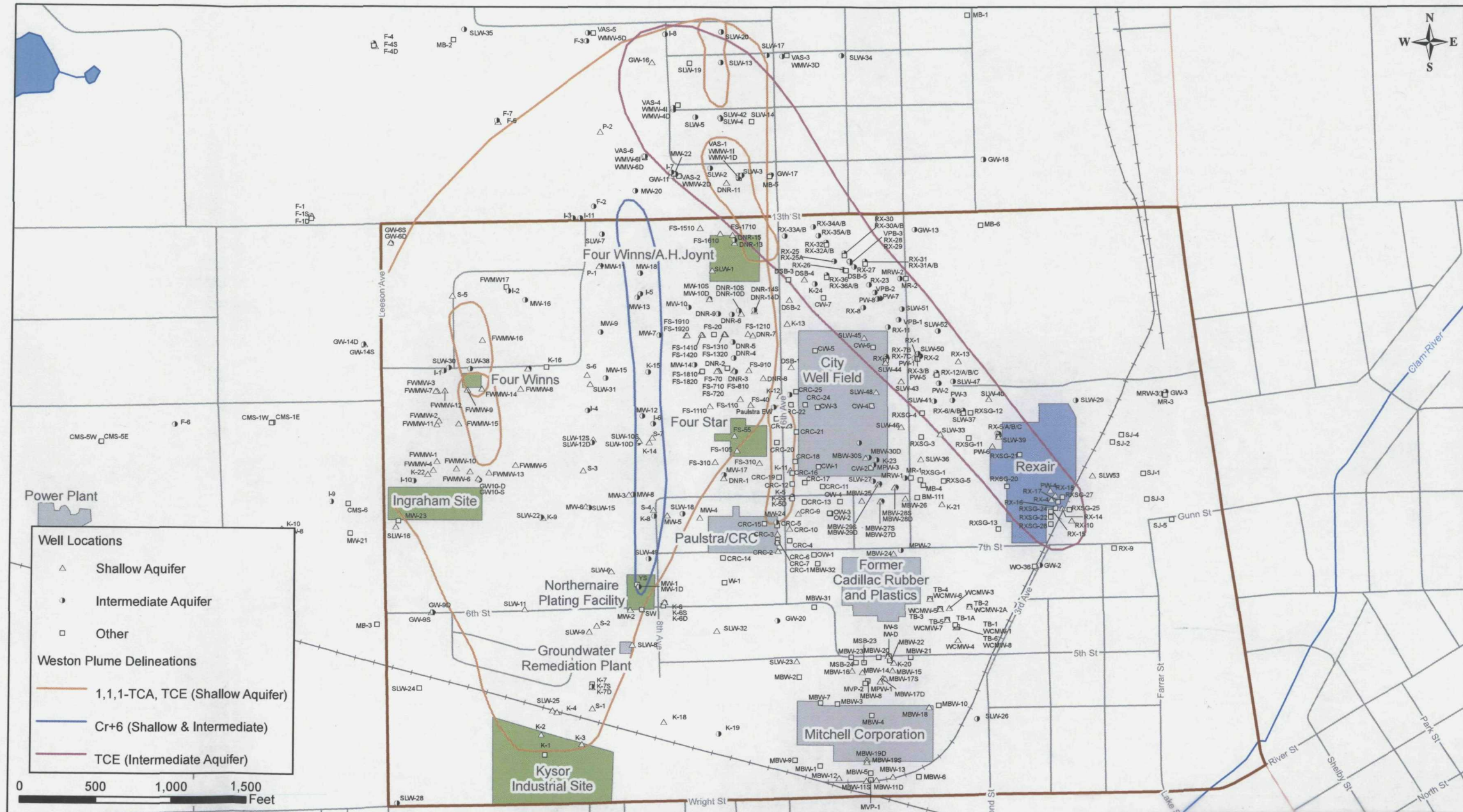
FIGURE 3

Sites of Contamination Near the City of Cadillac Municipal Wellfield



FIGURE 4

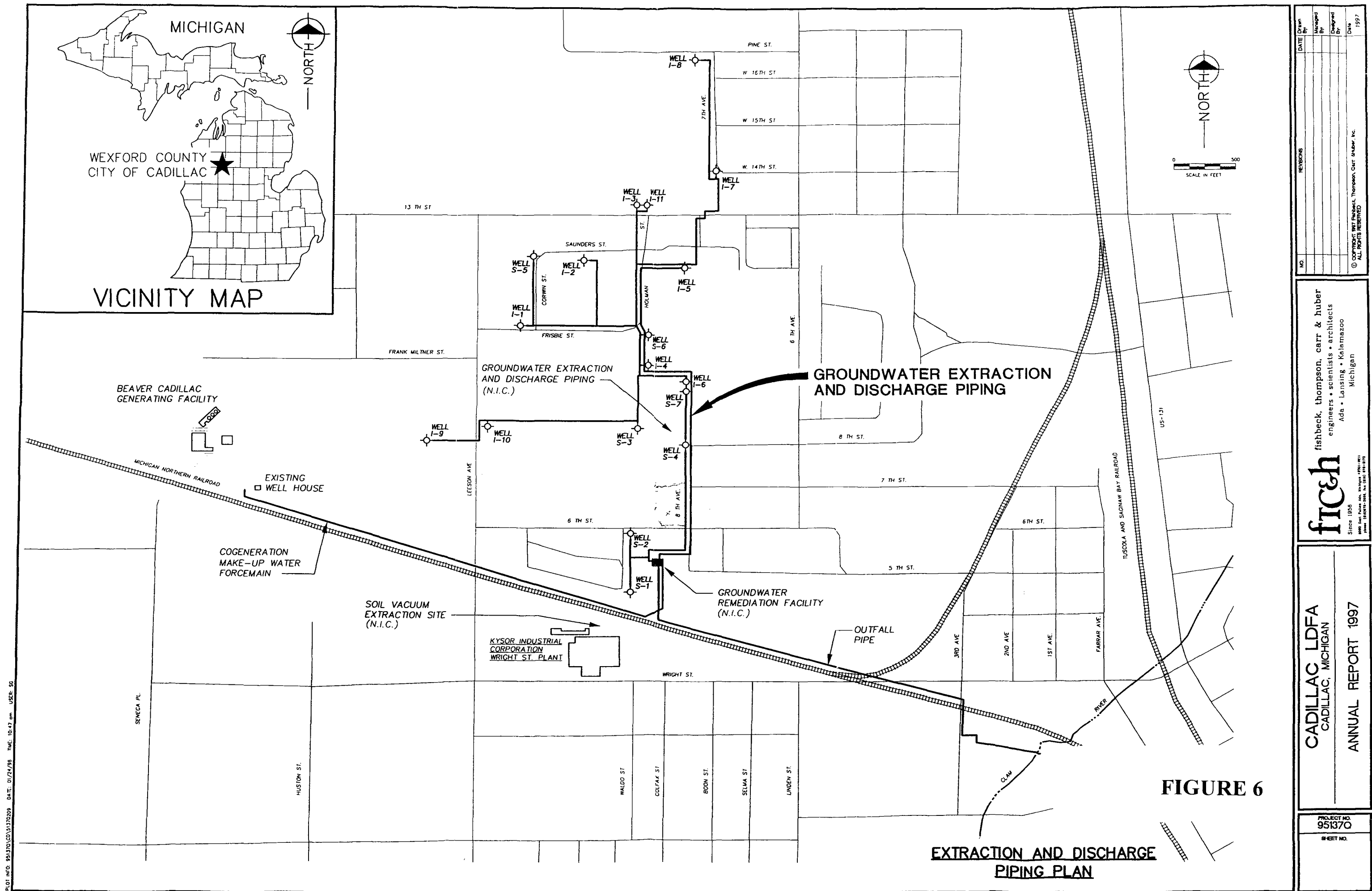
0 500 1,000 2,000 Feet



Note: Weston generalized plume depictions (Weston, 2003) are illustrated solely for the purpose of spatial reference since these have been produced previously; they form no basis of any evaluations in this report.
 Note: This figure illustrates all wells for which spatial data are available. Not all wells are (a) actively pumping, or (b) included in a monitoring program. See text for details.

General Extent of Groundwater Contamination and Locations of Wells

FIGURE 5



PLOT INFO: 951370(CD)1370209 DATE: 01/24/98 TIME: 10:47 am USER: SC

CADILLAC LDFA
CADILLAC, MICHIGAN

ANNUAL REPORT 1997

PROJECT NO.
951370

SHEET NO.

fic&h

fishbeck, thompson, carr & huber
engineers • scientists • architects
Ada • Lansing • Kalamazoo
Michigan

Since 1926

4000 East Fulton Ave., Suite 200, Kalamazoo, MI 49001
Phone: (269) 336-1111 Fax: (269) 336-1112

NO.	REVISIONS	DATE	Drawn By	Checked By	Designed By	Date

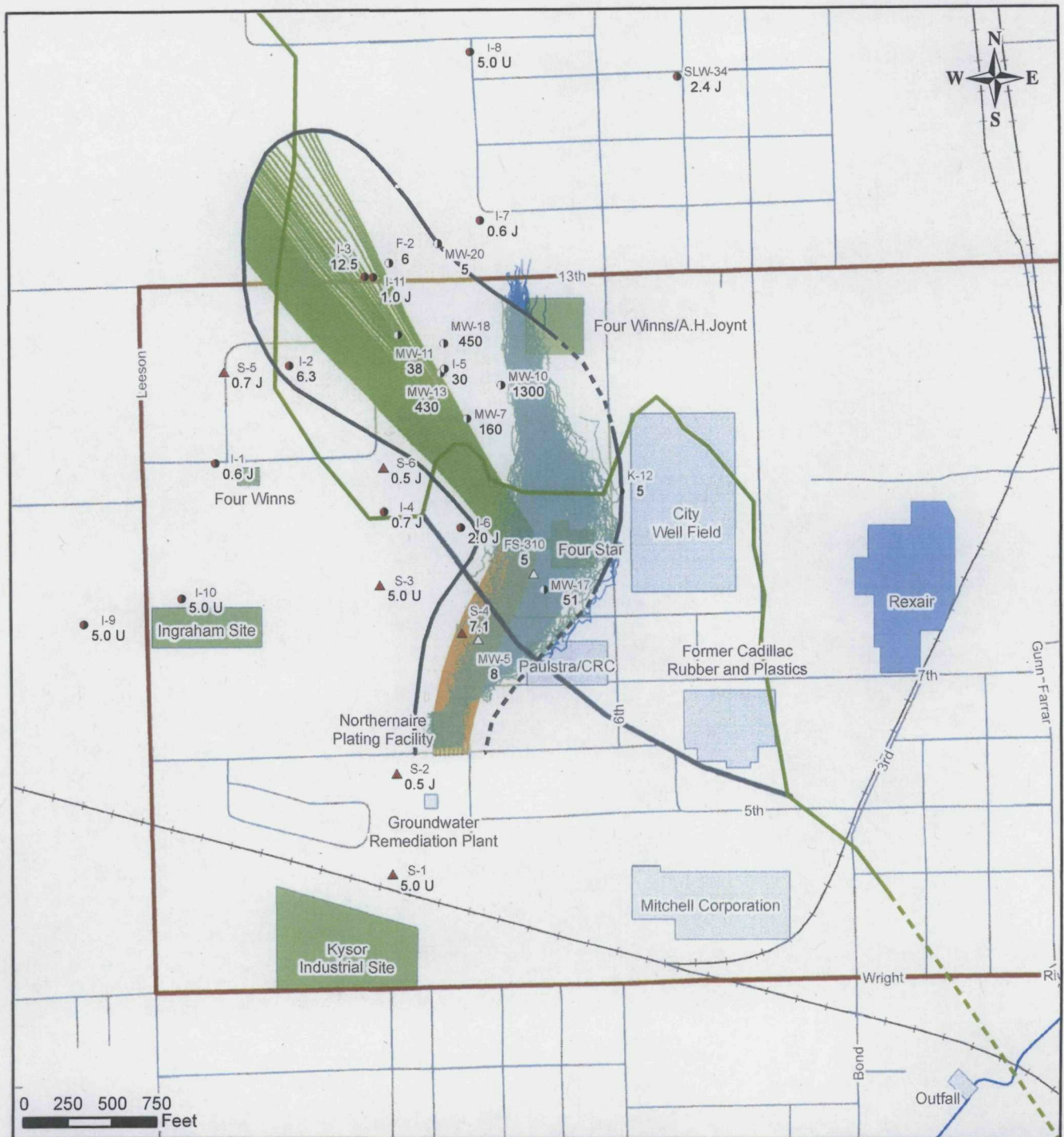
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North Park Subdivision - Residential Well Uses Cadillac, MI

Superfund
U.S. Environmental Protection Agency



FIGURE 7



Historical Maximum Since 1996

- MW-5
8 △ Shallow Zone Well and Cr(VI) (ug/L)
- F-2
6 ● Intermediate Zone Well and Cr(VI) (ug/L)

Maximum from March or June 2006

- S-4
7.1 ▲ Shallow Zone Well and Cr(VI) (ug/L)
- I-2
6.3 ● Intermediate Zone Well and Cr(VI) (ug/L)

Extent of Shallow Clay

Extent of Transition Zone

Pathlines Using Mapped
Water Levels (Figure 16)

Cr(VI) 5 ug/L Contour Based
on Quantile Kriging Interpolation

MDEQ Model Layer

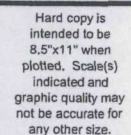
Layer 1 Pathline

Layer 2 Pathline

Layer 3 Pathline

Schematic of Cr(VI) Migration with Maximum Sampled Cr(VI) Shown
(June 1987, Pre-Remediation Conditions with MDEQ Model as External Drift)

FIGURE 8



DWRF Project Plan / Water System Improvements

FIGURE 9

ATTACHMENTS

Attachment 1 – Graphs

Attachment 2 – Additional Tables

Attachment 3 – Advertisement of Five-Year Review Start

Attachment 4 – List of Documents Reviewed

Attachment 5 – Site Inspection Checklist

Attachment 6 – Inspection Documentation Photos

ATTACHMENT 1 – GRAPHS

Graph 1 – LDFA Treatment Plant Total Chromium Effluent

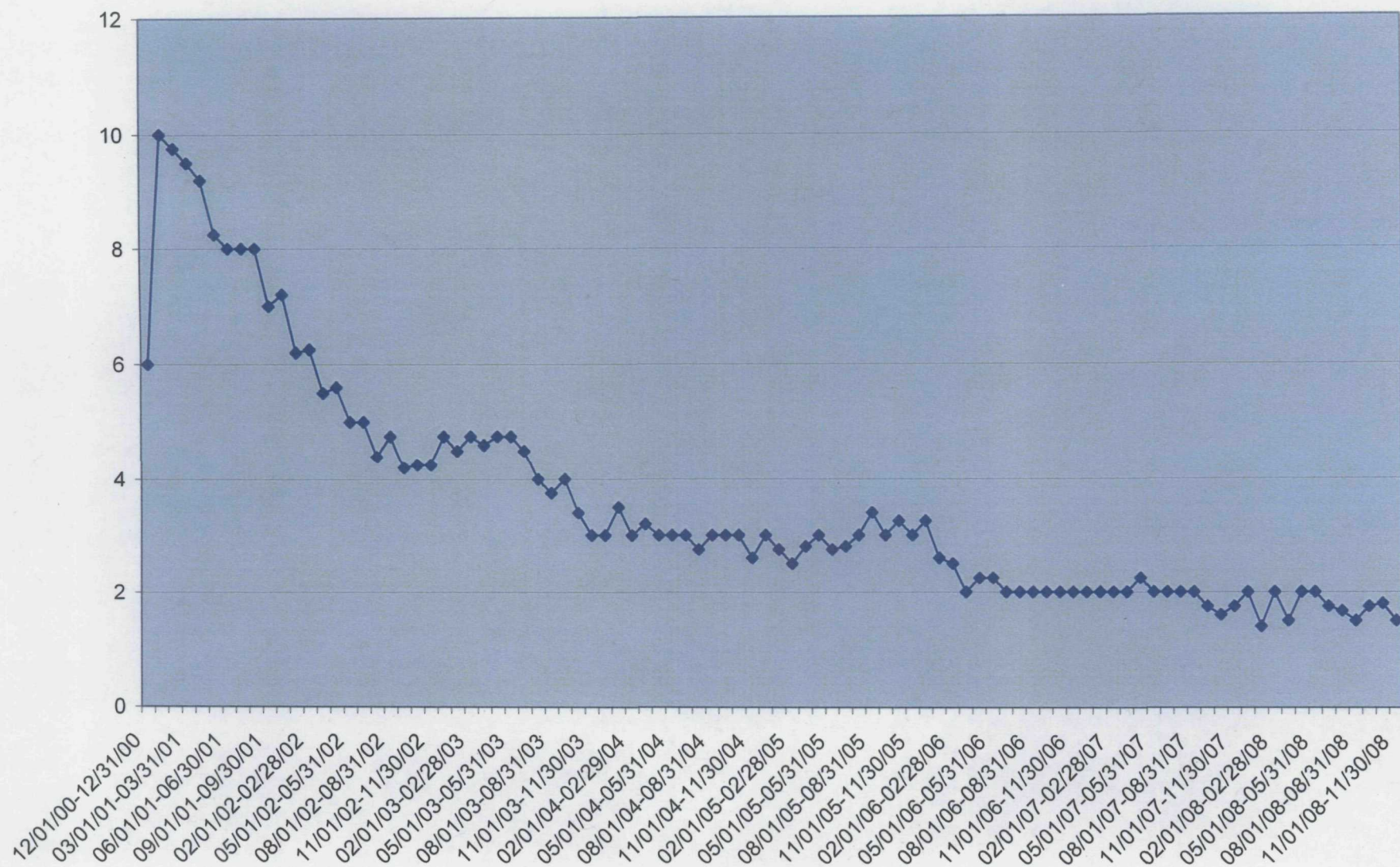
Graph 2 – LDFA Treatment Plant Hexavalent Chromium Effluent

Graphs 3 through 33 – Hexavalent Chromium and Total Chromium Concentrations
in Shallow and Intermediate Monitoring and Extraction
Wells

Graphs 34 and 35 – Estimated Hexavalent Mass Recovery at LDFA Wells

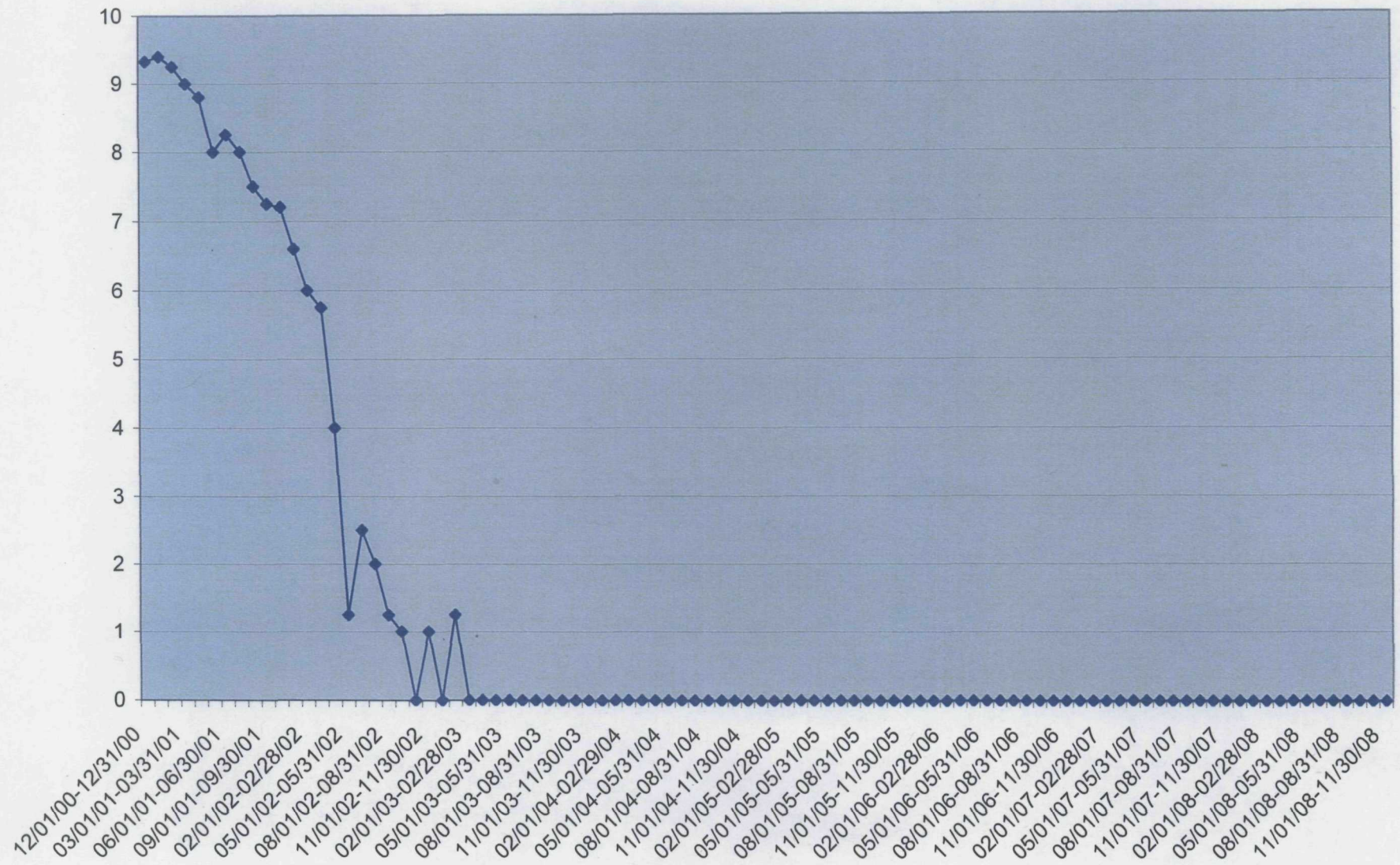
GRAPH 1

Effluent Total Chrome

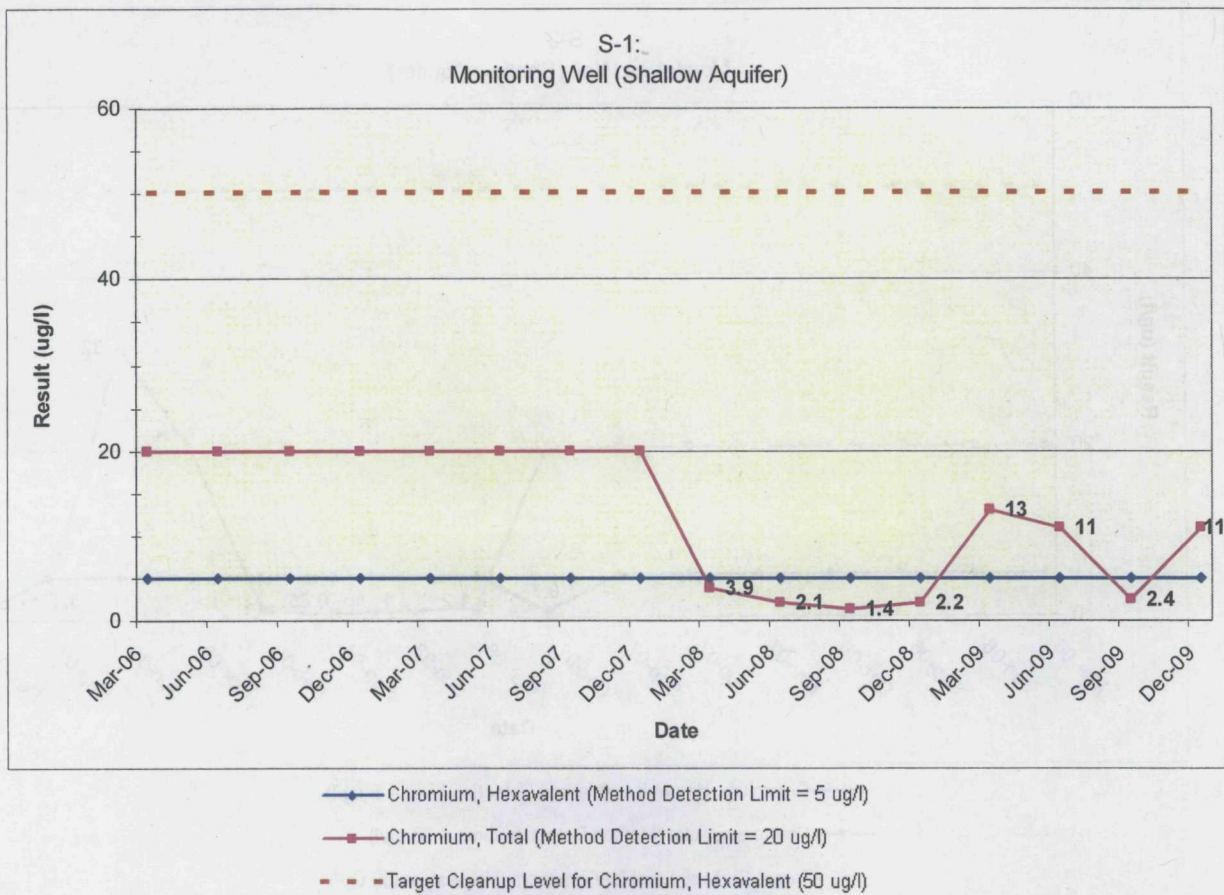


GRAPH 2

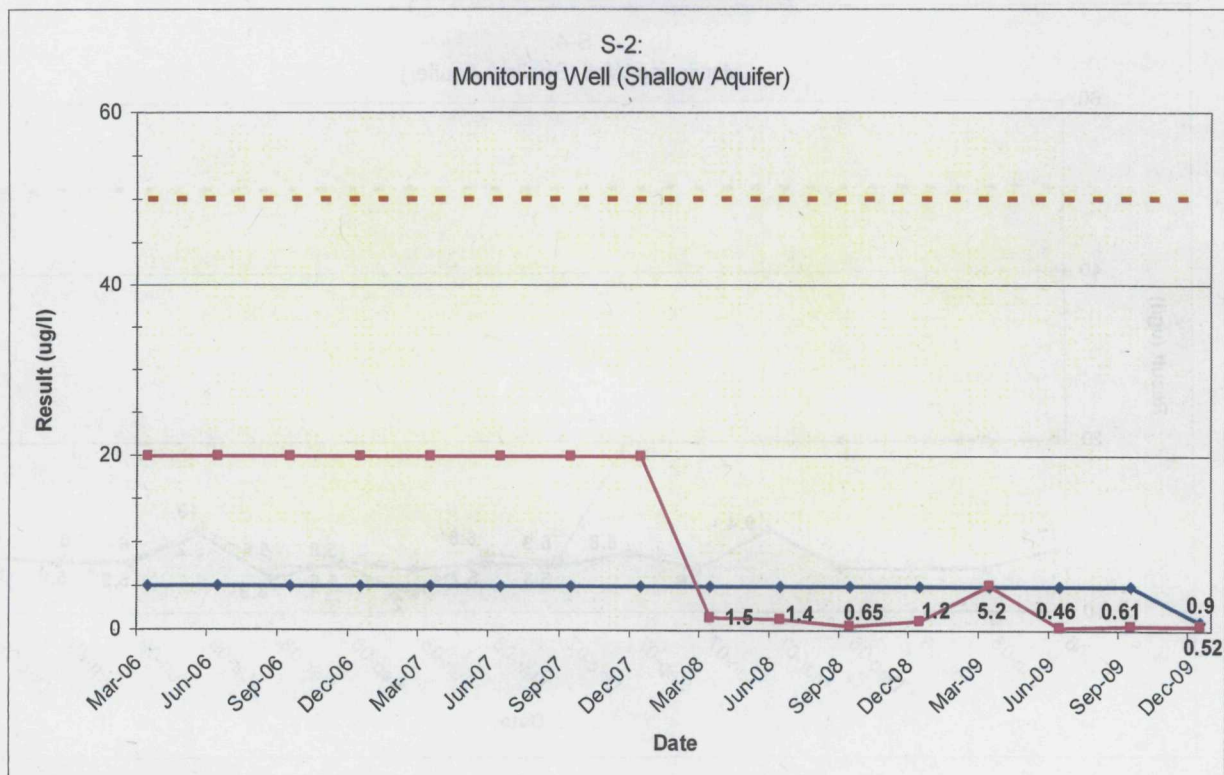
Effluent Hex Chrome

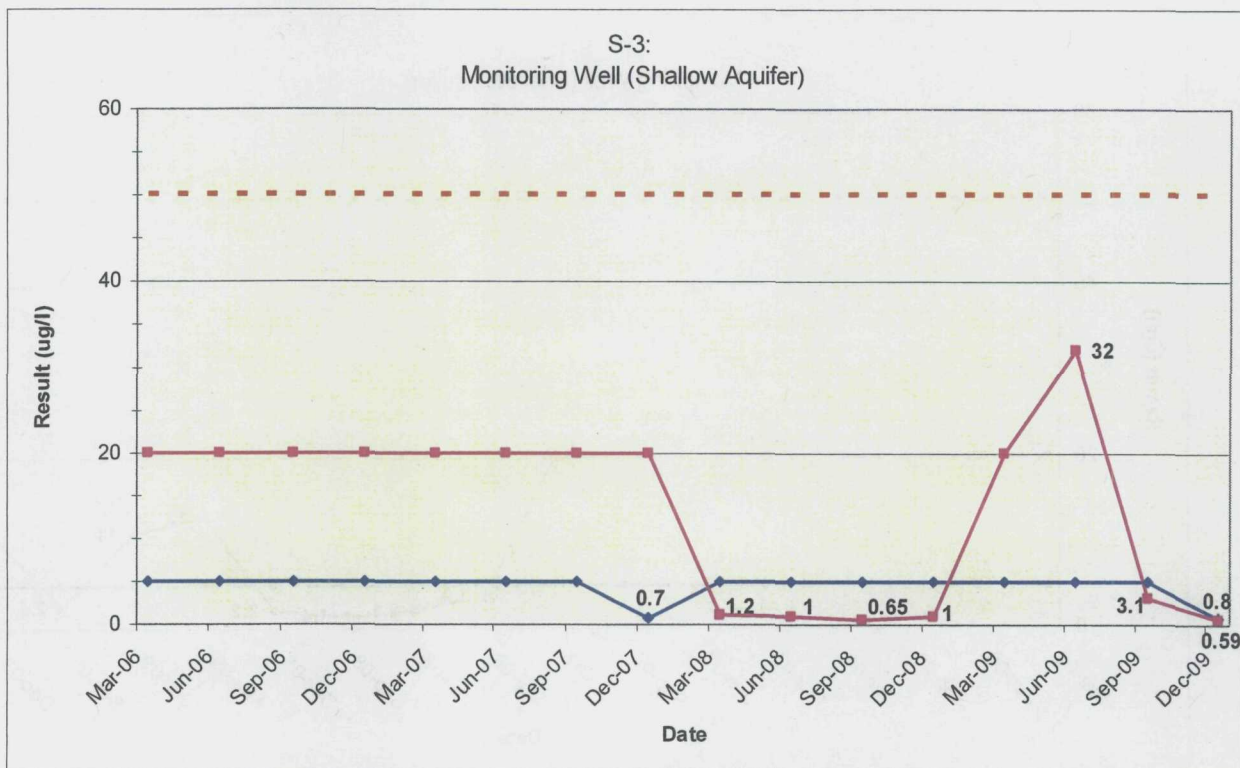


GRAPHS 3 – 33



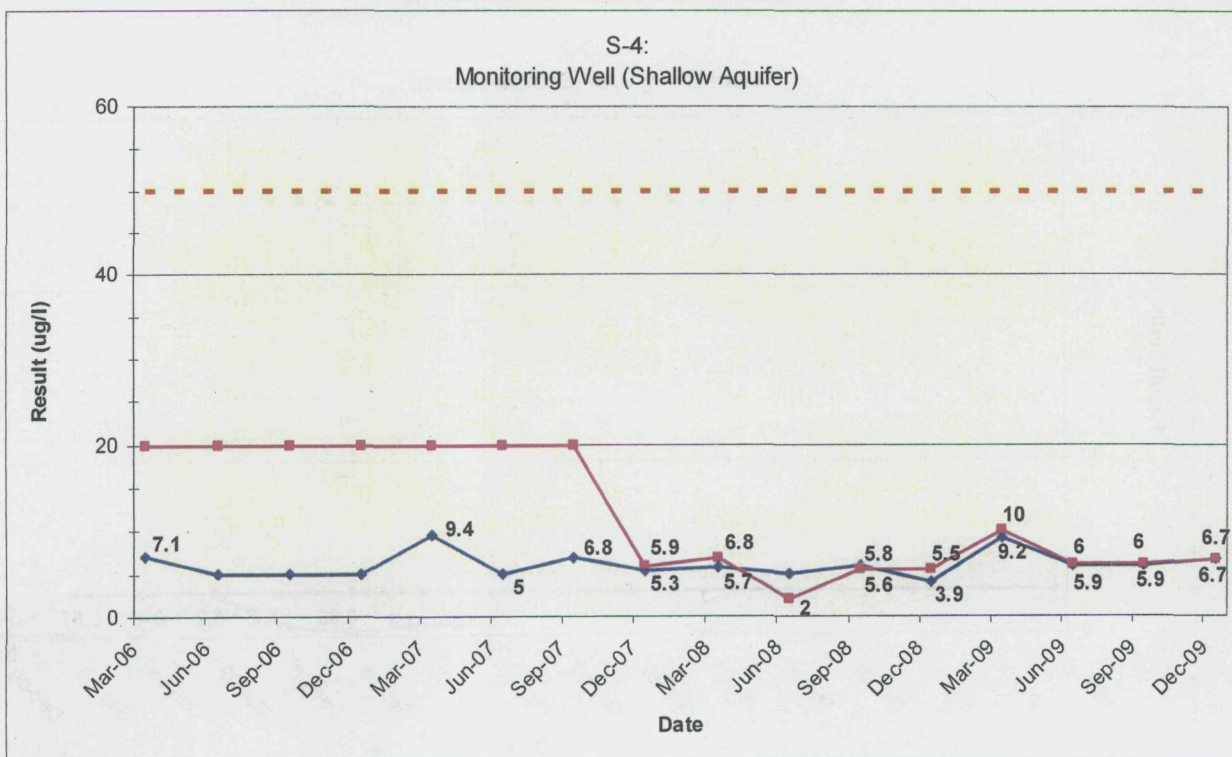
Listed Values Are Detected Results

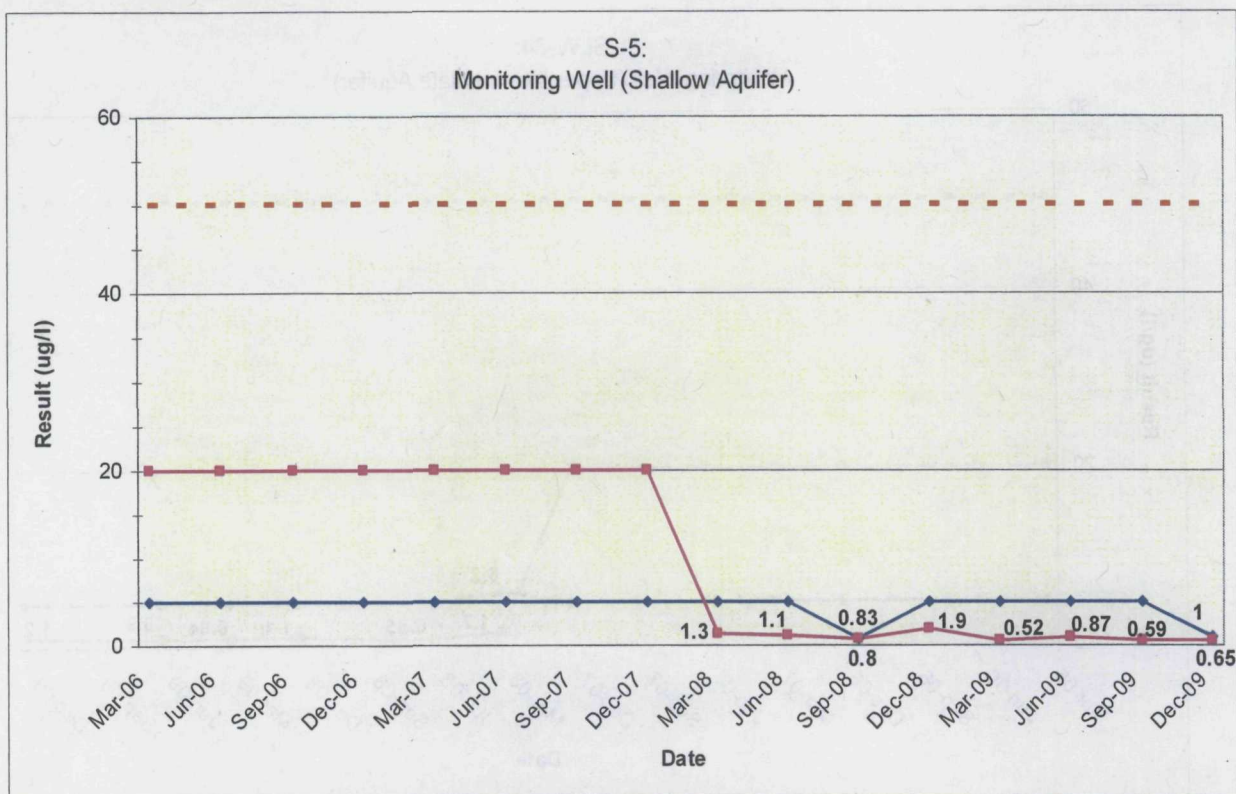




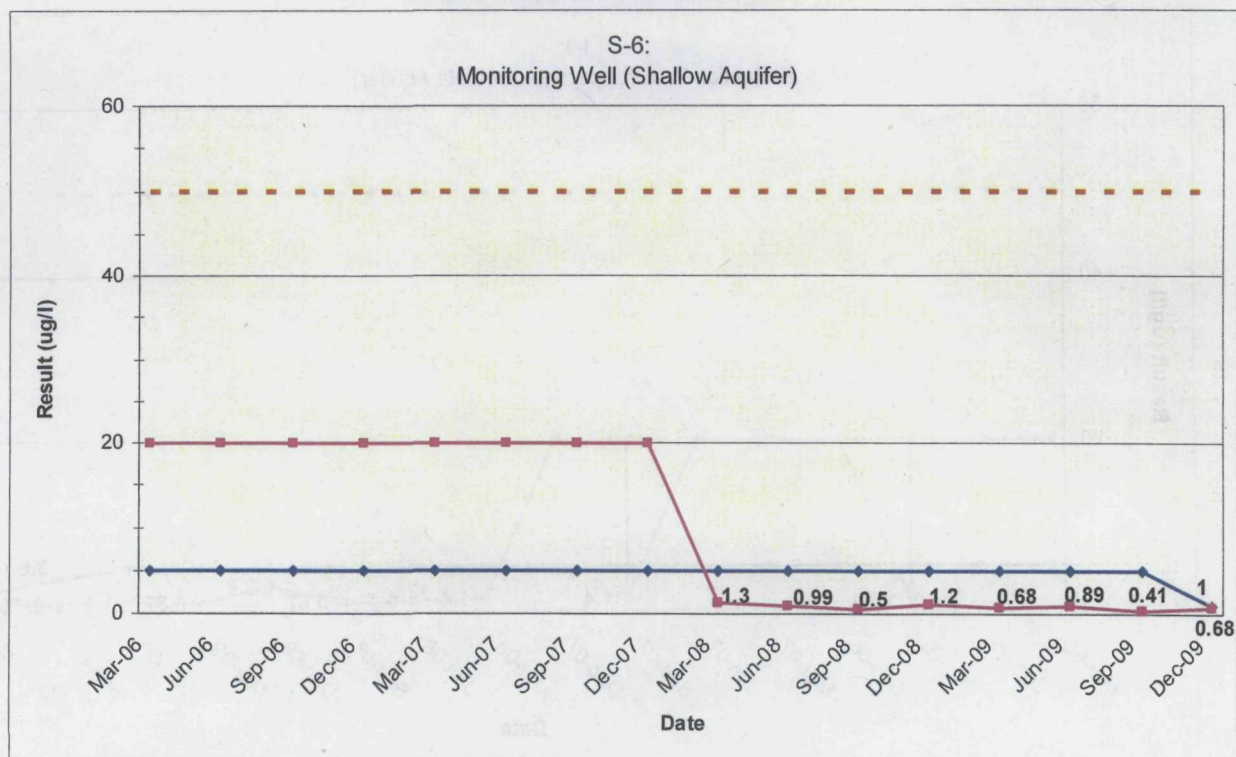
◆ Chromium, Hexavalent (Method Detection Limit = 5 ug/l)
 ■ Chromium, Total (Method Detection Limit = 20 ug/l)
 - - - Target Cleanup Level for Chromium, Hexavalent (50 ug/l)

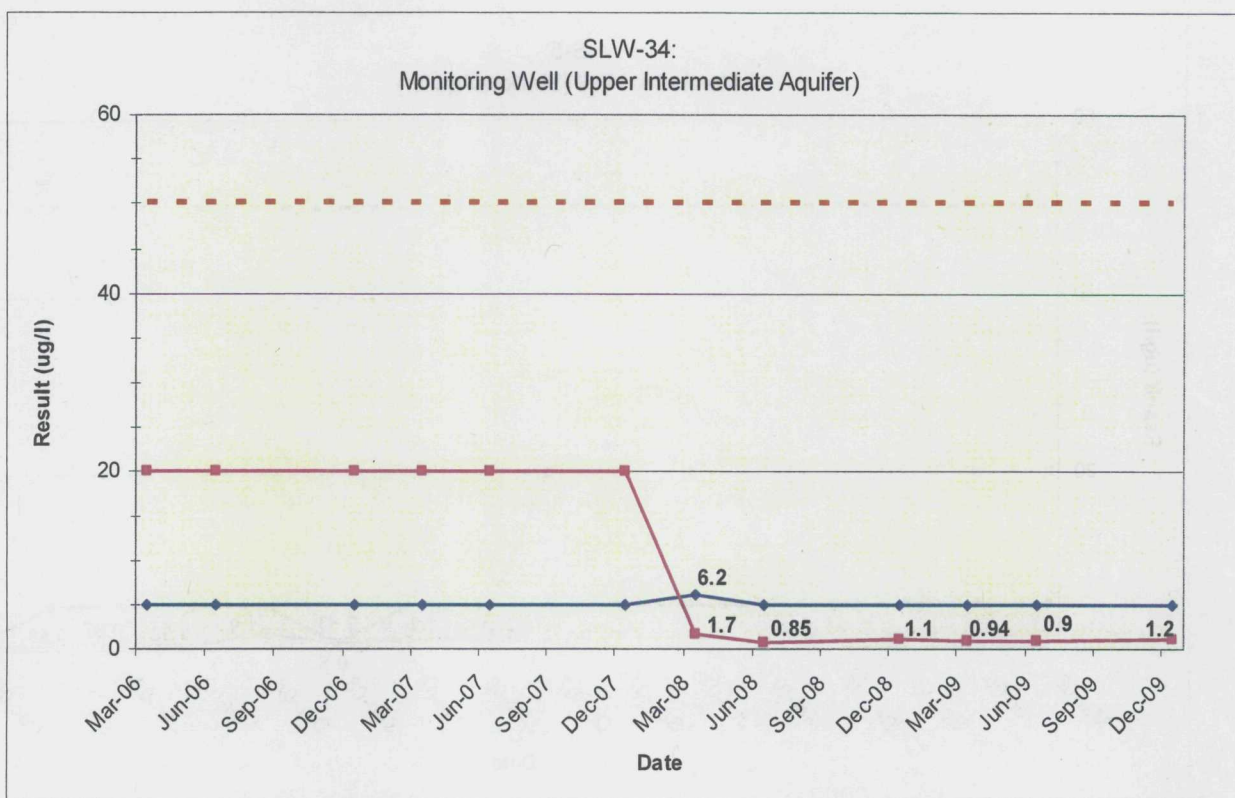
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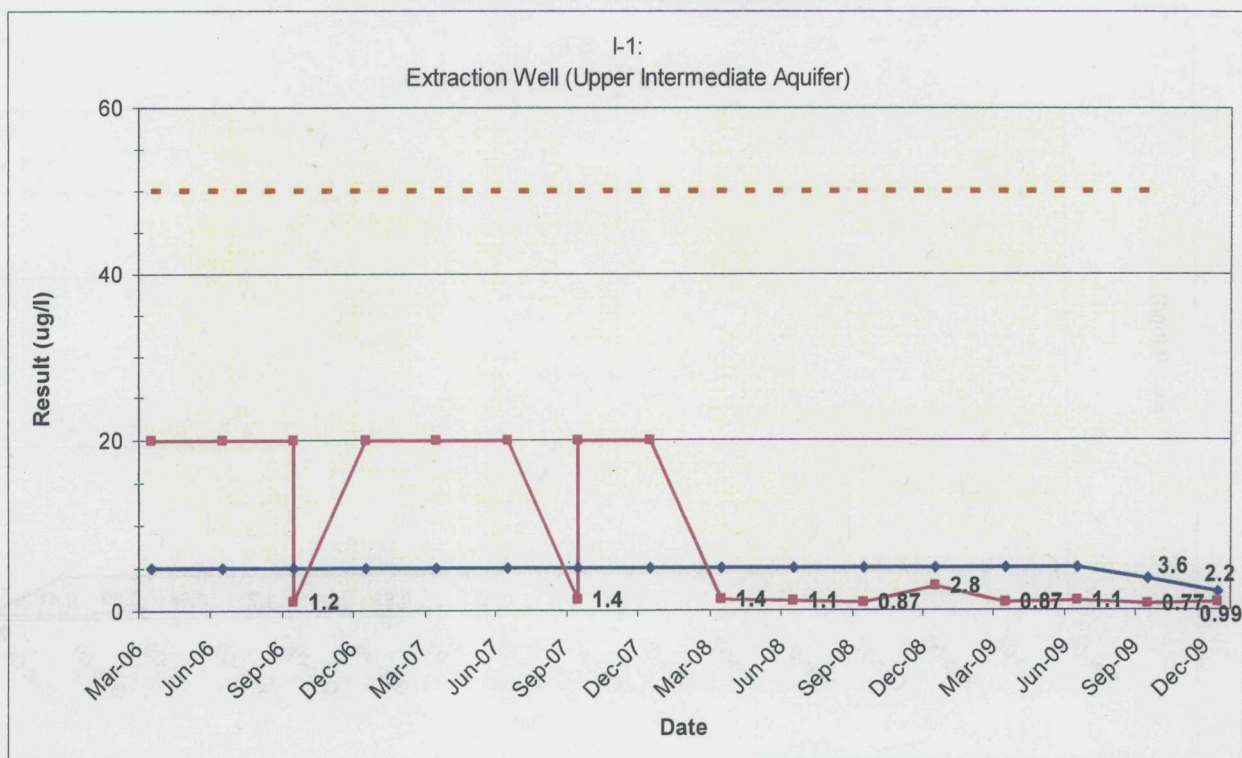
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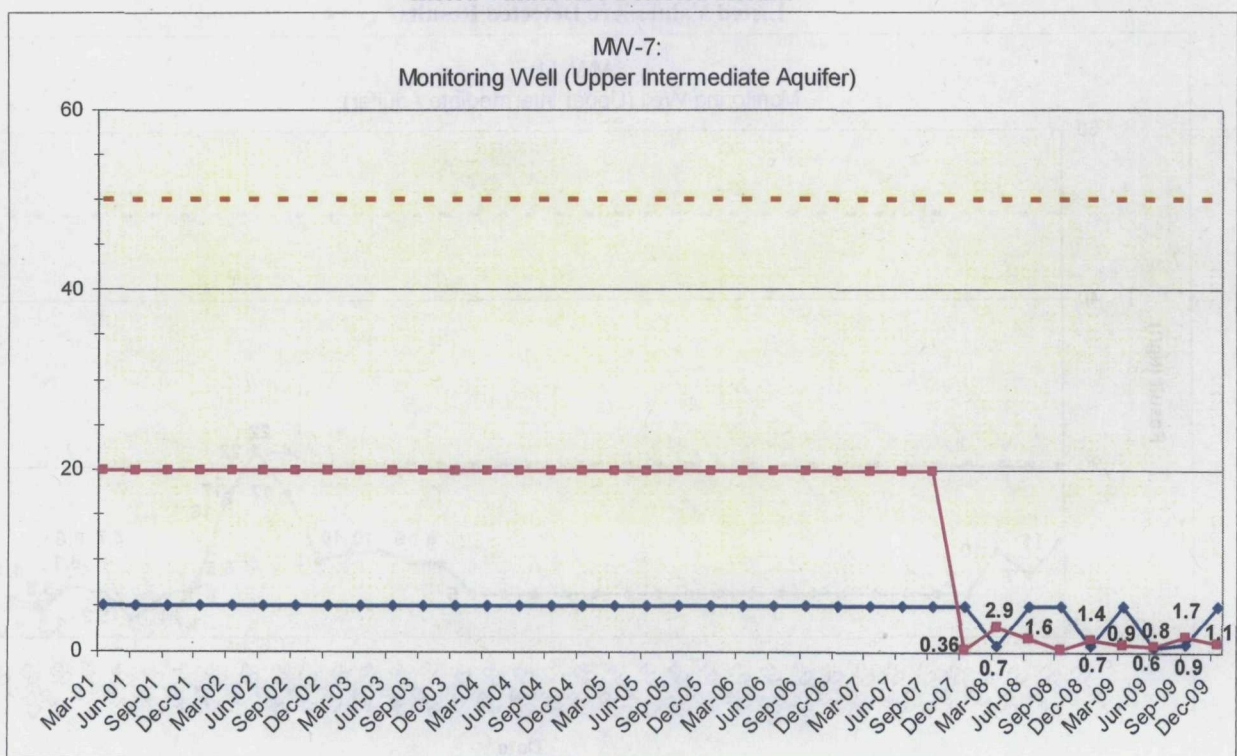
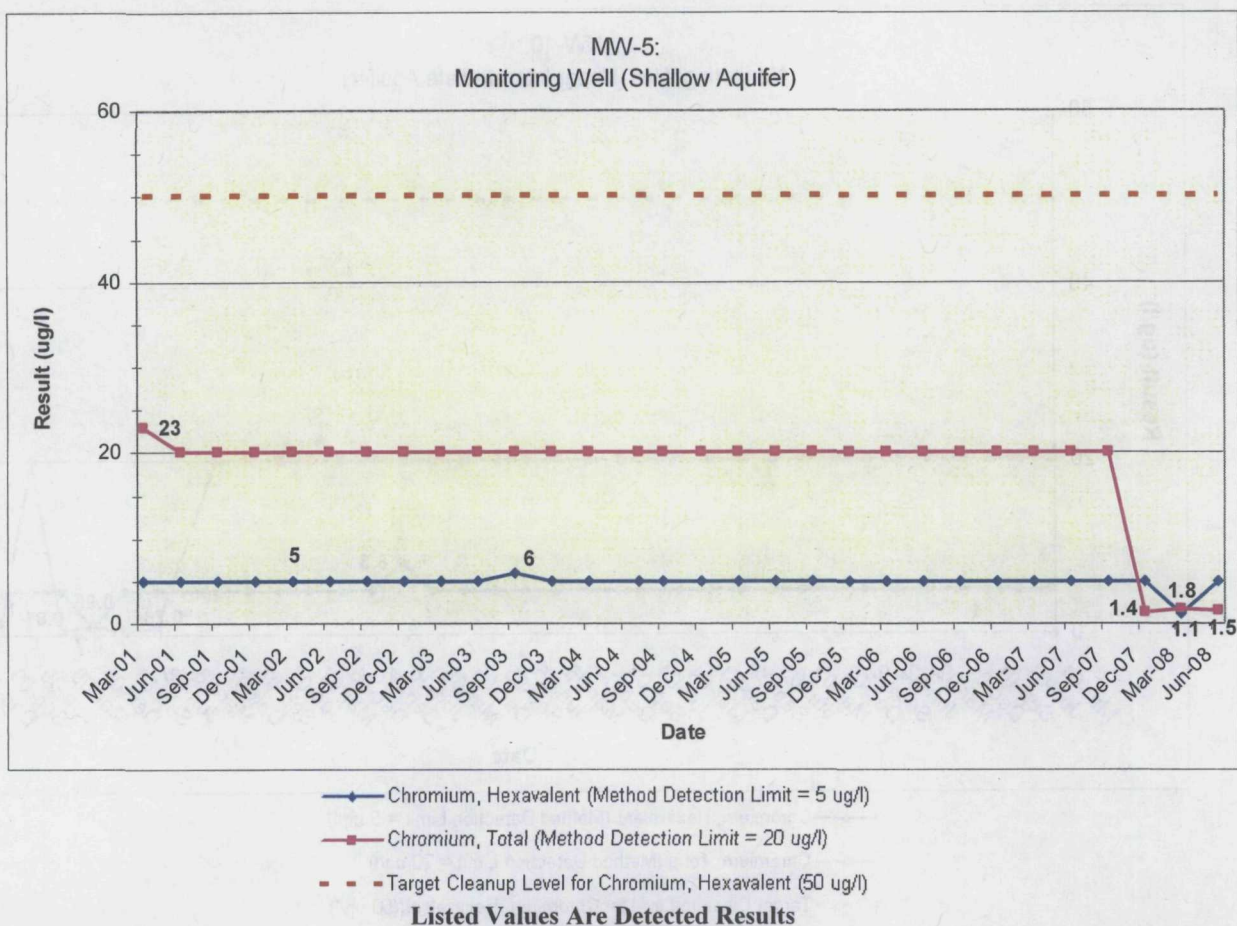


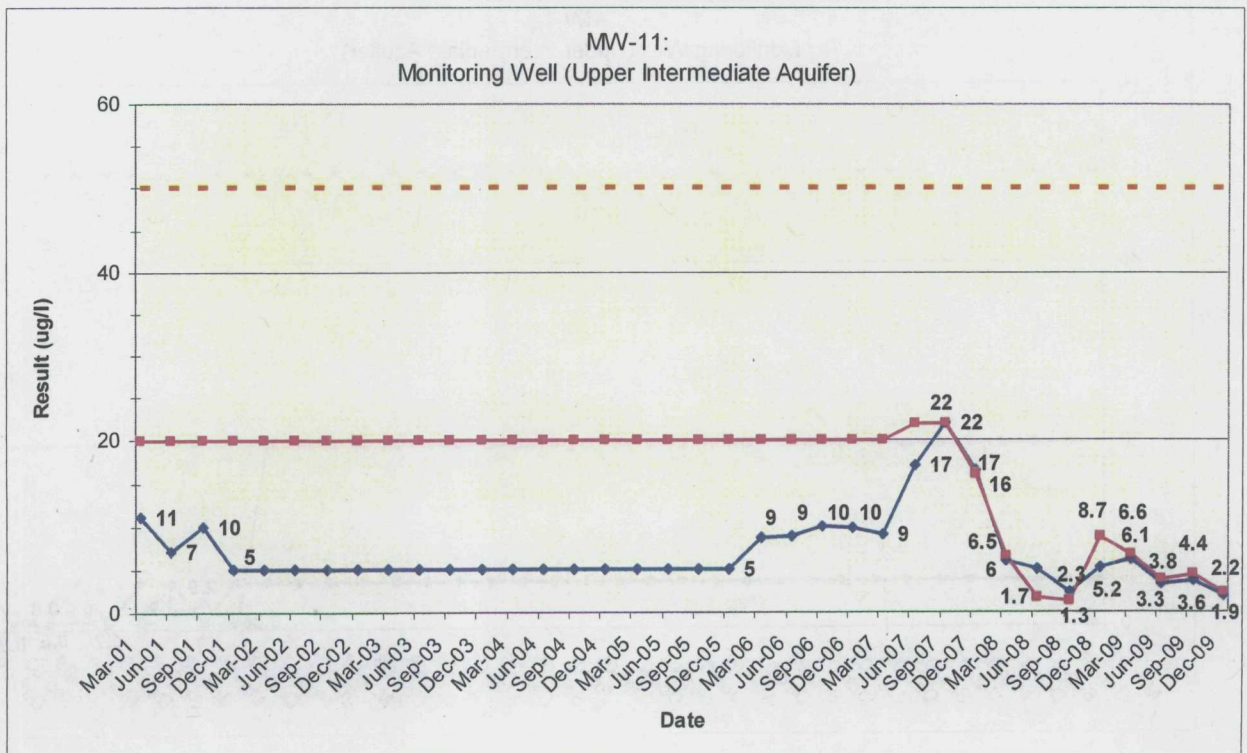
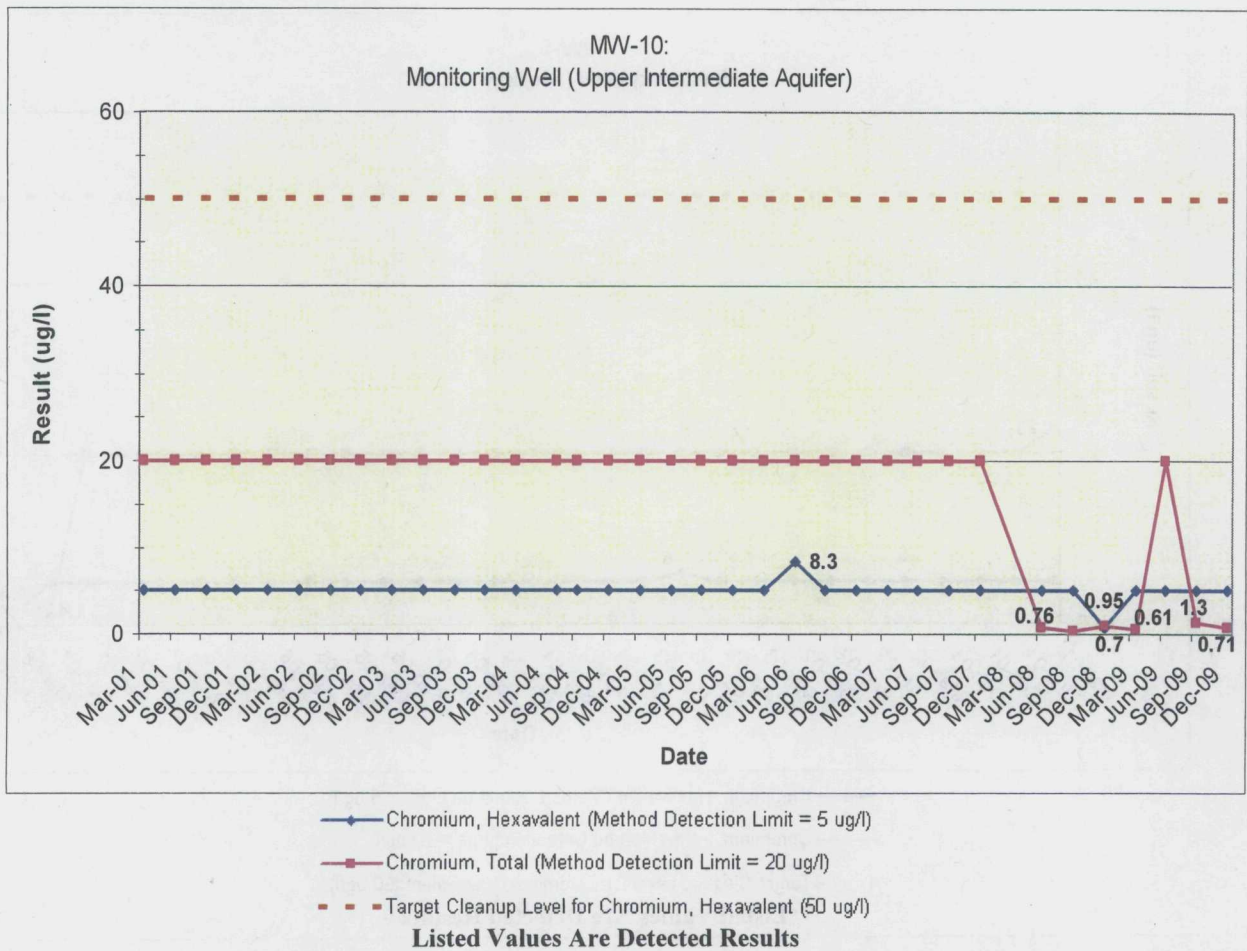


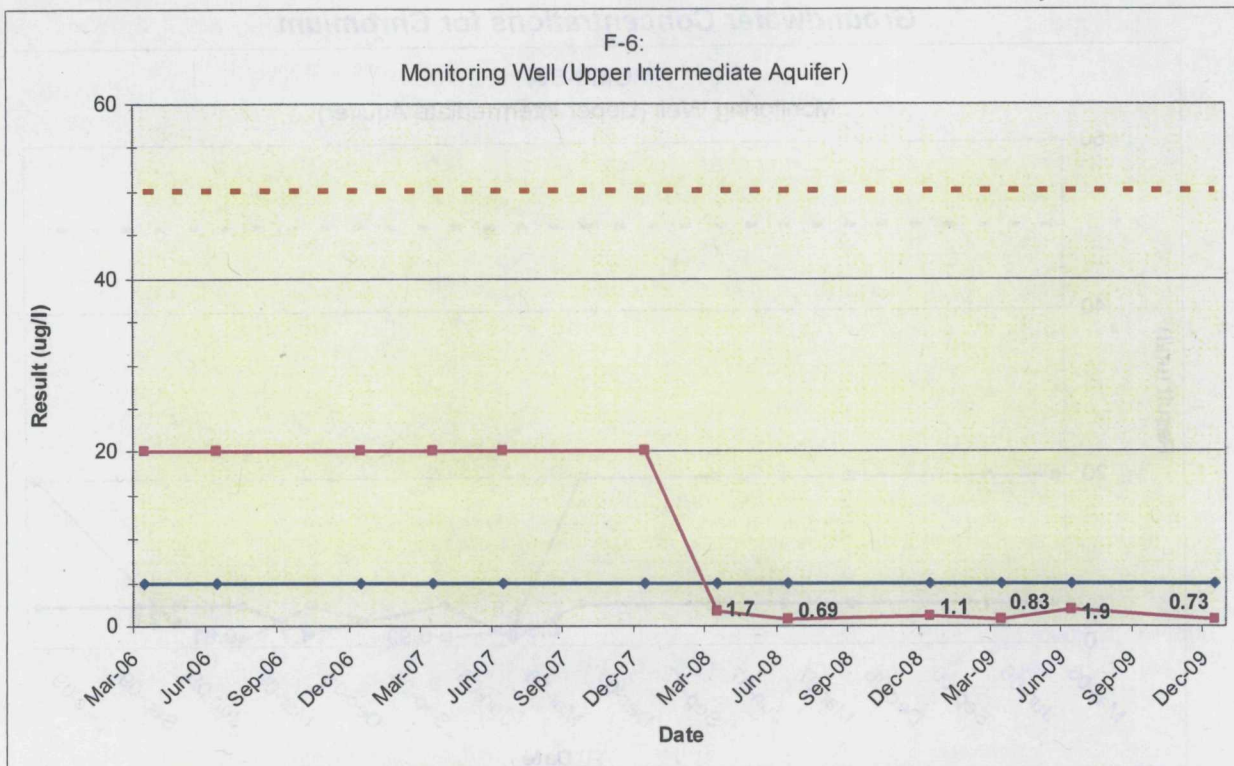
◆ Chromium, Hexavalent (Method Detection Limit = 5 ug/l)
 ■ Chromium, Total (Method Detection Limit = 20 ug/l)
 - - - Target Cleanup Level for Chromium, Hexavalent (50 ug/l)

Listed Values Are Detected Results



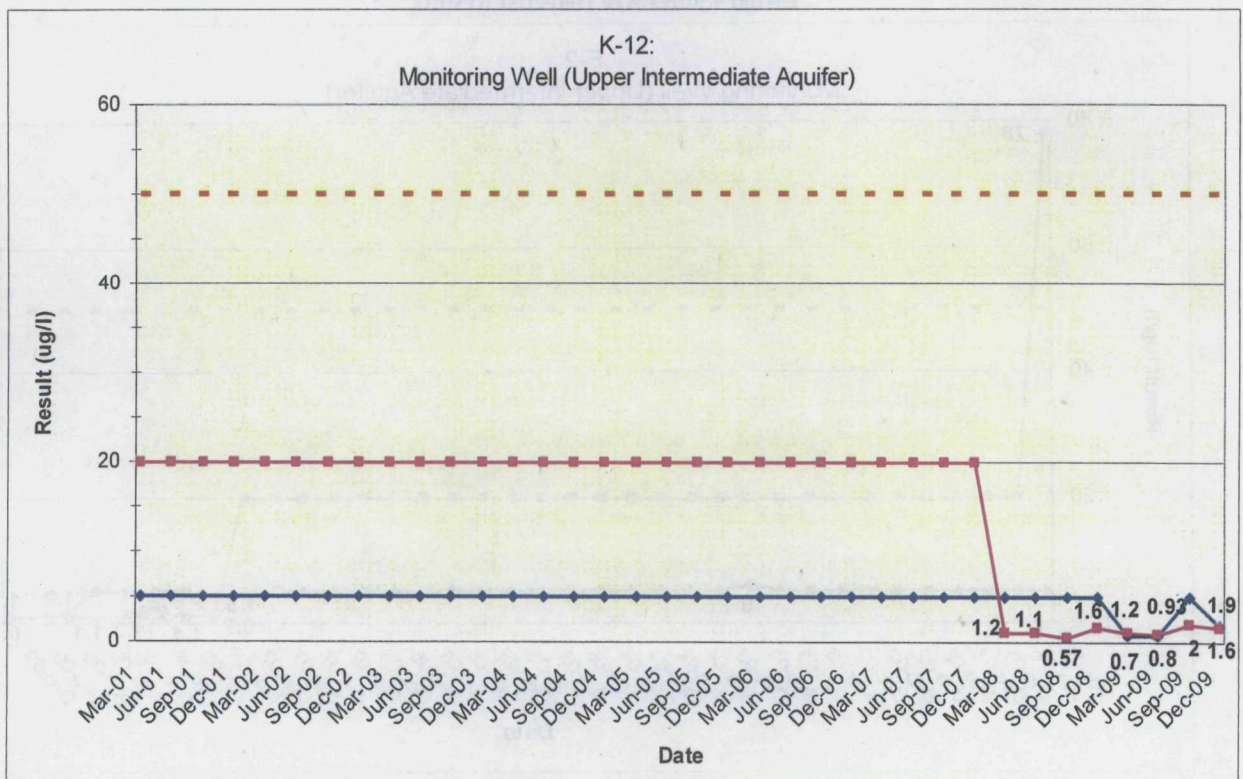




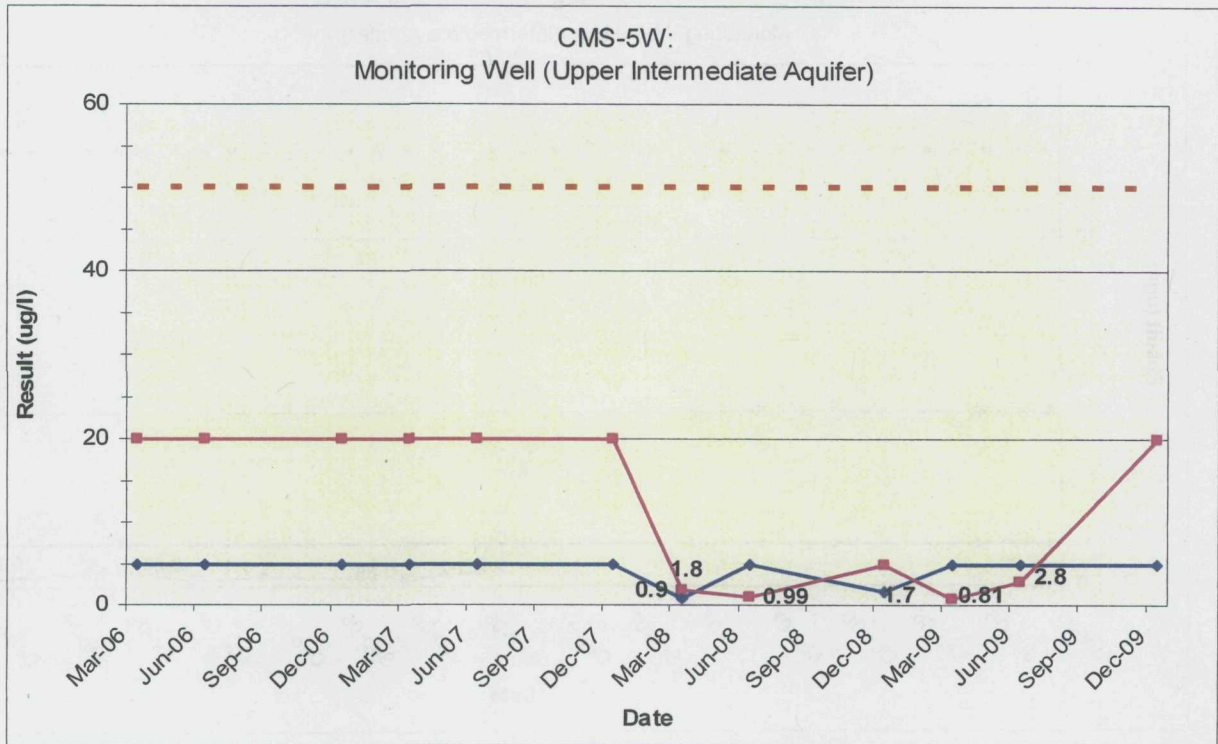


- Chromium, Hexavalent (Method Detection Limit = 5 ug/l)
- Chromium, Total (Method Detection Limit = 20 ug/l)
- - - Target Cleanup Level for Chromium, Hexavalent (50 ug/l)

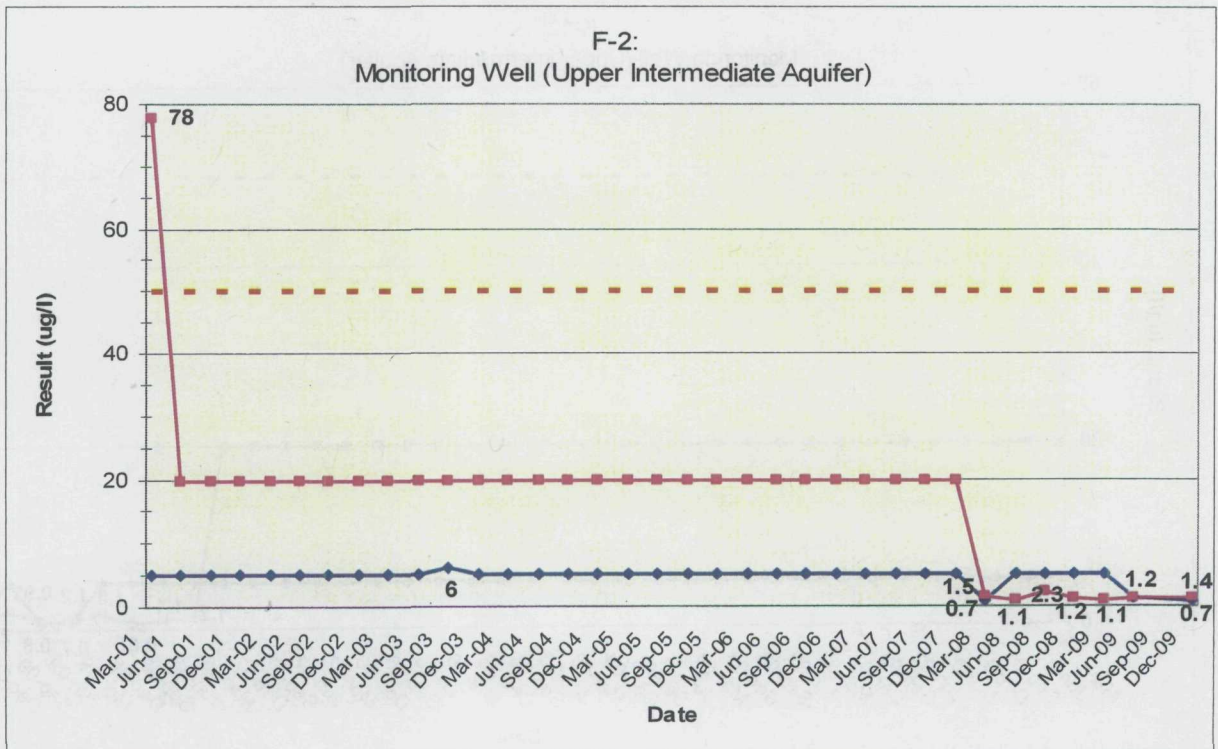
Listed Values Are Detected Results

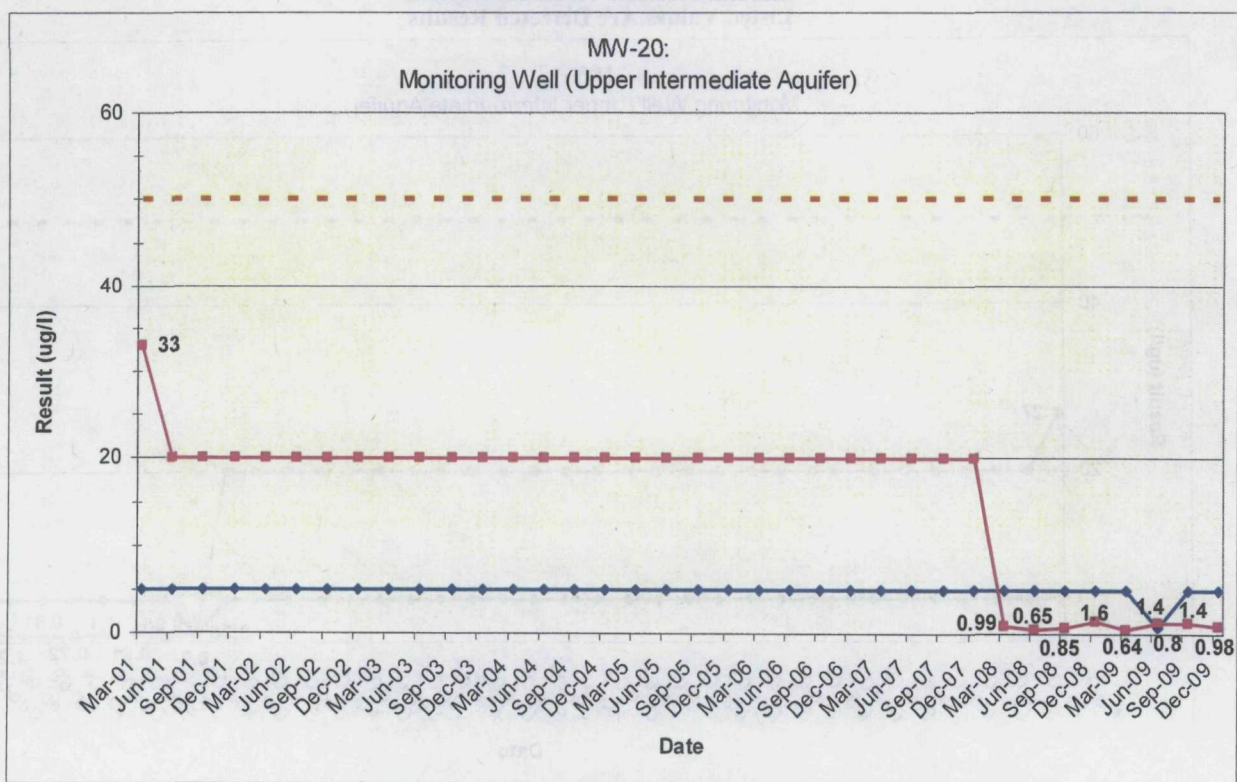
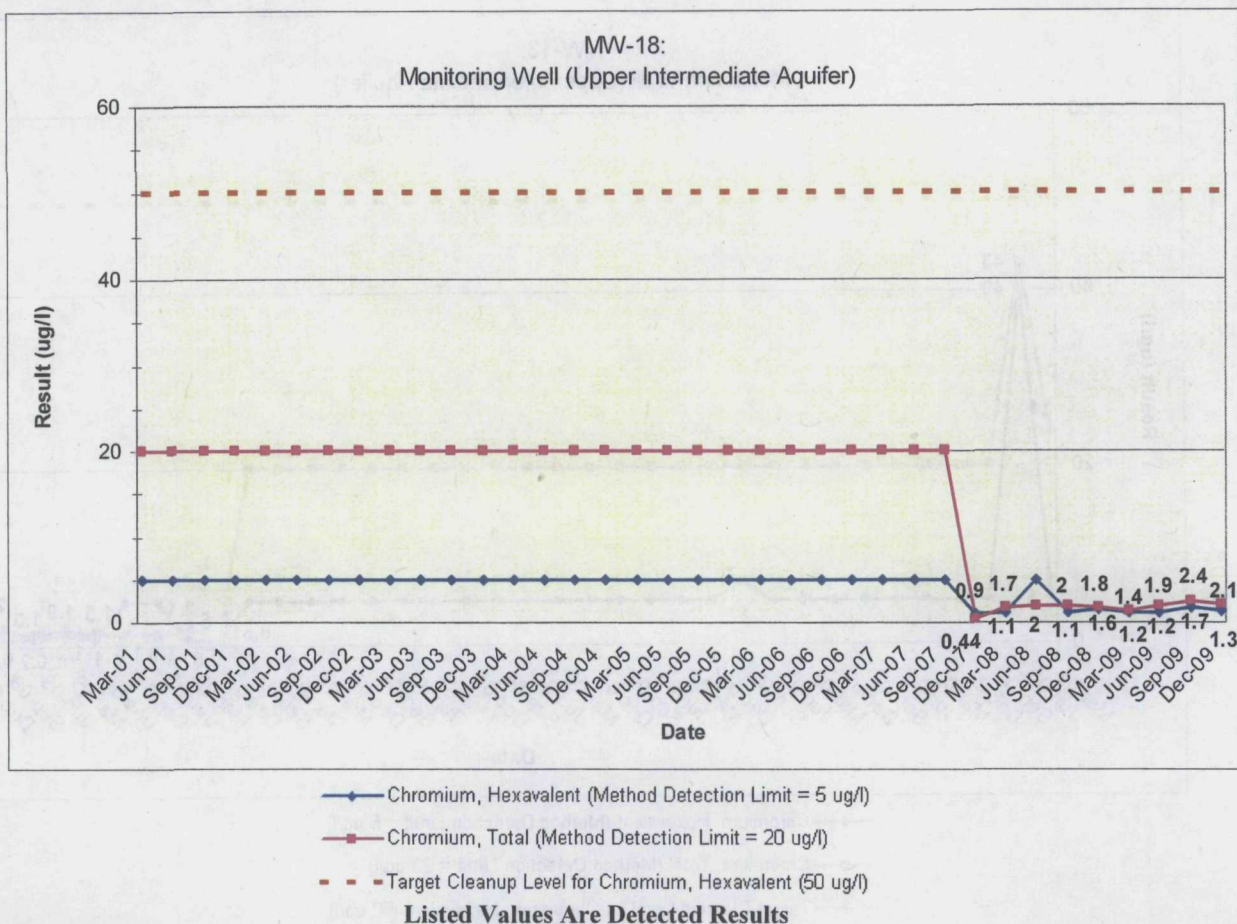


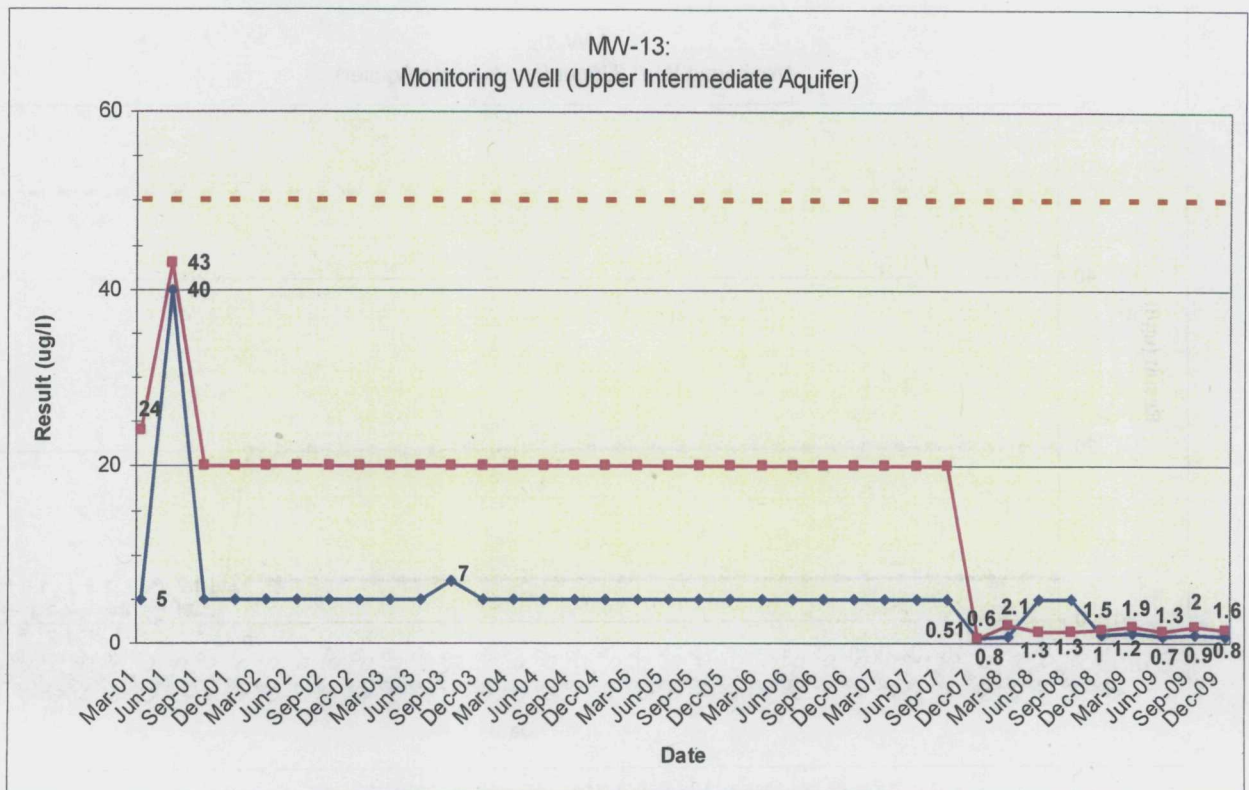
Groundwater Concentrations for Chromium



—●— Chromium, Hexavalent (Method Detection Limit = 5 ug/l)
 —■— Chromium, Total (Method Detection Limit = 20 ug/l)
 - - - Target Cleanup Level for Chromium, Hexavalent (50 ug/l)
Listed Values Are Detected Results

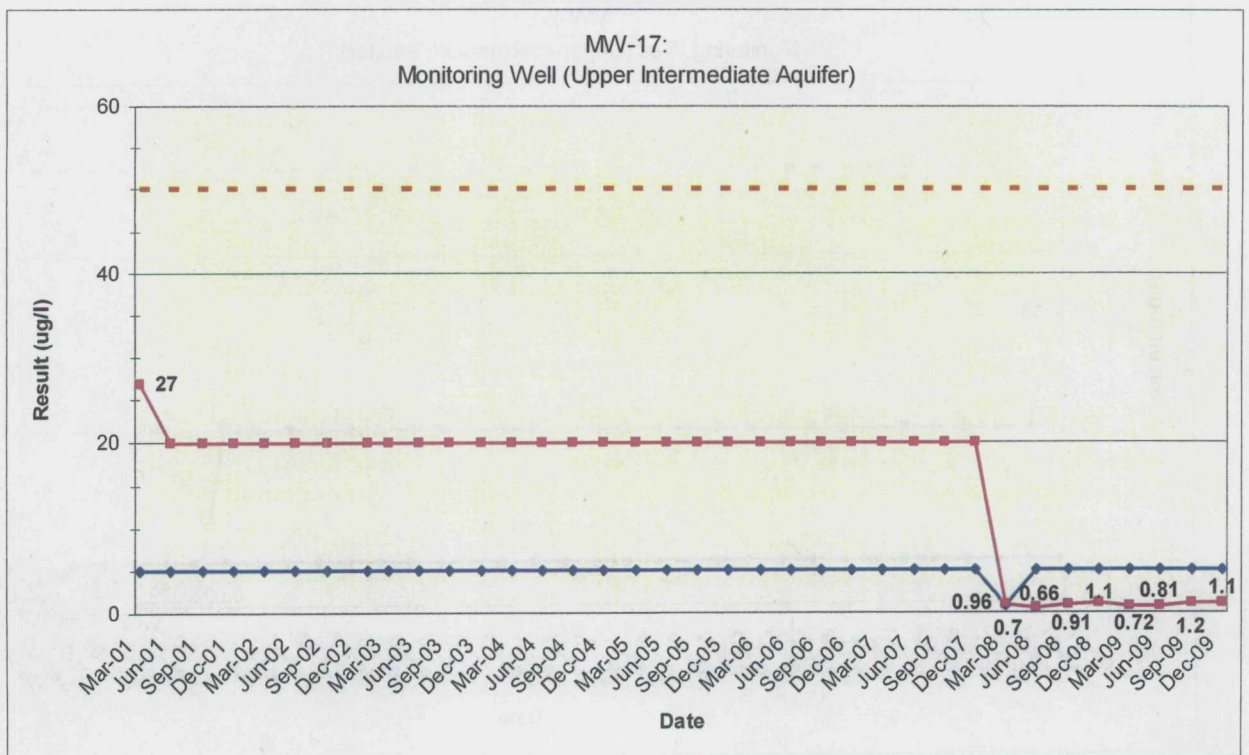


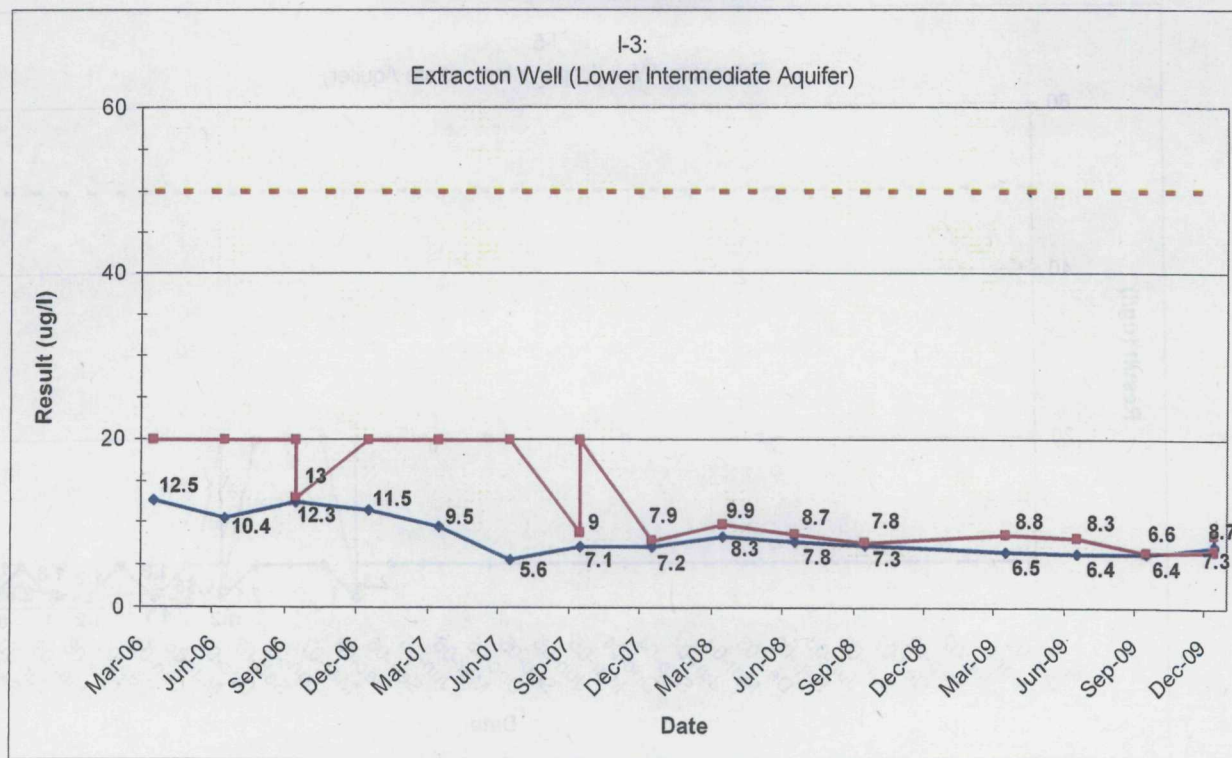
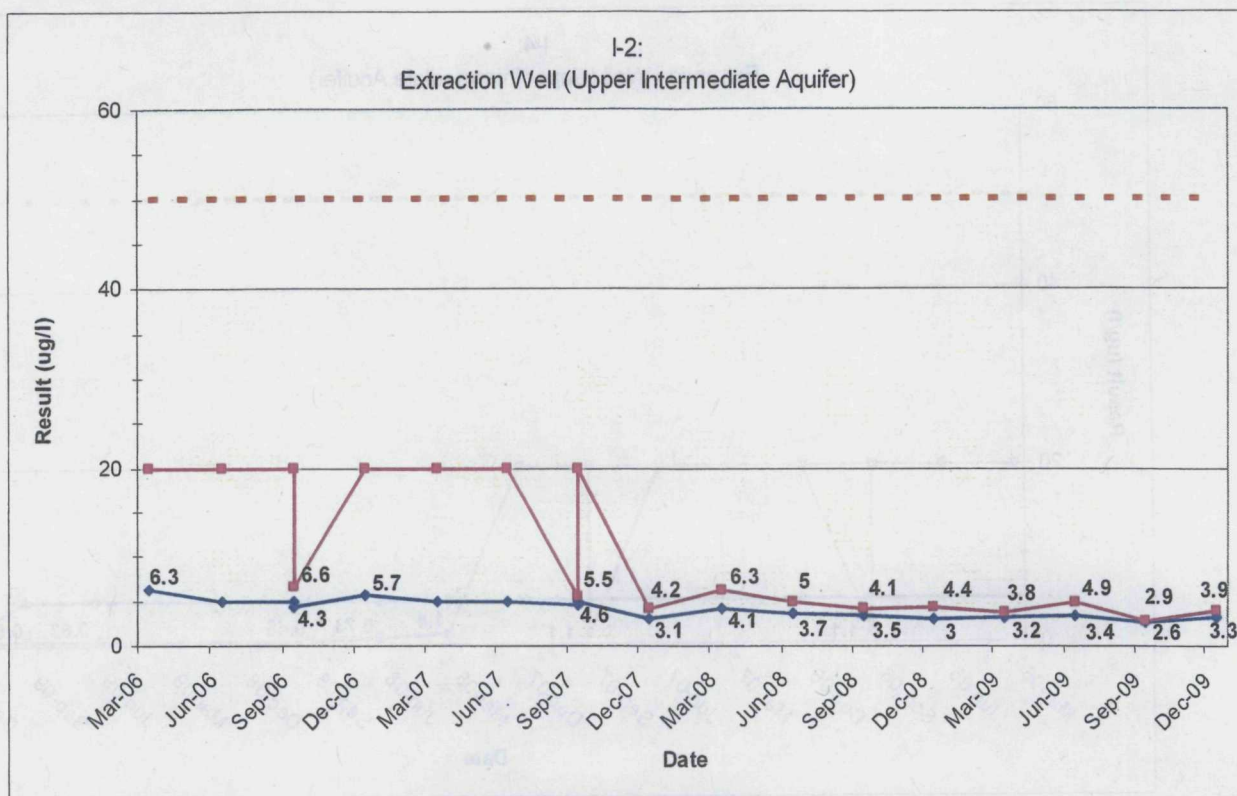


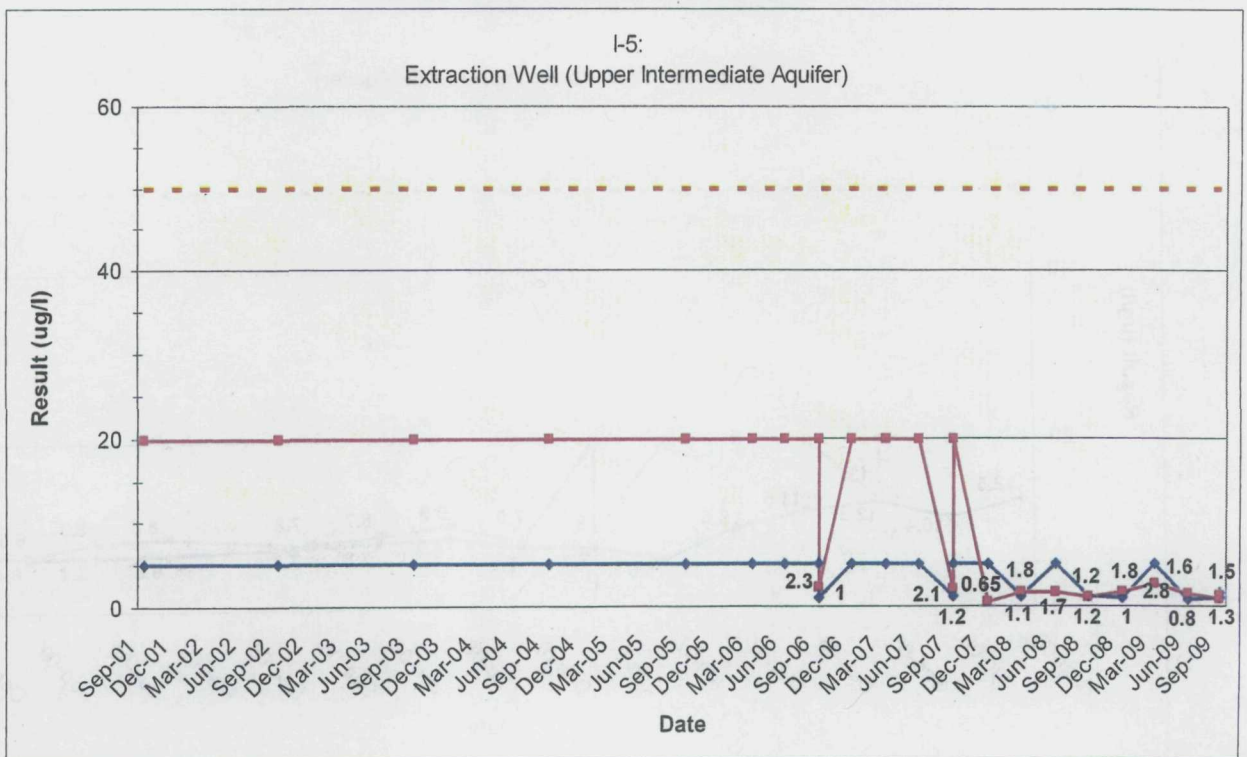
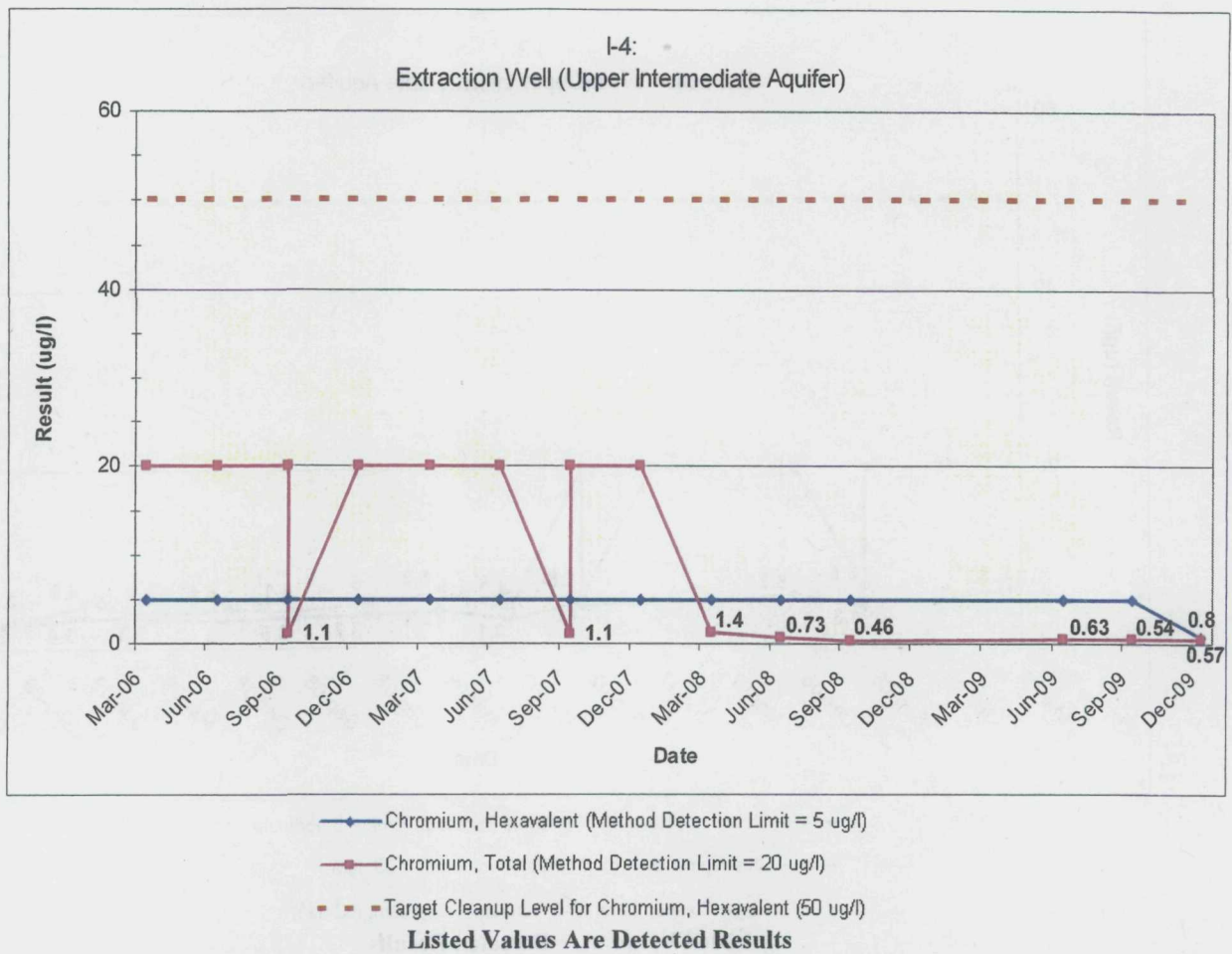


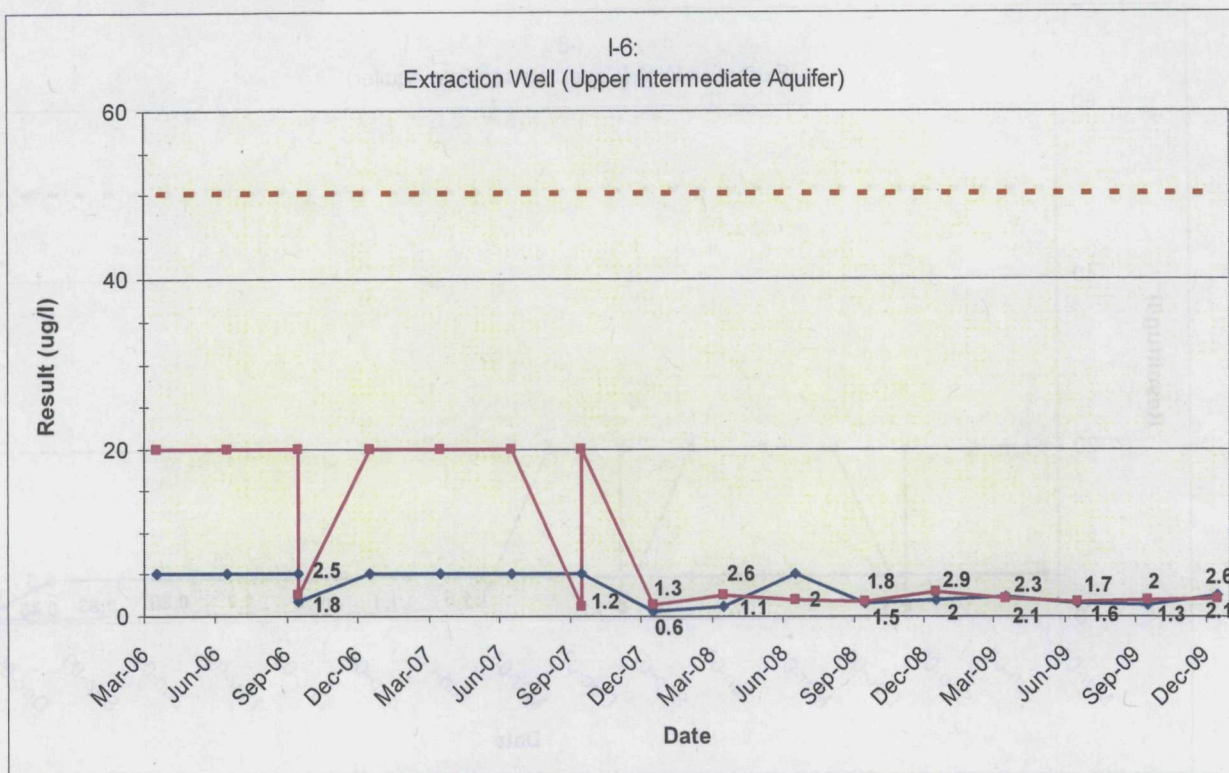
- Chromium, Hexavalent (Method Detection Limit = 5 ug/l)
- Chromium, Total (Method Detection Limit = 20 ug/l)
- - - Target Cleanup Level for Chromium, Hexavalent (50 ug/l)

Listed Values Are Detected Results



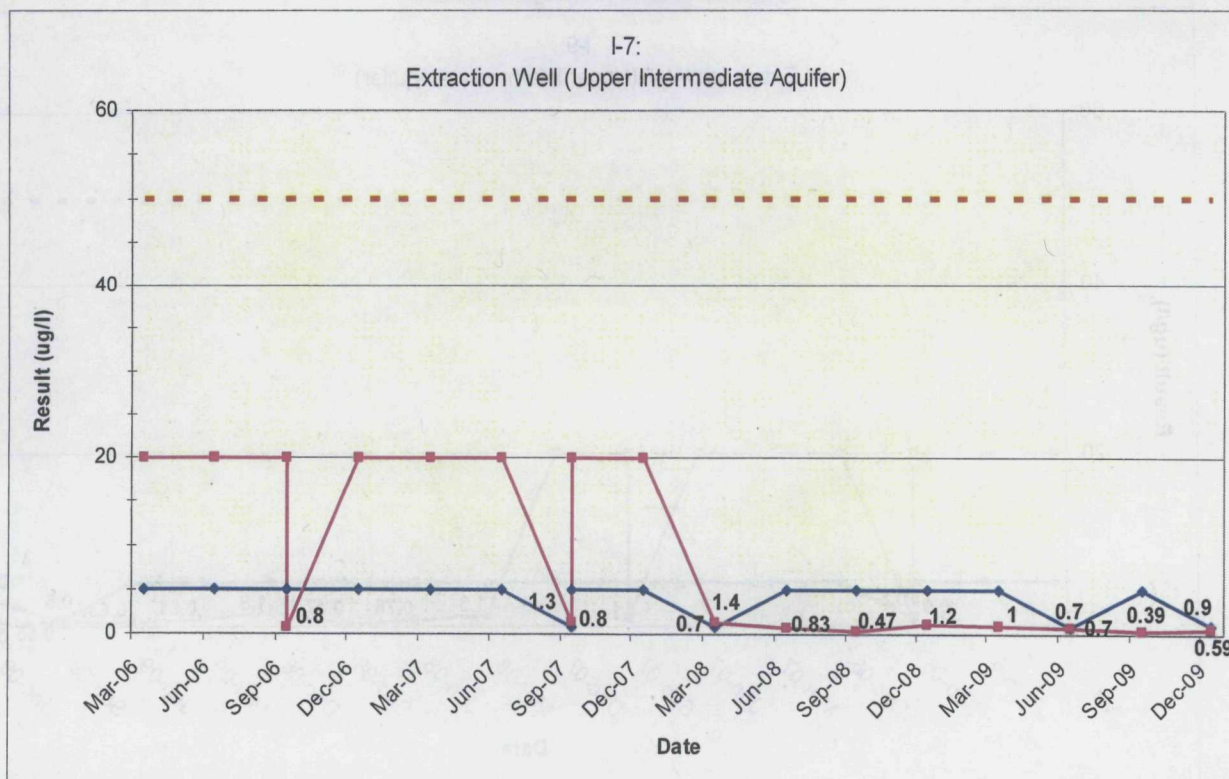


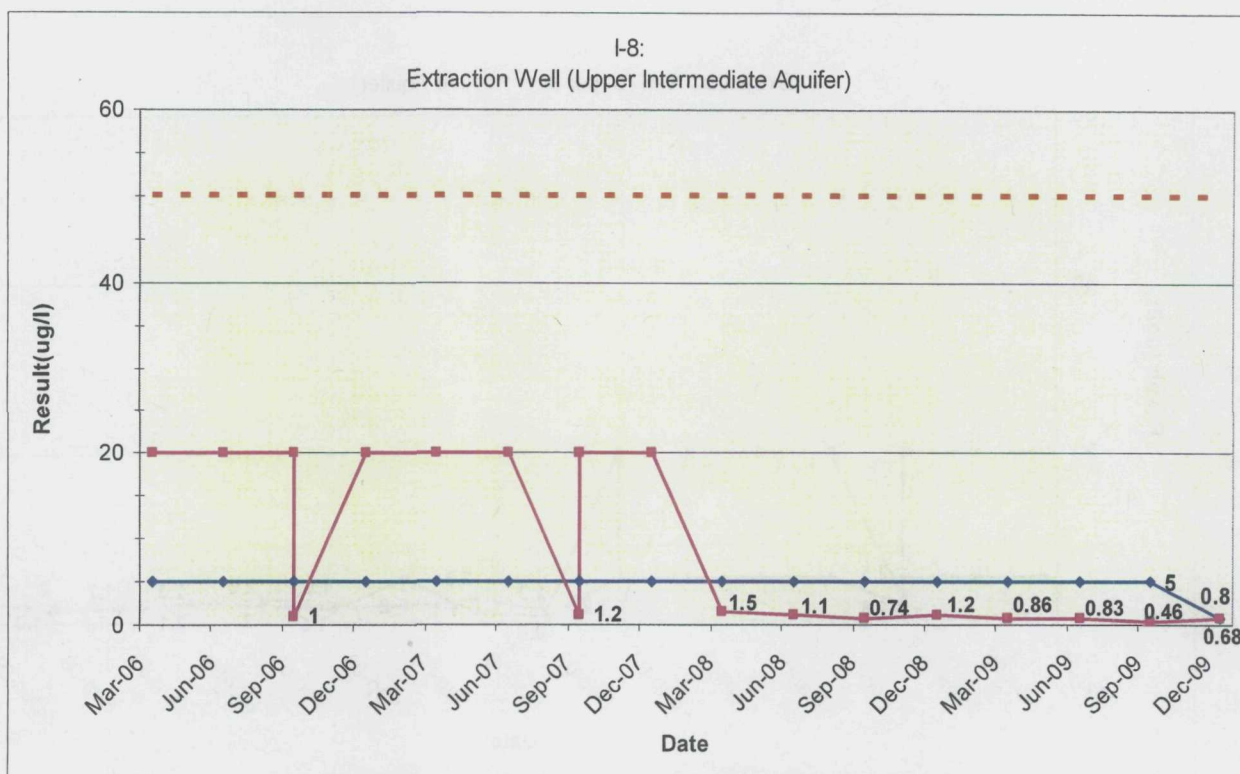




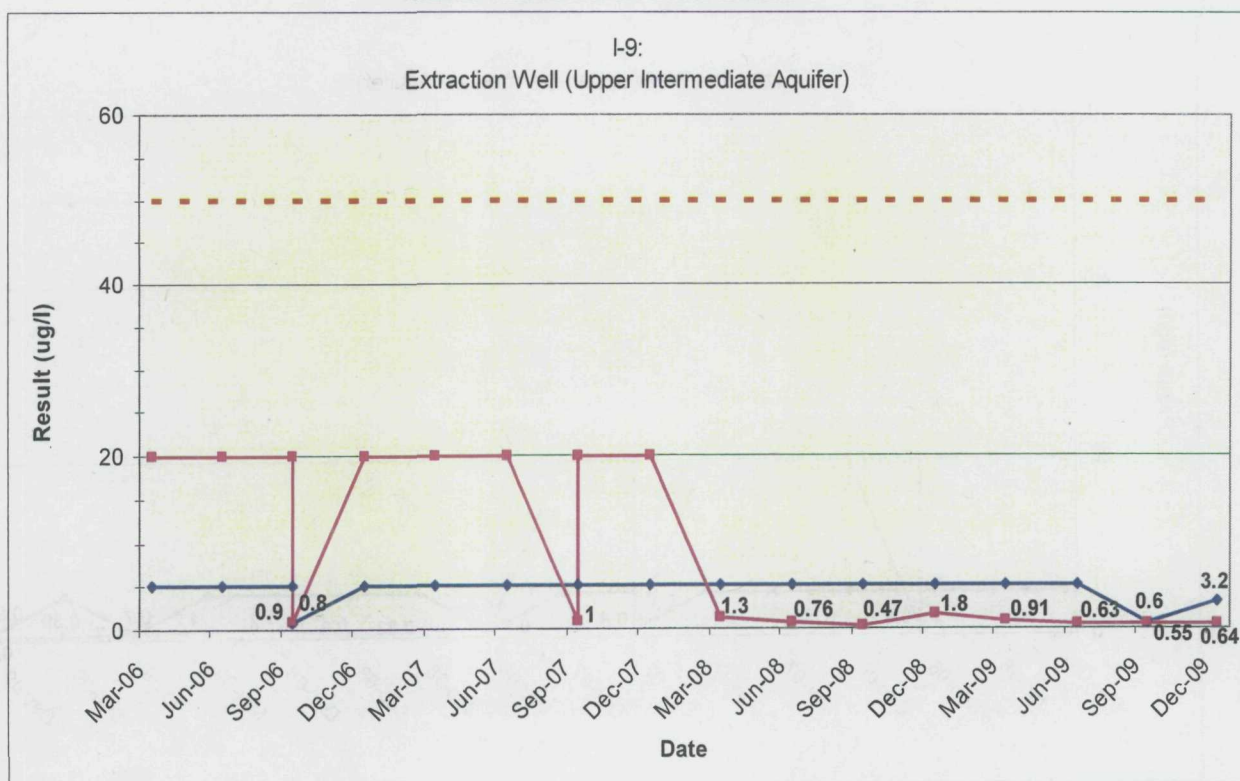
- Chromium, Hexavalent (Method Detection Limit = 5 ug/l)
- Chromium, Total (Method Detection Limit = 20 ug/l)
- - - Target Cleanup Level for Chromium, Hexavalent (50 ug/l)

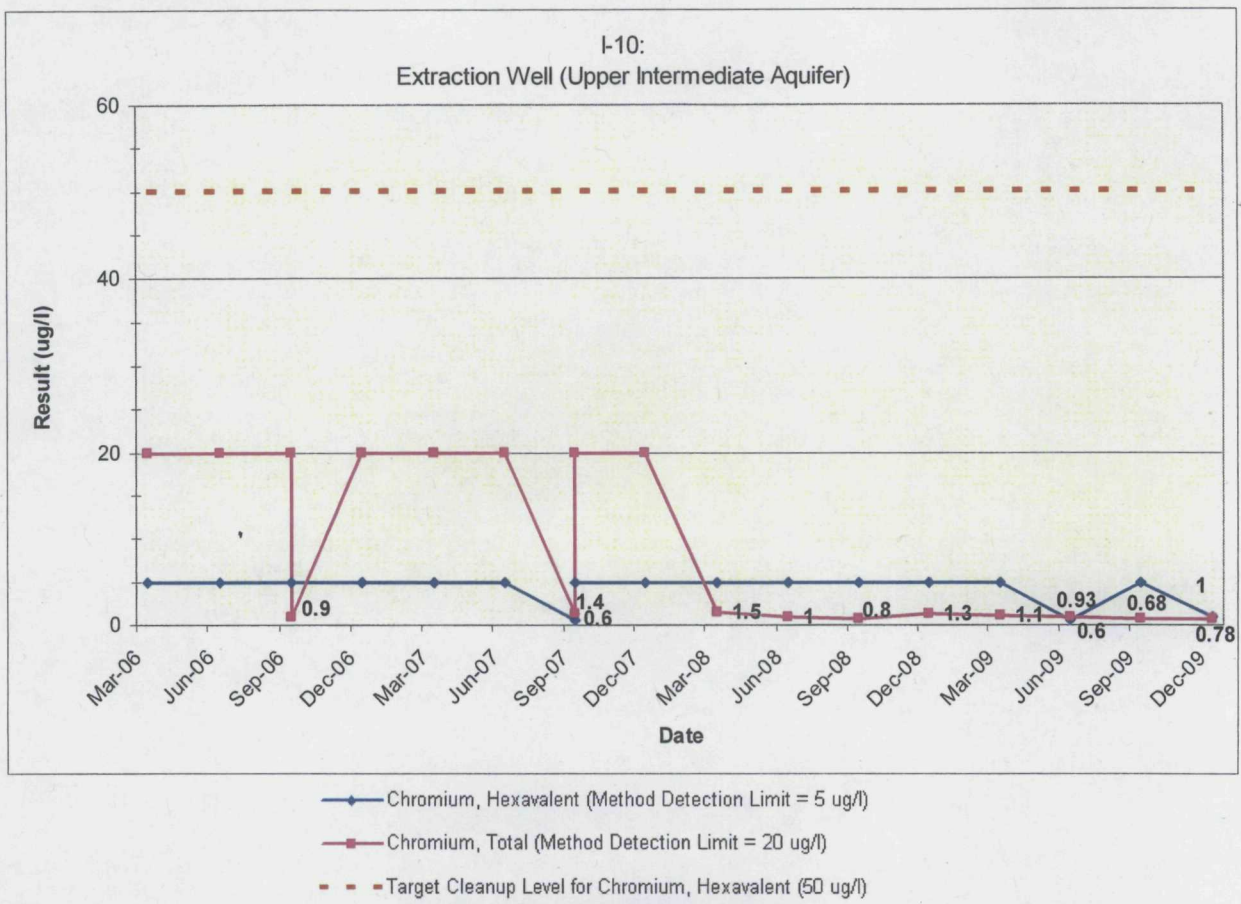
Listed Values Are Detected Results



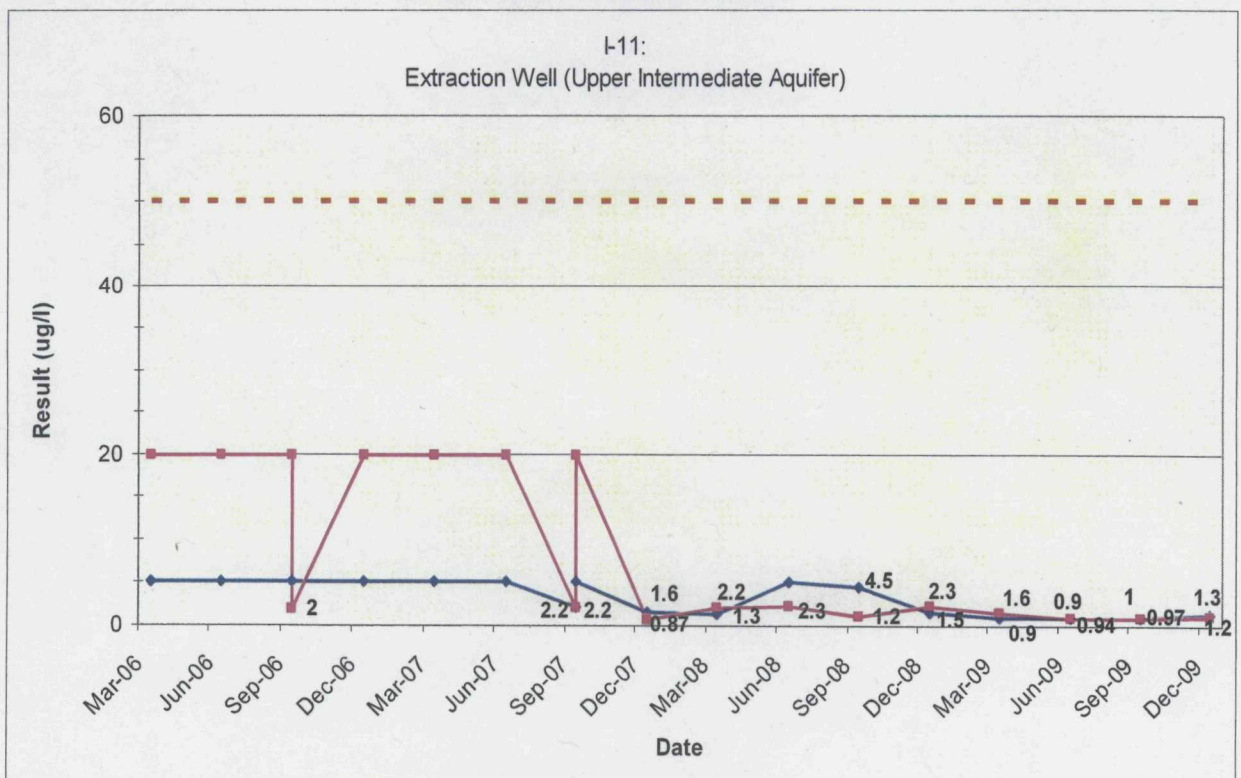


Listed Values Are Detected Results





Listed Values Are Detected Results



ATTACHMENT 2 – ADDITIONAL TABLES

Table 4 – LDFA Extraction Well Construction

Table 5 – LDFA Monitoring Plan

TABLE 4

LDFA Extraction Well Construction Summary			
Extraction Well No.	Date Completed	Well Depth (feet bgl)	Screened Interval (feet bgl)
S-1	7/7/95	92	77 - 92
S-2	7/8/95	71.5	56.5 - 71.5
S-3	8/20/95	67	52 - 67
S-4	7/10/95	50	35 - 50
S-5	7/9/95	67	57 - 67
S-6	7/24/95	53	43 - 53
S-7	8/5/95	46	36 - 46
I-1	7/10/95	187	162 - 187
I-2	7/19/95	185	160 - 185
I-3	9/29/95	221	201 - 221
I-4	8/20/95	177	152 - 177
I-5	9/8/95	150	125 - 150
I-6	8/2/95	179	154-179
I-7	10/5/95	164	139 - 164
I-8	8/30/95	158	133 - 158
I-9	7/25/95	190	165 - 190
I-10	8/15/95	191	166 -191
I-11	10/12/95	163	133 - 163

TABLE 5



S.S. PAPADOPULOS & ASSOCIATES, INC.

LDFA Monitoring Plan

Well	Easting ¹	Northing ¹	Aquifer ²	Compliance Wells ³	VOC Monitoring ³	Chromium Monitoring since March 2006 ⁴	Water Level Monitoring ⁵
S-1	2053830.4	16081779.7	S	X	A	X	
S-2	2053856.4	16082337.5	S	X	A	X	
S-3	2053763.0	16083387.3	S	X	A	X	
S-4	2054228.0	16083121.6	S	X	A	X	
S-5	2052889.8	16084571.7	S	X	A	X	
S-6	2053787.1	16084036.9	S	X	A	X	
S-7	2054218.2	16083613.7	S	X		X	
DNR-10S	2054800.5	16084453.6	S	X	A		
FS-110	2054796.2	16083874.3	S	X	A		
FS-310	2054632.6	16083448.5	S				
DNR-1	2054690.2	16083338.1	S				
F-5	2053184.7	16085742.6	S		A		X
GW-10S	2053066.9	16083334.0	S	X	A		X
GW-11	2054337.5	16085384.6	S		A		X
GW-14S	2052312.9	16084237.8	S		A		X
GW-6S	2052479.3	16084929.0	S				
GW-9S	2052760.4	16082424.0	S	X	A		X
K-10	2051760.3	16082990.2	S				
K-11	2055129.8	16083390.0	S		A		X
K-13	2055106.1	16084383.4	S		A		X
K-14	2054198.5	16083580.6	S				
K-17	2053660.2	16080928.6	S				
K-21	2056132.9	16083160.3	S				
K-22	2052732.9	16083361.1	S				X
K-6S	2054305.6	16082497.1	S				X
MW-1	2054133.9	16082616.0	S				
MW-2	2054078.8	16082446.3	S				
MW-3	2054067.1	16083225.3	S				
MW-5	2054319.4	16083085.2	S			X	X
MW-6	2053781.3	16083144.2	S				
P-1	2053864.0	16084771.5	S				
P-2	2053868.3	16085677.3	S				
SLW-1	2054608.9	16084745.6	S	X			
SLW-10S	2054132.2	16083590.4	S				
SLW-11	2053377.9	16082449.0	S				
SLW-12S	2053829.1	16083600.1	S				
SLW-16	2052526.6	16083004.9	S				
SLW-21S	2053498.2	16080917.9	S				
SLW-23	2055174.7	16082098.0	S		A		
SLW-25	2053563.5	16081767.8	S	X	A		X
SLW-31	2053808.2	16083974.3	S	X	A		X
SLW-32	2054646.0	16082304.1	S				
SLW-33	2056120.7	16083636.3	S				
SLW-37	2056269.8	16083780.7	S				
SLW-39	2056505.4	16083616.7	S				
SLW-40	2056442.9	16083872.6	S				
SLW-43	2055865.8	16083993.9	S				
SLW-44	2055761.4	16084121.9	S				
SLW-45	2055619.9	16084289.2	S				
SLW-46	2055864.5	16083685.5	S				
SLW-48	2055701.6	16083921.8	S				
SLW-6	2053954.7	16082704.8	S				
SLW-8	2054090.2	16082209.5	S				
SLW-9	2053806.3	16082296.8	S	X	A		X
FWMW-16	2053089.8	16084273.1	S	X	A		X
FWMW-9	2052993.9	16083935.2	S	X	A		X
F-1S	2051945.1	16085109.4	S		A		X
I-1	2052837.5	16084063.2	UI	X	A	X	
I-2	2053259.1	16084604.5	UI	X	A	X	
I-3	2053686.7	16085096.6	LI	X	A	X	
I-4	2053790.2	16083794.1	UI	X	A	X	



Table 1 LDFA Monitoring Plan

Well	Easting ¹	Northing ¹	Aquifer ²	Compliance Wells ³	VOC Monitoring ³	Chromium Monitoring since March 2006 ⁴	Water Level Monitoring ⁵
I-5	2054136.8	16084584.8	UI	X	A	X	
I-6	2054223.1	16083705.6	UI	X	A	X	
I-7	2054340.4	16085405.0	UI	X	A	X	
I-8	2054293.1	16086333.5	UI	X	A	X	
I-9	2052093.6	16083174.1	UI	X	A	X	
I-10	2052643.2	16083315.1	UI	X	A	X	
I-11	2053734.2	16085096.6	UI	X	A	X	
DNR-14D	2054888.0	16084477.7	UI	X	A		X
F-1D	2051939.6	16085106.1	UI		A		
F-3	2053772.5	16086286.1	UI		A		X
F-6	2051036.7	16083691.0	UI		Q	X	
F-7	2053177.1	16085752.5	LI		A		X
F-2	2053825.0	16085175.0	UI	X		X	X
GW-10D	2053072.2	16083321.3	UI	X			
GW-13	2055950.4	16085020.8	LI				
GW-14D	2052304.9	16084239.1	UI		A		X
GW-17	2054998.0	16085388.7	LI	X	A		X
GW-18	2056405.4	16085491.7	LI				
GW-1D	2056362.9	16080979.2	UI				
GW-2	2056782.2	16082745.3	UI				
GW-20	2055046.4	16082370.3	UI				
GW-3	2057615.9	16083922.1	LI				
GW-6D	2052481.7	16084922.4	UI				
GW-7	2054652.0	16087281.7	UI				
GW-8	2051762.7	16082982.3	UI				
GW-9D	2052770.3	16082424.4	UI	X			
CMS-5W	2050546.2	16083572.3	UI		Q	X	X
K-12	2055125.8	16083901.4	UI			X	X
K-15	2054191.2	16084056.3	UI				
K-16D	2053397.0	16084079.9	UI	X	A		X
K-23	2055704.7	16083460.6	UI				
K-24	2055291.3	16084651.9	UI				
K-6D	2054304.0	16082495.5	UI				
K-8	2054233.7	16083078.9	UI				
MRW-3	2057586.5	16083908.6	UI				
MW-10	2054455.8	16084494.1	UI			X	X
MW-11	2053875.5	16084774.8	UI			X	X
MW-12	2054151.2	16083757.5	UI				
MW-13	2054116.3	16084561.5	UI			X	X
MW-14	2054495.8	16084105.5	UI				
MW-15	2053911.6	16084014.0	UI				
MW-16	2053370.1	16084542.7	UI				
MW-17	2054690.8	16083359.9	UI			X	X
MW-18	2054134.9	16084725.6	UI	X		X	X
MW-20	2054101.4	16085280.0	UI			X	X
MW-7	2054262.3	16084305.6	UI			X	X
MW-8	2054095.2	16083225.0	UI	X	A		X
MW-9	2053877.8	16084325.6	UI				
SLW-10D	2054125.6	16083576.0	UI				
SLW-12D	2053822.6	16083577.3	UI				
SLW-13	2054660.3	16086146.2	UI				
SLW-15	2053805.5	16083134.4	UI				
SLW-17	2054965.8	16086196.2	UI				
SLW-18	2054420.5	16083095.0	UI				
SLW-2	2054594.0	16085434.2	UI				
SLW-20	2054661.5	16086350.7	UI				
SLW-22	2053483.4	16083070.4	UI				
SLW-26	2056366.5	16081707.5	UI				
SLW-27	2055681.4	16083316.3	UI				
SLW-29	2057019.1	16083863.8	UI				



Table 1 LDFA Monitoring Plan

Well	Easting ¹	Northing ¹	Aquifer ²	Compliance Wells ³	VOC Monitoring ³	Chromium Monitoring since March 2006 ⁴	Water Level Monitoring ⁵
SLW-3	2054800.3	16085386.6	UI				
SLW-34	2055462.8	16086189.8	UI		Q	X	
SLW-35	2052957.3	16086363.7	UI		A		X
SLW-4	2054656.9	16085775.1	UI				X
SLW-41	2056083.0	16083859.4	UI				
SLW-47	2056206.7	16083993.9	UI				
SLW-49	2054198.2	16082789.9	UI				
SLW-5	2054495.0	16085778.7	UI				
SLW-50	2055975.8	16084181.0	UI				
SLW-51	2055869.0	16084482.8	UI				
SLW-52	2056106.3	16084335.2	UI				
SLW-7	2053881.6	16084988.0	UI				

Notes

- 1 Coordinates sources.
 - a) December 2006 GPS City of Cadillac Survey in NAD 83, STATE Plane "City_Wells NAVD88.xls"
 - b) Extrapolated from "Dragun Northing_Easting Wells.xls," using a 2-point transformation to place in the NAD83 UTM z16N coordinate system. Extrapolated coordinates were corrected to match December 2006 survey by offset of UTM meters X=11.18 and UTM meters Y offset=-0.16 on estimates from "Dragun Northing_Easting Wells.xls" that were surveyed by Dragun in 2000/2002, and by offset of UTM meters X=10.79 and UTM meters Y offset=4.74 for estimate from "Dragun Northing_Easting Wells.xls" not surveyed by Dragun in 2000/2002
 - c) Extrapolated from EC Jordan, 1987, Remedial Investigation Report using a 2-point transformation to place in the NAD83 UTM z16N coordinate system
 - d) LDFA.xls
- 2 S = Shallow; UI = Upper Intermediate; LI= Lower Intermediate
- 3 Source = Tetra Tech MPS, 2005 - 2004 Annual Performance Monitoring Report for LDFA Northern Kysor Superfund Site. July. Q= Quarterly, A=Annual
- 4 Source = Chromium data collected on a quarterly basis since March 2006 in preparation for closure (Larry Campbell, LDFA, City of Cadillac, personal communication)
- 5 Based on data available in 2004



Cadillac High School MME Chief Administrator Penny Phelps works on putting together packets for proctors behind the closed door of a room at the high school. The MME was administered this week on Monday, Tuesday and today.

RICK CHARMOLI | CADILLAC NEWS

Testing process an all-consuming task spanning several days

BY RICK CHARMOLI
CADILLAC NEWS

CADILLAC — Here is a test question for you to ponder.

If there are three days of testing every March, how long would it take to

sigh of relief for people like Penny Phelps; there is only one more day to go. Phelps is the MME Chief Administrator for Cadillac High School, and she was charged with making sure the test went on with-

ated with the test.

From that point, the school received monthly updates from the state. Eventually, the names to be placed on the school's test roster had to be submitted to ensure the right amount

school student taking the test and two new students who were registered just last week.

"Getting their information processed and on the rolls is important," Phelps said.

at school during the three days of testing.

"It is a unique situation because of the spacing issues, and things have to be quiet. We can't have students passing in the halls and no bells," Bruggema

- **Acceptable photo ID** — must meet all of the following criteria: Examples: driver's license, passport, school ID; Current (not expired); Includes your first and last names; Issued by school, employer or government agency; Current photo clearly recognizable
- **A calculator for mathematics assessments only** (optional)
- **If you use a calculator**, be sure to bring a calculator that you are comfortable using, along with fresh batteries. Check the Prohibited Calculator List at www.act.org to make sure that your calculator is permitted for use on the MME mathematics tests.
- **A watch** (if you wish to pace yourself). Alarms must be turned off.

WHAT NOT TO BRING

- Cell phone, pager, iPod



EPA Begins Review of the Kysor Industrial Corp. and Northern Plating Superfund sites Cadillac, Michigan

U.S. Environmental Protection Agency is conducting a five-year review of the Kysor Industrial Corp. and Northern Plating Superfund sites. The sites are located in southern quarter of an industrial park on the northwest side of Cadillac. Kysor Industrial is adjacent to Wright Street and Northern Plating is on Sixth Street. The Superfund law requires regular checkups of sites that have been cleaned up — with waste managed on-site — to make sure the cleanup continues to protect people and the environment. This is the third five-year review of the Kysor site and the fourth of the Northern Plating site.

EPA will check to see if institutional controls are protecting the neighboring subdivision, including any private wells. In addition, EPA will review the adequacy of current local ordinances in place to restrict ground water (underground water) use, stop new well installation. The review will also look into any activities that would compromise the initial cleanup. A reevaluation of air emission data will also be done to determine the effectiveness of the cleanup.

More information is available at the Cadillac-Westford Library, 411 S. Lake St., Cadillac. The review should be completed by mid July 2010.

The five-year review is an opportunity for you to tell EPA about site conditions and any concerns you have. Contact:

Sheila Sullivan
Remedial Project Manager
312-886-5251
sullivan.sheila@epa.gov

Robert Paulson
Community Involvement Coordinator
312-886-0272
paulson.robert@epa.gov

You may also call Region 5 toll-free at 800-621-8431, 9:30 a.m. to 5:30 p.m., weekdays

EPA Region 5 (SI-7J)
77 W. Jackson Blvd.
Chicago, IL 60604



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Ellens Equipment

104 W. Maple, McBain, MI 49657 231-8
Mon.-Fri. 8 AM-5 PM, Sat. 8 AM-12

ATTACHMENT 3

Documents Used to Prepare Five-Year Review Report

Charter Township of Haring, Wexford Co., 1986. Michigan Ordinance no. 86-01 Zoning Ordinance (Ord. No. 86-01, Sect. 2.1) 1986.

City of Cadillac, 1997. Ordinance No. 97-10, ch. 25, sect. 2.300), Title II, Chpt. 24, Water Supply and Wastewater Systems, recertification in August 2005.

City of Cadillac, March 25, 2010. Presentation to the LDFA Board.

Cadillac Local Development Finance Authority Remediation Project: Summary, via fax on January 18, 2005.

Fishbeck, Thompson, Carr & Huber, April 1995. Final Operation and Maintenance Manual, Volume I, Northernnaire/Kysor Sites Remediation.

Fishbeck, Thompson, Carr & Huber, April 1995. Final Performance Monitoring Plan for Northernnaire/Kysor Sites Remediation.

Fishbeck, Thompson, Carr & Huber Ingraham Property, January 1997. Remedial Action Report.

GeoTrans, Inc., August 17, 2009. Soil Vapor Extraction System Closure report, prepared for the City of Cadillac.

Longshore Environmental Services, 2000, 2002, and 2003. Annual Reports to City of Cadillac.

Mannik & Smith Group, Inc. April 2009. Technical Memorandum Supplemental Hydrogeologic Investigation, Rexair Inc., Cadillac, Michigan. Prepared for the Michigan Department of Natural Resources and Environment.

Michigan Department of Natural resources, 1996. Final Discharge Monitoring Permit, Discharge Requirements to Clam River.

Roy F. Weston Inc., February 2002. Additional Hydrogeologic Investigation Report, Northwest of the Rexair, Inc. Site, Cadillac Industrial Park.

S.S. Papadopoulos and Associates, Inc., April 2007. Hydraulic Capture Zone Evaluation, Kysor Northernnaire Superfund Site, Cadillac, Michigan, prepared for U.S. EPA.

TetraTech, Ann Arbor, MI, February 8, 2006. Kysor Preliminary Soil Investigation, prepared for LDFA.

U.S. EPA, September 29, 1989. Record of Decision, Northernnaire/Kysor sites, OU2.

U.S. EPA, March 3, 1994. Explanation of Significant Differences, Northernaire/Kysor Sites, OU2.

U.S. EPA, January 30, 1995. Unilateral Administrative Order for Remedial Action, RA-V-W-95-C-282.

U.S. EPA, September 23, 1996. Preliminary Site Close-Out Report, Kysor Industrial Corp.

U.S. EPA, July 26, 2000. First Five-Year Review Report, Kysor Industrial Corp. Site, Cadillac, Michigan.

U.S. EPA, September 2005. Remedial Action Work Plan for the Northernaire/ Kysor Sites, Cadillac, Michigan.

U.S. EPA, September 2005. Second Five-Year Review Report, Kysor Industrial Corp. Site, Cadillac, Michigan.

Various memoranda and correspondence between U.S. EPA, MDEQ, MDNRE, City of Cadillac, Haring Township, and others.

OSWER No. 9355.7-03B-P

Five-Year Review Site Inspection Checklist (Template)

I. SITE INFORMATION			
Site name: <u>Northernaire/Kysor</u>		Date of inspection: <u>May 21, 2010</u>	
Location and Region: <u>Cadillac, MI, Reg-5</u>		EPA ID: <u>MI0043681840(Kysor) MI002088</u> 3609 (North)	
Agency, office, or company leading the five-year review: <u>U.S. EPA</u>		Weather/temperature: <u>Sunny, warm ~60°F</u>	
Remedy Includes: (Check all that apply) ☐ Landfill cover/containment ☒ Access controls ☒ Institutional controls ☒ Groundwater pump and treatment ☐ Surface water collection and treatment ☒ Other <u>SVE</u> ☐ Monitored natural attenuation ☐ Groundwater containment ☐ Vertical barrier walls			
Attachments: ☐ Inspection team roster attached ☐ Site map attached			
II. INTERVIEWS (Check all that apply)			
1. O&M site manager <u>Jeff Dietlin</u> <u>Cadillac Utilities Director</u> <u>5/21/2010</u> Name Title Date Interviewed ☒ at site ☒ at office ☒ by phone Phone no. _____ Problems, suggestions; ☐ Report attached <u>No Problems</u>			
2. O&M staff _____ Name Title Date Interviewed ☐ at site ☐ at office ☐ by phone Phone no. _____ Problems, suggestions; ☐ Report attached _____			

3. **Local regulatory authorities and response agencies** (i.e., State and Tribal offices, emergency response office, police department, office of public health or environmental health, zoning office, recorder of deeds, or other city and county offices, etc.) Fill in all that apply.

Agency Wexford County, Dept. of Pub. Works

Contact John DiVozzo

Name

Director

Title

6/17/10

Date Phone no.

Problems; suggestions; ☐ Report attached

Agency MDNRE

Contact Nick Swiger

Name

Title

5/21/10

Date Phone no.

Problems; suggestions; ☐ Report attached

Agency MDNRE

Contact Steven Cunningham

Name

Cadillac
District Enf. Coord.

Title

5/21/10

Date Phone no.

Problems; suggestions; ☐ Report attached

Agency

Contact

Name

Title

Date Phone no.

Problems; suggestions; ☐ Report attached

4. **Other interviews (optional)** ☐ Report attached.

III. ON-SITE DOCUMENTS & RECORDS VERIFIED (Check all that apply)

- | | | | | |
|----|---|---|--|------------------------------|
| 1. | O&M Documents | | | |
| | ☒ O&M manual office/plant | ☒ Readily available | ☒ Up to date | ☐ N/A |
| | ☒ As-built drawings | ☒ Readily available | ☒ Up to date | ☐ N/A |
| | ☒ Maintenance logs - LDEFA | ☒ Readily available | ☒ Up to date | ☐ N/A |
| | Remarks <u>Daily Bench logs kept at LDEFA</u> | | | |
| 2. | Site-Specific Health and Safety Plan | ☒ Readily available | ☒ Up to date | ☐ N/A |
| | ☐ Contingency plan/emergency response plan | ☒ Readily available | ☒ Up to date | ☐ N/A |
| | Remarks <u>A Risk Management Program plan kept at Cadillac Utilities office</u> | | | |

3.	O&M and OSHA Training Records	☒ Readily available	☒ Up to date	☐ N/A
Remarks <u>In-house OSHA training records kept at PTK plant</u> <u>Eight-hour Incident Response training</u>				
4.	Permits and Service Agreements	☒ Readily available	☒ Up to date	☐ N/A
	☒ Air discharge permit	☒ Readily available	☒ Up to date	☐ N/A
	☒ Effluent discharge	☒ Readily available	☒ Up to date	☐ N/A
	☒ Waste disposal, POTW	☒ Readily available	☒ Up to date	☐ N/A
	☐ Other permits _____	☐ Readily available	☐ Up to date	☐ N/A
Remarks _____				
5.	Gas Generation Records	☐ Readily available	☐ Up to date	☐ N/A
Remarks _____				
6.	Settlement Monument Records	☐ Readily available	☐ Up to date	☐ N/A
Remarks _____				
7.	Groundwater Monitoring Records	☒ Readily available	☒ Up to date	☐ N/A
Remarks <u>Records kept at Cadillac utilities office</u>				
8.	Leachate Extraction Records	☐ Readily available	☐ Up to date	☐ N/A
Remarks _____				
9.	Discharge Compliance Records	☐ Readily available	☐ Up to date	☐ N/A
	☐ Air	☐ Readily available	☐ Up to date	☐ N/A
	☒ Water (effluent)	☒ Readily available	☒ Up to date	☐ N/A
Remarks _____				
10.	Daily Access/Security Logs	☒ Readily available	☐ Up to date	☐ N/A
Remarks <u>The treatment buildings are kept locked.</u> <u>Fences are locked</u>				
IV. O&M COSTS				
1.	O&M Organization	☐ State in-house	☐ Contractor for State	
		☐ PRP in-house	☐ Contractor for PRP	
		☐ Federal Facility in-house	☐ Contractor for Federal Facility	
	☒ Other <u>L DFA City of Cadillac, see report for costs</u>			

2.	O&M Cost Records	☒ Readily available ☒ Up to date ☐ Funding mechanism/agreement in place Original O&M cost estimate _____	☒ Breakdown attached	
Total annual cost by year for review period if available				
	From _____ Date	To _____ Date	_____ Total cost	☐ Breakdown attached
	From _____ Date	To _____ Date	_____ Total cost	☐ Breakdown attached
	From _____ Date	To _____ Date	_____ Total cost	☐ Breakdown attached
	From _____ Date	To _____ Date	_____ Total cost	☐ Breakdown attached
	From _____ Date	To _____ Date	_____ Total cost	☐ Breakdown attached

3.	Unanticipated or Unusually High O&M Costs During Review Period Describe costs and reasons:	<u>New Telemetry system installed for Monitoring</u> <u>Well system</u> _____ _____ _____
----	--	---

V. ACCESS AND INSTITUTIONAL CONTROLS ☐ Applicable ☐ N/A			
A. Fencing			
1.	Fencing damaged Remarks _____	☐ Location shown on site map ☐ Gates secured ☐ N/A <u>The SVE system is fenced, All wells and electrical systems</u> <u>on Monitoring wells are locked</u>	
B. Other Access Restrictions			
1.	Signs and other security measures Remarks _____	☐ Location shown on site map ☐ N/A	
C. Institutional Controls (ICs)			

1. Implementation and enforcement			
Site conditions imply ICs not properly implemented	☐ Yes	☒ No	☐ N/A
Site conditions imply ICs not being fully enforced	☒ Yes	☐ No	☐ N/A
Type of monitoring (e.g., self-reporting, drive by) _____			
Frequency <u>Periodically</u>			
Responsible party/agency <u>City of Cadillac, Haring Township, Wexford Co. DPA</u>			
Contact <u>Jeff Dietlin (Cadillac)</u>	<u>Utilities Mgr</u>	<u>5/21/10</u>	
Name <u>John Divo</u>	Title <u>Wexford Co</u>	Date	Phone no.
Reporting is up-to-date ☒ Yes ☐ No ☐ N/A			
Reports are verified by the lead agency ☒ Yes ☐ No ☐ N/A			
Specific requirements in deed or decision documents have been met ☒ Yes ☐ No ☐ N/A			
Violations have been reported ☐ Yes ☐ No ☐ N/A			
Other problems or suggestions: ☐ Report attached <u>Somewhat</u>			
2. Adequacy ☐ ICs are adequate ☐ ICs are inadequate ☐ N/A			
Remarks <u>ICs are somewhat adequate. Township and City are upgrading ordinances to require connection to public water</u>			
D. General			
1. Vandalism/trespassing ☐ Location shown on site map ☒ No vandalism evident			
Remarks _____			
2. Land use changes on site ☐ N/A			
Remarks <u>None</u>			
3. Land use changes off site ☐ N/A			
Remarks <u>None in general</u>			
VI. GENERAL SITE CONDITIONS			
A. Roads ☐ Applicable ☐ N/A			
1. Roads damaged ☐ Location shown on site map ☒ Roads adequate ☐ N/A			
Remarks _____			
B. Other Site Conditions			
Remarks _____			
VII. LANDFILL COVERS ☐ Applicable ☐ N/A			
A. Landfill Surface			

1.	Settlement (Low spots) Areal extent _____ Remarks _____	☐ Location shown on site map Depth _____	☐ Settlement not evident
2.	Cracks Lengths _____ Widths _____ Depths _____ Remarks _____	☐ Location shown on site map	☐ Cracking not evident
3.	Erosion Areal extent _____ Remarks _____	☐ Location shown on site map Depth _____	☐ Erosion not evident
4.	Holes Areal extent _____ Remarks _____	☐ Location shown on site map Depth _____	☐ Holes not evident
5.	Vegetative Cover ☐ Trees/Shrubs (indicate size and locations on a diagram) Remarks _____	☐ Grass ☐ Cover properly established	☐ No signs of stress
6.	Alternative Cover (armored rock, concrete, etc.) Remarks _____	☐ N/A	
7.	Bulges Areal extent _____ Remarks _____	☐ Location shown on site map Height _____	☐ Bulges not evident
8.	Wet Areas/Water Damage ☐ Wet areas ☐ Ponding ☐ Seeps ☐ Soft subgrade Remarks _____	☐ Wet areas/water damage not evident ☐ Location shown on site map ☐ Location shown on site map ☐ Location shown on site map ☐ Location shown on site map	Areal extent _____ Areal extent _____ Areal extent _____ Areal extent _____
9.	Slope Instability Areal extent _____ Remarks _____	☐ Slides ☐ Location shown on site map	☐ No evidence of slope instability
B. Benches (Horizontally constructed mounds of earth placed across a steep landfill side slope to interrupt the slope in order to slow down the velocity of surface runoff and intercept and convey the runoff to a lined channel.)			
1.	Flows Bypass Bench Remarks _____	☐ Location shown on site map	☐ N/A or okay

2.	Bench Breached	☐ Location shown on site map	☐ N/A or okay
Remarks _____			
3.	Bench Overtopped	☐ Location shown on site map	☐ N/A or okay
Remarks _____			
C. Letdown Channels ☐ Applicable ☐ N/A (Channel lined with erosion control mats, riprap, grout bags, or gabions that descend down the steep side slope of the cover and will allow the runoff water collected by the benches to move off of the landfill cover without creating erosion gullies.)			
1.	Settlement	☐ Location shown on site map	☐ No evidence of settlement
Areal extent _____		Depth _____	
Remarks _____			
2.	Material Degradation	☐ Location shown on site map	☐ No evidence of degradation
Material type _____		Areal extent _____	
Remarks _____			
3.	Erosion	☐ Location shown on site map	☐ No evidence of erosion
Areal extent _____		Depth _____	
Remarks _____			
4.	Undercutting	☐ Location shown on site map	☐ No evidence of undercutting
Areal extent _____		Depth _____	
Remarks _____			
5.	Obstructions	Type _____	☐ No obstructions
☐ Location shown on site map		Areal extent _____	
Size _____			
Remarks _____			
6.	Excessive Vegetative Growth	Type _____	
☐ No evidence of excessive growth			
☐ Vegetation in channels does not obstruct flow			
☐ Location shown on site map		Areal extent _____	
Remarks _____			
D. Cover Penetrations ☐ Applicable ☐ N/A			
1.	Gas Vents	☐ Active	☐ Passive
☐ Properly secured/locked		☐ Functioning	☐ Routinely sampled
☐ Evidence of leakage at penetration		☐ Good condition	
☐ N/A		☐ Needs Maintenance	
Remarks _____			
2.	Gas Monitoring Probes	☐ Properly secured/locked	☐ Functioning
☐ Evidence of leakage at penetration		☐ Routinely sampled	☐ Good condition
		☐ Needs Maintenance	☐ N/A
Remarks _____			

3.	Monitoring Wells (within surface area of landfill)	☐ Properly secured/locked	☐ Functioning	☐ Routinely sampled	☐ Good condition
		☐ Evidence of leakage at penetration		☐ Needs Maintenance	☐ N/A
Remarks _____					
4.	Leachate Extraction Wells	☐ Properly secured/locked	☐ Functioning	☐ Routinely sampled	☐ Good condition
		☐ Evidence of leakage at penetration		☐ Needs Maintenance	☐ N/A
Remarks _____					
5.	Settlement Monuments	☐ Located	☐ Routinely surveyed	☐ N/A	
Remarks _____					
E. Gas Collection and Treatment		☐ Applicable	☐ N/A		
1.	Gas Treatment Facilities	☐ Flaring	☐ Thermal destruction	☐ Collection for reuse	
		☐ Good condition	☐ Needs Maintenance		
Remarks _____					
2.	Gas Collection Wells, Manifolds and Piping	☐ Good condition	☐ Needs Maintenance		
Remarks _____					
3.	Gas Monitoring Facilities (e.g., gas monitoring of adjacent homes or buildings)	☐ Good condition	☐ Needs Maintenance	☐ N/A	
Remarks _____					
F. Cover Drainage Layer		☐ Applicable	☐ N/A		
1.	Outlet Pipes Inspected	☐ Functioning	☐ N/A		
Remarks _____					
2.	Outlet Rock Inspected	☐ Functioning	☐ N/A		
Remarks _____					
G. Detention/Sedimentation Ponds		☐ Applicable	☐ N/A		
1.	Siltation	Areal extent _____	Depth _____	☐ N/A	
		☐ Siltation not evident			
Remarks _____					
2.	Erosion	Areal extent _____	Depth _____		
		☐ Erosion not evident			
Remarks _____					
3.	Outlet Works	☐ Functioning	☐ N/A		
Remarks _____					
4.	Dam	☐ Functioning	☐ N/A		
Remarks _____					

H. Retaining Walls		☐ Applicable	☐ N/A
1.	Deformations	☐ Location shown on site map	☐ Deformation not evident
	Horizontal displacement _____	Vertical displacement _____	
	Rotational displacement _____		
	Remarks _____		
2.	Degradation	☐ Location shown on site map	☐ Degradation not evident
	Remarks _____		
I. Perimeter Ditches/Off-Site Discharge		☐ Applicable	☐ N/A
1.	Siltation	☐ Location shown on site map	☐ Siltation not evident
	Areal extent _____	Depth _____	
	Remarks _____		
2.	Vegetative Growth	☐ Location shown on site map	☐ N/A
	☐ Vegetation does not impede flow		
	Areal extent _____	Type _____	
	Remarks _____		
3.	Erosion	☐ Location shown on site map	☐ Erosion not evident
	Areal extent _____	Depth _____	
	Remarks _____		
4.	Discharge Structure	☐ Functioning	☐ N/A
	Remarks _____		
VIII. VERTICAL BARRIER WALLS		☐ Applicable	☐ N/A
1.	Settlement	☐ Location shown on site map	☐ Settlement not evident
	Areal extent _____	Depth _____	
	Remarks _____		
2.	Performance Monitoring	Type of monitoring _____	
	☐ Performance not monitored		
	Frequency _____	☐ Evidence of breaching	
	Head differential _____		
	Remarks _____		
IX. GROUNDWATER/SURFACE WATER REMEDIES		☒ Applicable	☐ N/A
A. Groundwater Extraction Wells, Pumps, and Pipelines		☒ Applicable	☐ N/A
1.	Pumps, Wellhead Plumbing, and Electrical	☒ All required wells properly operating ☐ Needs Maintenance ☐ N/A	
	☐ Good condition		
	Remarks <i>Shallow well S-7 is dry and has not been operating</i>		
	<i>Well S-3 platform was crushed on one edge. Electrical panels and</i>		
	<i>telemetry in good shape</i>		
2.	Extraction System Pipelines, Valves, Valve Boxes, and Other Appurtenances	☒ Good condition ☐ Needs Maintenance	
	Remarks _____		

3.	Spare Parts and Equipment		
	☒ Readily available	☐ Good condition	☐ Requires upgrade ☐ Needs to be provided
	Remarks _____		
B. Surface Water Collection Structures, Pumps, and Pipelines ☐ Applicable ☐ N/A			
1.	Collection Structures, Pumps, and Electrical		
	☒ Good condition	☐ Needs Maintenance	
	Remarks _____		
2.	Surface Water Collection System Pipelines, Valves, Valve Boxes, and Other Appurtenances		
	☒ Good condition	☐ Needs Maintenance	
	Remarks _____		
3.	Spare Parts and Equipment		
	☒ Readily available	☐ Good condition	☐ Requires upgrade ☐ Needs to be provided
	Remarks _____		
C. Treatment System ☐ Applicable ☐ N/A			
1.	Treatment Train (Check components that apply)		
	☒ Metals removal	☐ Oil/water separation	☐ Bioremediation
	☒ Air stripping	☐ Carbon adsorbers	
	☐ Filters _____		
	☐ Additive (e.g., chelation agent, flocculent) _____		
	☐ Others _____		
	☒ Good condition	☐ Needs Maintenance	
	☒ Sampling ports properly marked and functional		
	☒ Sampling/maintenance log displayed and up to date		
	☒ Equipment properly identified		
	☐ Quantity of groundwater treated annually <u>~2.7 MG/D</u>		
	☐ Quantity of surface water treated annually _____		
	Remarks _____		
2.	Electrical Enclosures and Panels (properly rated and functional)		
	☐ N/A	☒ Good condition	☐ Needs Maintenance
	Remarks _____		
3.	Tanks, Vaults, Storage Vessels		
	☐ N/A	☒ Good condition	☐ Proper secondary containment ☐ Needs Maintenance
	Remarks _____		
4.	Discharge Structure and Appurtenances		
	☐ N/A	☒ Good condition	☐ Needs Maintenance
	Remarks _____		

5.	Treatment Building(s) ☐ N/A ☒ Good condition (esp. roof and doorways) ☐ Needs repair ☒ Chemicals and equipment properly stored Remarks _____
6.	Monitoring Wells (pump and treatment remedy) ☒ Properly secured/locked ☐ Functioning ☒ Routinely sampled ☒ Good condition ☒ All required wells located ☐ Needs Maintenance ☐ N/A Remarks _____
D. Monitoring Data	
1.	Monitoring Data ☒ Is routinely submitted on time ☒ Is of acceptable quality
2.	Monitoring data suggests: ☒ Groundwater plume is effectively contained ☒ Contaminant concentrations are declining
D. Monitored Natural Attenuation	
1.	Monitoring Wells (natural attenuation remedy) ☐ Properly secured/locked ☐ Functioning ☐ Routinely sampled ☐ Good condition ☐ All required wells located ☐ Needs Maintenance ☒ N/A Remarks _____
X. OTHER REMEDIES	
If there are remedies applied at the site which are not covered above, attach an inspection sheet describing the physical nature and condition of any facility associated with the remedy. An example would be soil vapor extraction.	
XI. OVERALL OBSERVATIONS	
A.	Implementation of the Remedy Describe issues and observations relating to whether the remedy is effective and functioning as designed. Begin with a brief statement of what the remedy is to accomplish (i.e., to contain contaminant plume, minimize infiltration and gas emission, etc.). <i>Remedy is to contain plume and remove source of Contam from Soil.</i> <i>SVE system effective and is being decommissioned</i> <i>Groundwater extraction and treatment system is well-maintained and effectively capturing the groundwater plume.</i> <i>The city is in process of planning a new well field due to age of the existing system.</i>
B.	Adequacy of O&M

Describe issues and observations related to the implementation and scope of O&M procedures. In particular, discuss their relationship to the current and long-term protectiveness of the remedy.

O&M practices are effective

*The remedy is protective but needs stricter ordinances
regarding private well use*

C. Early Indicators of Potential Remedy Problems

Describe issues and observations such as unexpected changes in the cost or scope of O&M or a high frequency of unscheduled repairs, that suggest that the protectiveness of the remedy may be compromised in the future.

None -

D. Opportunities for Optimization

Describe possible opportunities for optimization in monitoring tasks or the operation of the remedy.

**ATTACHMENT 6 – INSPECTION DOCUMENTATION
PHOTOS**

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
May 21, 2010



Figure 1 – Electrical panel for Local Development Finance Authority (LDFA) extraction well I-7, located about 1,200 feet northwest of the edge of the Cadillac city well field. The antenna on top right side of panel is radio telemetry control system.



Figure 2 – View of shallow aquifer well GW-11 located about 500 feet northwest of the Four Winns/A.H. Joynt facility. The well is used for water level and VOC monitoring.

Five-Year Review Site Inspection – Kysor and Northernaire Superfund Sites,
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Figure 3 – View of the Four Winns facility as seen from the intersection of Seventh Avenue and Fourteenth Street, where well I-7 is located.



Figure 4 – View of electrical panel for LDFA intermediate depth extraction well I-8, located at the intersection of Seventh Avenue and Grahek Street in the North Park subdivision area. The well is about 2,250 feet northwest of, and the farthest distance from, the Cadillac well field.

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Figure 5 – View of well SLW-35. This shallow-aquifer well is located about 1,025 feet north of Thirteenth Street and northwest of the North Park subdivision area.



Figure 6 – View of electrical panel for LDFA intermediate extraction well I-5, located north of the Michigan Rubber facility and 1,000 feet northwest of the Four Star facility. Well I-5 is the dedicated chromium extraction well.

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Figure 7 – A view of the back of the Four Winns Frisbie facility, which manufactures recreational boats. The electrical panel for the LDFA shallow-aquifer extraction well S-6 is visible at the far right background of the photo.



Figure 8 - A closer view of the back side of the Four Winns Frisbie facility.

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Figure 9 – A view of LDFA shallow-aquifer extraction well S-3, located about 1,250 feet west of the south west corner of the Cadillac city well field. The well cover in the middle of the photograph is cracked around the metal cover. The electrical panel is at the top right corner of the photo.



Figure 10 – A view of the south side of the LDFA groundwater treatment plant. The two air stripping towers are visible in the center mid-ground of the photo. The plant is in front of the towers.

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Figure 11 –View from atop the air-stripping towers looking south toward the Kysor plant (now owned by Borg-Warner) just south of the railroad tracks. The Cadillac city Department of Public Works yard is immediately north (below) the railroad tracks. The soil vapor extraction (SVE) system (not visible in photo) is located between the Kysor plant and the railroad tracks.

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Figure 12 – View of one of the two GAC treatment tanks inside the LDFA treatment plant. The chromium stream from dedicated extraction well I-5 and the VOCs that may also be present in the extracted groundwater from I-5 are processed in these tanks. Most of VOC stream does not enter these vessels.

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Figure 13 – View inside the treatment plant of the joining of the dedicated chromium well (I-5) extraction stream with the VOC extraction wells. The thin vertical pipe carries the combined streams to the GAC filter tanks before entering the air strippers.

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Figure 14 – View from atop the air stripping towers looking north. The Northernnaire facility is in the mid-ground. The Paulstra/CRC facility (long low red building) is to the right and immediately northeast of Northernnaire.



Figure 15 – View from atop the air stripping facility looking east. The Mitchell Corporation is in the background.

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Figure 16 – View from atop the air stripping tower looking west. The Co-Gen power plant is barely visible in the far back center of the photo.



Figure 17 – View of the SVE system, now decommissioned after meeting soil cleanup goals. The system is located directly north of the Kysor plant and south of the railroad tracks. The foreground area is capped with a PVC liner. The concrete-asphalt pad is visible behind the blue GAC tanks.